Development of a Microfluidic Nozzle to Generate Water Sheet Jets for Cooling Sharp Leading Edges

Research Thesis

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The author of this thesis states that the research, including the collection, processing and presentation of data, addressing and comparing to previous research, etc., was done entirely in an honest way, as expected from scientific research that is conducted according to the ethical standards of the academic world. Also, reporting the research and its results in this thesis was done in an honest and complete manner, according to the same standards.

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements. This dissertation contains fewer than 65,000 words including appendices, bibliography, tables and equations and has fewer than 100 figures.

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Nomenclature

Acronyms / Abbreviations

CFD	Computational Fluid Dynamics
CHF	Critical Heat FLux
CW	Continuous Wave
DRIE	Deep Reactive Ion Etching
GCI	Grid COnvergence Index
LS	Liquid Sheet
PIV	Particle Image Velocimetry
RIE	Reactive Ion Etching
TPS	Thermal Protection System
UHTC	Ultra High Temperature Ceramics

Greek Symbols

α	Aspect ratio
μ	Dynamic viscosity [Pa·s]
μ_{eff}	Effective viscosity
ν	Kinematic viscosity [m^2/s]
ϕ	Azimuthal angle [θ]
ϕ_l	Any general variable
ρ	Density [kg/m^3]
σ	Surface tension [N/m]

θ Angle [°]

Latin Symbols

S Surface integral [m^2]

V Volume integral [m^3]

Ca Capillary Number [-]

$d_\circ$ Droplet diameter [m]

d_L Diameter of detaching ligament [m]

f Total wavegrowth [-]

h Thickness of the liquid sheet [m](from literature)[57]

K hx

k ht

L Characteristic length [m]

l_j Length of the liquid sheet[m](from literature)[57]

l_s Length of the liquid sheet [m]

$l_{c,a}$ Calculated actual length-after polishing [m]

$l_{c,n}$ Calculated nominal length-before polishing [m]

N_l Shape function for node l

n_j Outward normal vector

N_{nodes} Total number of nodes [-]

p Pressure Field [Pa]

Q Volumetric flow rate [m^3/s]

r Radial distance [m]

R_j Jet radius [m]

Re Reynolds Number [-]

s Ratio of gas density to liquid density [-]

t Thickness of the nozzle [m]

t_a	Actual thickness [m]
t_n	Nominal thickness [m]
U	Velocity [m/s]
u_j	Velocity in j_{th} spatial direction [m/s]
u_n	Normal component of velocity [m/s]
u_i	Velocity component in i_{th} spatial direction [m/s]
w_o	Outlet width [m]
w_s	Width of the liquid sheet [m]
$w_{c,a}$	Calculated actual width-after polishing [m]
$w_{c,n}$	Calculated nominal width-before polishing [m]
We	Weber Number [-]
x	Distance from the apex of the primary sheet [m]
X_b	Breakup length [m]
x_i	Spatial coordinate [m]
x_j	j_{th} spatial coordinate [m]

Abstract

This thesis presents a comprehensive investigation into the dynamics of liquid sheets produced by a specialized microfluidic device, engineered for effective impingement on the high-temperature surfaces of sharp leading edges. The study thoroughly examines the influence of varying nozzle exit aspect ratios, geometrical configurations, and flow rates on the hydrodynamic characteristics, stability, and overall performance of these liquid sheets under different conditions.

The research integrates numerical simulations to model the complex internal flow dynamics within the nozzle, with a particular focus on the interaction between the working liquid and the geometrical features of the nozzle. These simulations are meticulously validated against experimental data, which includes precise measurements of velocity profiles and flow uniformity. By addressing critical flow instabilities and optimizing operational conditions, the study provides in-depth insights into the behavior of liquid sheets, offering a reliable framework for predicting and controlling their dynamics.

Furthermore, this work delves into the detailed fabrication process of the microfluidic device, offering a step-by-step methodology that highlights material selection, microfabrication techniques, and assembly procedures. This ensures not only the reproducibility of the device but also its potential applicability in future studies. The findings of this thesis significantly advance the understanding of liquid sheet behavior, paving the way for the development of cutting-edge cooling technologies capable of withstanding extreme thermal environments, such as those encountered in aerospace and hypersonic applications.

Keywords- microfluidics, sharp leading edge, hypersonic cooling, liquid sheets

Chapter 1

Introduction

1.1 Aerodynamic heating in hypersonic vehicles

High-speed transport vehicles operating in the hypersonic regime have the potential to revolutionize travel by significantly reducing flight times over long distances.[1] Due to their high maneuverability and ability to cover vast distances in short periods, hypersonic vehicles have also become increasingly important in military applications.[2] However, the aerodynamic, thermal, and propulsive performance of hypersonic vehicles is an active area of research, as the challenges associated with achieving high Mach numbers have yet to be fully overcome.[3]

In the 1990s, advancements in higher Mach number aircraft made significant progress in understanding the physics of hypersonic flows through computational methods. Limited experimental facilities were, however, available to validate these results. The development of Gladden's Hot Gas Facility at NASA's Lewis Research Center marked a significant advancement in experimental results. This facility could provide heat fluxes up to 450 W/cm^2 on flat surfaces and up to 5000 W/cm^2 at the leading edge stagnation point of a structure in a supersonic flow. This enabled the evaluation of coating erosion, oxidation, and cooling techniques, providing a means for conducting hypersonic ground testing.[4] Another example of flight testing for hypersonic vehicles is NASA's X-43 which utilized innovative scramjet technologies to achieve flight speeds ranging from Mach 7 to 10.

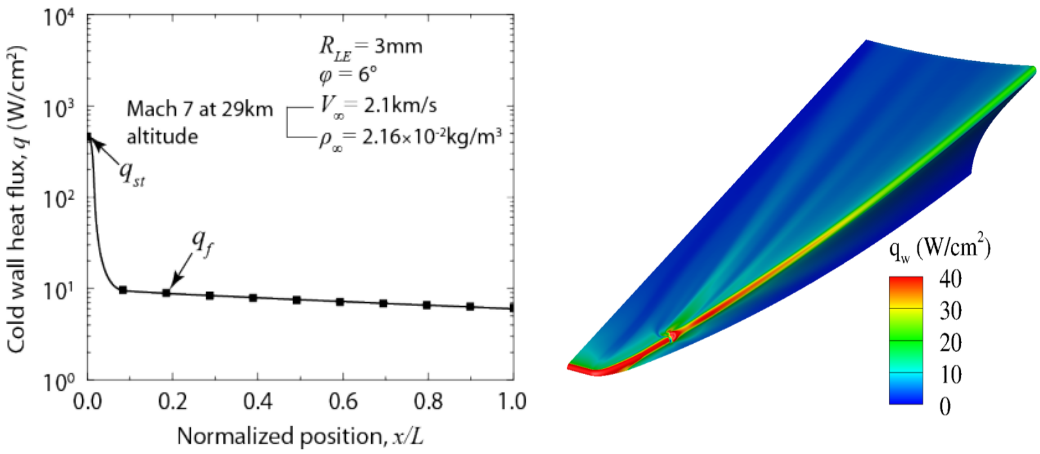
Figure 1.1(a) shows the X-43 vehicle, depicted in an animation illustrating the vehicle's separation from the Pegasus booster.[5] Hypersonic aircraft usually use kerosene as fuel. However, recent advancements favor hydrogen (H2) as a more environmentally friendly option [6]. For instance, the Swiss-based startup Destinus[7] is working on maximizing efficiency using hydrogen as a cryogenic fuel, as shown in Figure 1.1(b).

These vehicles typically cruise in the stratosphere, where the ambient temperature is approximately 250 K.[8] As the Mach number increases and the vehicle accelerates, significant aerodynamic heating occurs.

(a) (b)

Fig. 1.1 Hypersonic vehicles operating at speeds above Mach 5. (a) X-43 hypersonic vehicle (Image from NASA)[5] (b) Destinus's hypersonic vehicle (Destinus D) [7]

A major challenge is therefore managing the heat load that accumulates during flight, which must be precisely controlled to ensure reliable and efficient operation.[9] When the Mach number exceeds 5, the vehicle enters the hypersonic regime, where aerodynamic heating predominantly influences the flow physics.[10] This heating arises from viscous dissipation, friction and compression, which occur due to the formation and interaction of shock waves around the vehicle.[11] [12]



(a) (b)

Fig. 1.2 Heat flux around a hypersonic vehicle: (a) heat flux for a wedge with a leading edge radius of 3 mm at Mach 7,[12]and (b) numerical simulation of surface heat on the leading edge[11]

Figure 1.2a shows the heat flux distribution around a vehicle's leading edge traveling at Mach 7 at an altitude of 29 km above sea level, considering the temperature of the gas approaching the leading edge is lower than the wall temperature. A peak heat flux of 460 W/cm^2 occurs at the tip and rapidly decreases along the curved radius toward the flat surface.[13] Due to the vehicle's high speed, even small impacts by droplets can significantly enhance heat flux in a given region, as illustrated in Figure 1.2b. Managing this heat load is crucial for vehicles operating in the hypersonic regime while keeping in consideration the weight of these vehicles.[14]

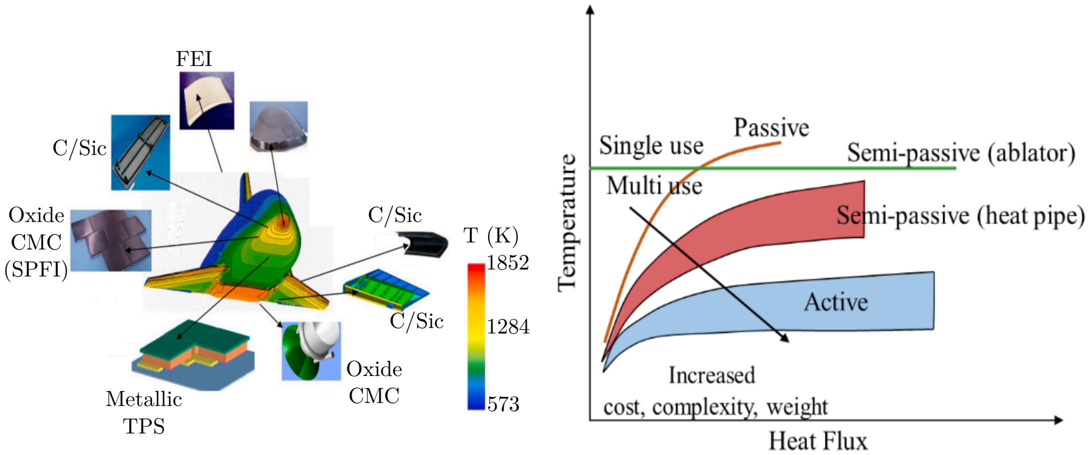
In extreme environmental conditions, the primary aero-thermal concerns involve several critical factors: the transition of the boundary layer from laminar to turbulent flow [15], shock-shock interactions at the leading edge [16], and the materials used for the vehicle's structure.[17] Delaying the transition in the hypersonic boundary layer can significantly reduce aerodynamic heating on the vehicle and decrease skin friction on the exterior surface.[18] Additionally, shock-shock interactions pose significant challenges as they lead to increased drag, resulting in substantial aerodynamic heating. Since drag is dependent on surface area, one straightforward way to mitigate this issue is by reducing the surface area, often achieved by designing sharp leading edges. Designing a leading edge involves the challenge of making it sharp enough to achieve acceptable aerodynamic and propulsion efficiency [19]. However, this comes with a trade-off: as the leading-edge radius decreases, aerodynamic heating intensifies because heat flux is inversely proportional to the square root of the leading-edge radius,[20] which is typically on the order of 1 to 5 mm.

The wall heat flux is significantly influenced by both the vehicle's speed and altitude. For instance, a vehicle flying at an altitude of 50 km at Mach 6 will generate substantially more heat compared to a similar flight at 100 km. This difference is primarily due to the reduced ambient pressure and density encountered at higher altitudes, which affects the overall thermal environment.

Considering all the physical phenomena that arise in the hypersonic flight range, especially at the leading edges, the small thickness required for high aerodynamic efficiency leaves no room for bulky active cooling systems. **The objective of this MSc thesis is to develop a microfluidic nozzle capable of generating a very thin impingement liquid sheet jet for thermal management purposes**.

1.2 Cooling methods for hypersonic leading edges

Thermal management in hypersonic leading edges is crucial due to the extreme temperatures caused by aerodynamic heating. To mitigate the intense heat generated during hypersonic flights, especially on leading edges exposed to high heat fluxes, advanced cooling techniques need to be applied. In general, both **active**



(a) (b)

Fig. 1.3 Thermally heated zones and cooling methods: (a) Thermal protection system (TPS) and hot structure hardware proposed by ESA [22] (b) Thermal management techniques[22]

and **passive cooling** methods are employed to handle the intense thermal loads, especially on leading edges and other high-heat areas.[21]

Passive cooling relies on materials and design features that naturally absorb, distribute, and dissipate heat without external power or mechanical systems.[22] These methods are generally simpler, more reliable, and require less maintenance. Active cooling, on the other hand, involves the use of mechanical systems and external power sources to manage heat actively using coolants. These systems are often more complex but can provide more precise temperature control and increased heat transfer capabilities. There are a number of factors that have to be taken into consideration while deciding which method of cooling is to be utilized for any particular geometry.[23] As shown in Figure 1.3(a), there are zones with different thermal ranges, so these regions are to be accessed through different cooling methods in order to achieve the required heat removal, as shown in Figure 1.3(b).

1.2.1 Passive Cooling

For passive cooling, Thermal Protection Systems (TPS) are typically developed using insulating and ablative materials as well as thermal shields. High heat flux-prone areas are constructed from materials with extremely low thermal conductivity, such as ceramics, which have very high melting points (greater than 3473.15 K), and by using ablative coatings.[24] Ablative coatings are materials that erode under gas shear. A commonly used ablative material is quartz, which softens, melts, and flows under high-enthalpy environments.[25] Generally, the heat flux to the wall is used to determine the thickness of passive cooling materials.[26] These cooling systems are cost-effective because they do not require additional control inputs or external energy. However passive cooling methods offer a lower degree of reusability for high-performance thermal protection and their capabilities are limited by the material's properties.

1.2.2 Active Cooling

Although passive techniques can isolate a considerable amount of heat, the aerodynamic heat transmitted to the interior of the structure must still be addressed. Active cooling is a more efficient and dynamic method for cooling highly heated surfaces.[27] Compared to the heavy heat shields and insulation used in passive cooling, active cooling offers much higher efficiency.[28]. Active cooling involves directing a coolant onto the heated surface, and this cooling fluid should ideally cool the surfaces primarily through convection. The major types of active cooling schemes used for hypersonic vehicles are[29]:

- Jet impingement
- Spray cooling
- Film cooling
- Transpiration cooling
- Regenerative active
- Radiative cooling

Although the above mentioned cooling schemes can remove greater amount of heat compared to passive cooling methods, limitations include coolant consumption and low heat dissipation.[30]

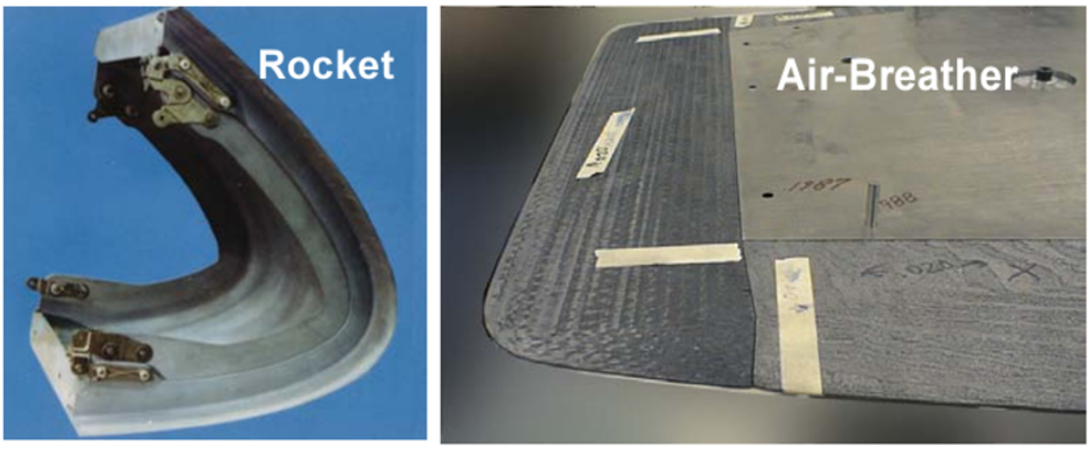


Fig. 1.4 Blunt edge of a rocket based vehicle vs sharp LE of X-43 [23]

1.2.3 Hybrid cooling schemes

The choice of cooling method depends on the thermal load the surface must endure. Figure 1.4 shows two different hypersonic leading edges. Although both leading edges are oriented for hypersonic applications, there is a difference in how the leading edge is designed. The figure shows the difference in the radius of curvature between blunt and sharp leading edges, emphasizing that the cooling requirements differ for each type.

Developing novel, reusable, and efficient thermal protection systems for sharp leading edges is challenging. Today, a combination of active and passive cooling methods is often used to take advantage of the strengths of both approaches. For instance, recent innovations include using Ultra High-Temperature Ceramics (UHTC) porous media for transpiration cooling, which can potentially reduce or even eliminate the need for ablative materials [31].

Previous studies have shown that the heat transfer coefficient, total drag, and total lift coefficient are sensitive to the thickness of the leading edge. Significant differences are observed when the leading edge radius is reduced [32]. Although materials for leading edges can withstand higher temperatures, experiments indicate that deformation can occur under high thermal loads. This deformation compromises structural integrity and leads to impaired flight performance [33], underscoring the importance of using more active cooling schemes.

To address these challenges, hybrid cooling techniques are increasingly popular for bridging the gaps between existing methods. Approaches such as combining film cooling or jet impingement with thermal coatings offer promising solutions for enhancing thermal management in hypersonic vehicles [34]. By integrating these strategies, the efficiency and reliability of the cooling system can be improved.

Chapter 2

Literature Review

Aerodynamic heating requires an effective cooling mechanism to protect materials from damage. This thesis explores the development of an internal active microfluidic cooling system utilizing a liquid sheet spray that impinges on the internal surfaces of a sharp leading edge. This chapter delves into the fundamental principles of impingement cooling, spray cooling, and liquid sheet jets which have not previously been applied for thermal management.

2.1 Jet Impingement Cooling and Spray Cooling

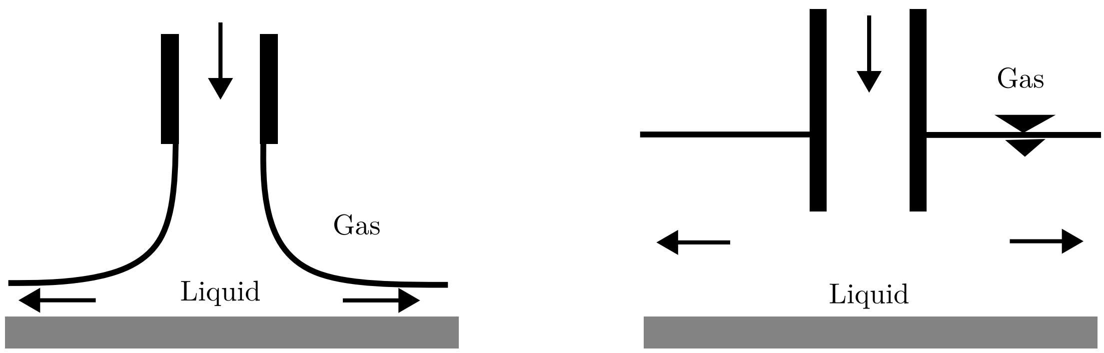
2.1.1 Impingement Cooling

Impingement cooling has been utilized for cooling the internal surface of leading edges, including hydrogen-cooled systems [23]. In this section, a brief overview of the physics governing impingement cooling is provided.

An impingement cooling system typically comprises an array of high-velocity jets directed at a target surface. For high-speed aerospace applications, hydrogen and helium are commonly used as coolants, although they exhibit relatively poor cooling performance [35]. In contrast, impingement cooling with supercritical CO_2 has been shown to achieve higher heat transfer rates compared to film and transpiration cooling methods [36]. Upon impact, the jets expand radially across the target surface, creating a thin boundary layer. Additionally, large-scale secondary flow structures formed around the leading edge further enhance the cooling effect. The roughness of the target wall must also be considered, as it can disrupt local flow characteristics.

Impingement jets for cooling applications are commonly divided into two types:

i) **Free jet**: A free jet is a fluid jet that is discharged into a medium with different properties, such as air or another gas, without any immediate confinement or interaction with surrounding fluid of similar properties.[37]



(a) (b)

Fig. 2.1 Different types of impingement cooling techniques are illustrated as follows: (a) Schematic of a free impingement jet [37] (b) Schematic of a submerged impingement jet [37]

In a free jet, the fluid spreads and expands freely as it moves away from the nozzle exit, with the jet's velocity and momentum gradually decreasing due to interaction with the surrounding medium. In this case the fluid stream is introduced into a surrounding medium with distinct properties. The boundary of the jet mixes with the surrounding fluid, leading to shear and entrainment of the ambient fluid into the jet. Free jets are commonly observed in various applications such as exhaust systems, spray nozzles, and ventilation systems. A schematic representation of a free impingement jet is shown in Figure 2.1a.

ii) **Submerged Jets**: A submerged impingement jet occurs when a fluid jet is introduced into a surrounding fluid medium that has the same or similar density and thermophysical properties as the jet itself. The jet then impinges onto a target surface, typically in a quiescent or uniformly moving fluid environment. In this case, the jet remains "submerged" in the surrounding fluid, meaning the jet does not break into a different medium (such as air) but instead interacts with the surrounding fluid. This type of impingement jet is often used in cooling applications, where the interaction between the jet and the target surface enhances heat transfer. A schematic representation of a submerged impingement jet is shown in Figure 2.1(b).

It is important to consider that impingement cooling may result in non-uniform cooling patterns due to the high-velocity jets striking the surface. This can create localized areas with high heat transfer rates, leading to uneven cooling if the distribution of the jets is not optimized. To achieve uniform thermal management, it is essential to avoid hotspots and minimize thermal gradients, which in turn reduces the risk of thermal stresses and potential material failure.

2.1.2 Spray Cooling

Spray cooling is a high heat flux removal technique, with heat fluxes reported as high as 1000 W/cm^2 when using water.[38] It involves a complex process in which droplet impingement, fluid thermal coupling, and droplet-solid collision coexist. The concept of mist cooling was first proposed by Japanese scientist Toda in 1972.[39]. In this method, the liquid medium is atomized into fine droplets by a specific mechanism and ejected from a nozzle, impinging the heated surface at high velocity. These droplets form a thin liquid film on the surface, where convection, evaporation, and boiling processes occur, leading to enhanced heat transfer.

As shown in Figure 2.2a, the use of coolant such as water as it has a much higher specific heat capacity than fluorinated fluids (Fluorinerts), which allows it to absorb more heat per unit mass. This property enhances its ability to dissipate heat effectively when used in spray cooling.

Spray cooling can be categorized into three regimes based on the heat transfer mode: the single-phase regime, the two-phase regime, and the critical heat flux (CHF) regime [40]. In the single-phase regime, the liquid film and droplets do not boil or evaporate, making this regime suitable for cooling surfaces with relatively low temperatures. On the other hand, in the two-phase regime, phase change processes occur, leading to film evaporation or the boiling of droplets when the surface reaches a superheated state.

The critical heat flux (CHF) is reached when the heat flux attains its peak value, beyond which the cooling performance no longer increases, as depicted in Figure 2.2b. The latent heat absorbed during the phase change process significantly enhances heat transfer [41]. The finite cooling capacity of a spray-cooling system is defined by the CHF, which represents the upper heat flux limit. Beyond this point, excessive heating of the surface occurs, and a static spray can no longer effectively cool it. Although many theoretical models have been proposed to explain the triggering mechanism for CHF, the current understanding remains limited.

(a) (b)

Fig. 2.3 Spray Cooling in Different Gravity Environments: (a) Normal Gravity [39] (b) Microgravity [39]

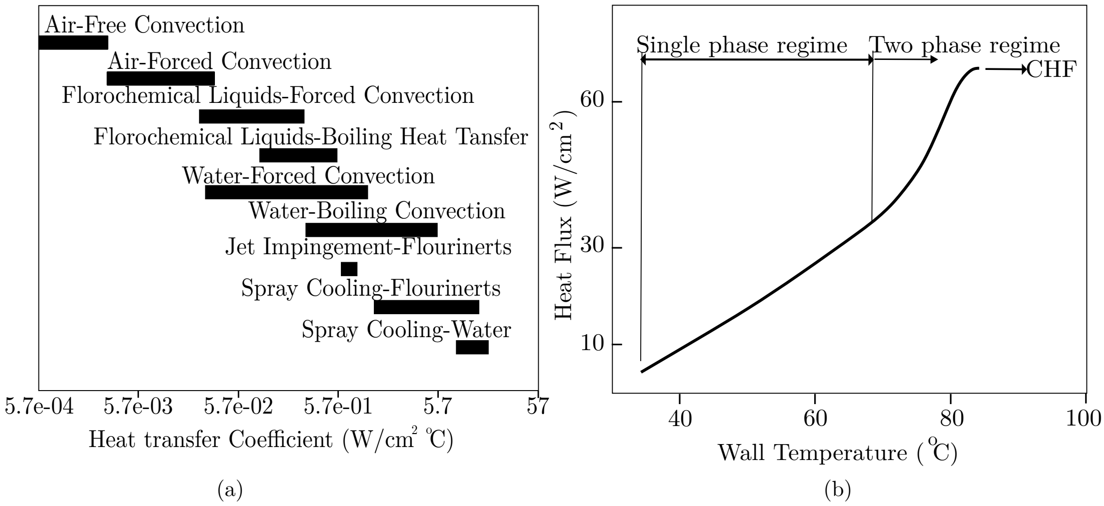


Fig. 2.2 (a) Heat transfer coefficients of various cooling mechanisms. [41] (b) Various regimes of spray cooling highlighting the CHF regime [41]

Furthermore, spray cooling with boiling behaves differently under normal gravitational conditions on the ground compared to in space or high-altitude environments, as illustrated in Figure 2.3. Therefore, the specific environment in which the cooling technique is applied must be carefully considered to ensure optimal performance.

The heat transfer rate in spray cooling is influenced by the size and velocity of the droplets. Inconsistent droplet distribution and potential evaporation can reduce overall heat transfer efficiency. Precise control over spray patterns and droplet sizes is also challenging to achieve. Additionally, nozzles and spray mechanisms are prone to clogging from particulates or contaminants, leading to diminished performance and increased maintenance requirements.

Spray cooling offers a more uniform cooling profile due to the high surface-area-to-volume ratio of fine droplets, which increases the available area for heat exchange. Unlike impingement cooling, which concentrates cooling effects at specific impact points and can lead to uneven temperature gradients and potential hotspots.

2.1.3 Concluding remarks

It is clear that both spray and impingement cooling techniques fall short in providing uniform heat transfer across surfaces and precise control over cooling parameters. While these methods offer distinct advantages, they also face limitations, particularly in demanding applications such as cooling the sharp leading edges of hypersonic vehicles. To overcome these challenges, a hybrid cooling technique that combines the principles of spray and impingement cooling is necessary. **Liquid sheet jet cooling offers a sophisticated thermal management solution**

that integrates the benefits of both spray and impingement cooling. This approach harnesses the uniform coverage of spray cooling along with the targeted heat transfer capabilities of impingement cooling, resulting in enhanced overall cooling performance.

2.2 Liquid Sheet Jets

Conventionally, a flat, leaf-like structure bounded by relatively thick cylindrical rims, formed in a plane perpendicular to the axes of two jets colliding at an oblique angle, is referred to as a liquid sheet [42]. Experimental investigations into the generation of liquid sheets date back to the 19th century, where Savart studied the impact of two opposing jets, as shown in Figure 2.4. [43][44]

Numerous studies have investigated the stability of liquid sheets over the years, as planar liquid sheet instability plays a significant role in spray technology, coating, food processing, medical applications, and more. Some pioneering work includes the temporal analysis by Squire [45], which shows that the Weber number (We) > 1 leads to increased instability. Taylor's study on the distribution of thickness and pressure within the liquid sheet provides insights into its shape.[46] Taylor also demonstrated the dependency of sheet expansion on collision angle, as shown in Figure 2.5(a). A smaller collision angle between two cylindrical jets reduces the sheet's extension in the opposite direction until it gradually disappears. Building on Taylor's work, Choo and Kang [47] investigated the effect of impinging jet velocity on sheet thickness and velocity, both of which directly influence sheet stability.

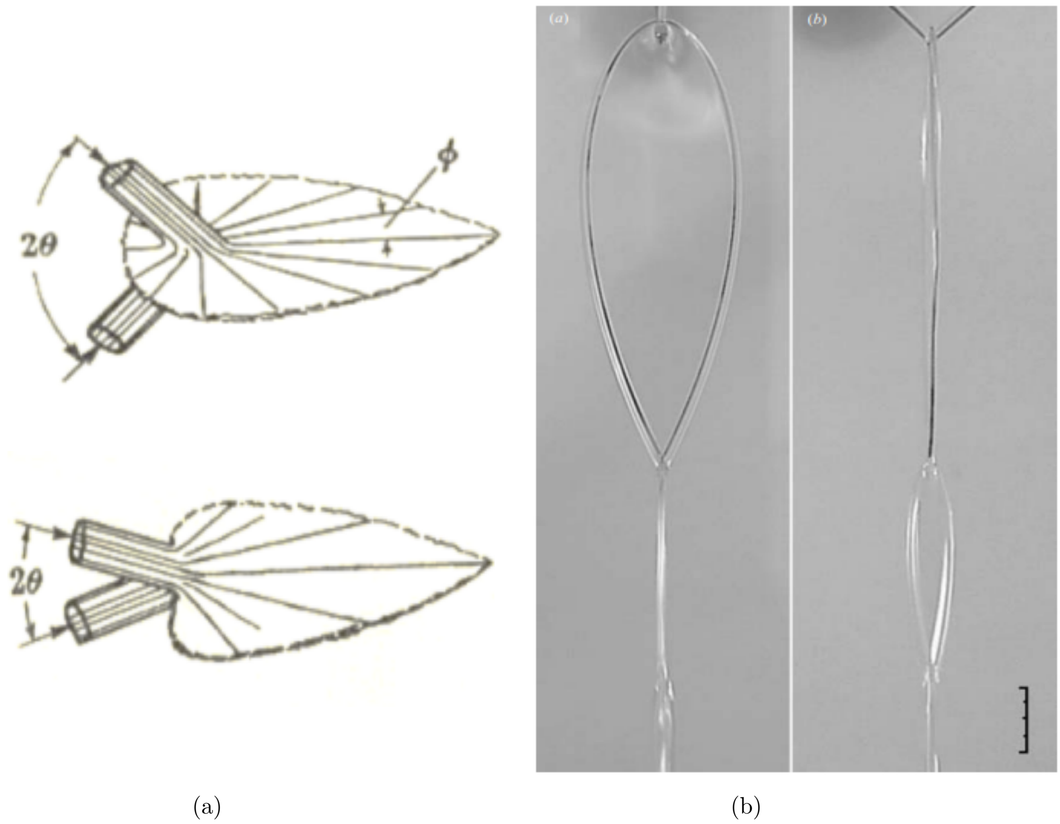


Fig. 2.5 Liquid sheet from colliding liquid jets: (a) Different sheet structure caused due to variation in collision angle [45] (b) Stable liquid chain formed from colliding liquid jets [42]. Scale bar: 1 cm

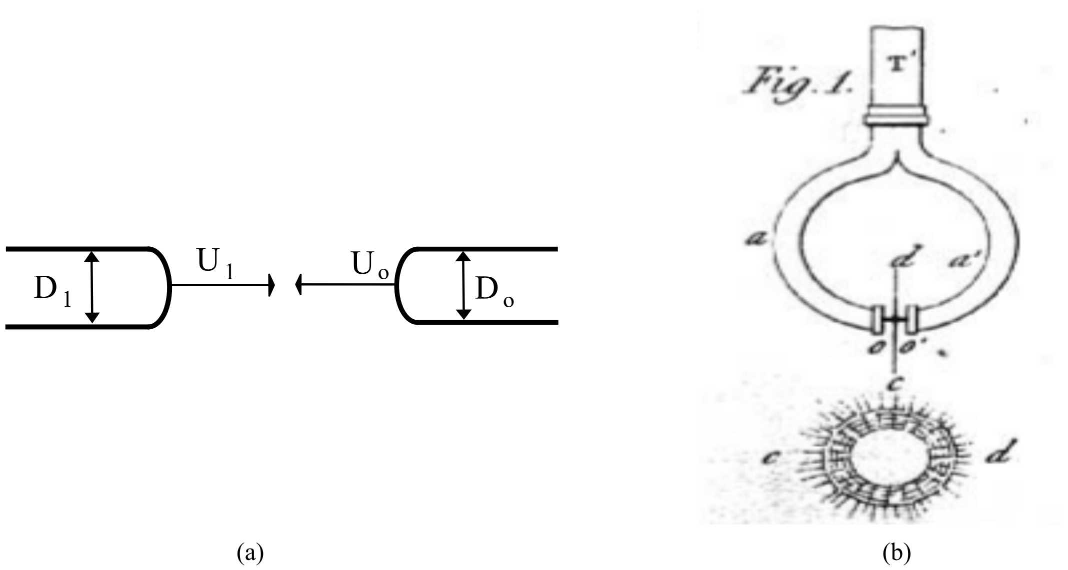


Fig. 2.4 Work done by Savart to generate a liquid sheet through the collision of jets: (a) Two jets impact [44] (b) Axisymmetric flat liquid sheet [44]

Bush and Hasha contributed further theoretical and experimental results by using more viscous liquids to address the issue of unstable rims observed in Taylor's work, as shown in Figure 2.5(b) [42]. Their approach aimed to stabilize the edges of the liquid sheet, which was a limitation in previous studies.

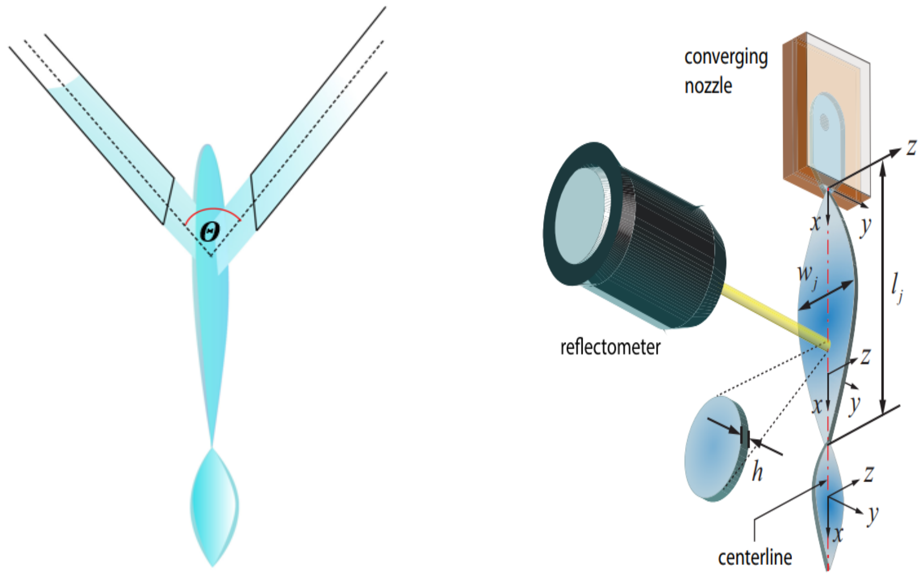
The atomization process, which occurs after the disintegration of ligaments from the rims of a liquid sheet, was explored in detail by Dombrowski and Johns [48]. Their study characterizes the drop size distribution and successfully demonstrates that the diameter of the drops formed from the breaking ligaments is always smaller than the ligament diameter. Additionally, Villermaux and Clanet [49] conducted experiments focusing on the detailed mechanisms of how drops are formed from the rims of the liquid sheet. Most experimental studies have been conducted with low-viscosity Newtonian fluids. However, Miller et al. [50] investigated the behavior of non-Newtonian fluids and demonstrated that the flapping instability typically caused by initial perturbations is significantly reduced. As a result, the morphology of the liquid sheet is noticeably different, with no occurrence of fishbone structures. One of the major analytical studies conducted to validate the findings of Miller and Taylor is the work by Ibrahim and Przekwas.[51] In this study, they confirm that the theoretical equations are in good agreement with the experimental conclusions regarding the shape and thickness of the liquid sheet.

2.2.1 Current applications of liquid sheets

The use of ultra-thin liquid sheet jets is a novel approach for cooling the sharp leading edges of hypersonic vehicles. These ultra-thin sheets also have diverse applications in modern technologies, such as infrared, X-ray, and electron spectroscopy, as well as high-resolution, time-resolved techniques [52]. In liquid propellant rocket engines, they serve as highly efficient enhancers for fluid mixing. When impinging-jet injectors release liquid jet streams, these jets collide at oblique angles, forming liquid sheets [53]. By introducing reacting fluids, these liquid sheets can act as continuous, wall-free reactors, enhancing their mixing capabilities. A deeper understanding of the interaction between the reacting fluids' properties and the liquid sheet's characteristics enables precise control of micro-mixing scales and the selection of specific induced reactions [54]. Furthermore, the exceptional mixing properties of liquid sheets are being applied in life sciences, particularly for studying the kinetics of protein folding [55].

2.2.2 Liquid Sheet Generation

Numerous studies have focused on experiments related to generating a free-standing thin liquid sheet from the collision of two jets. The generation of the liquid sheet



(a)

(b)

Fig. 2.6 (a) Oblique Collision of Two Laminar Jets: Two laminar jets collide at an angle, forming a thin liquid sheet whose dynamics depend on jet velocity, collision angle, viscosity, and surface tension. (b) Convergent 2D Microfluidic Nozzle: A liquid sheet is formed via a microfluidic nozzle, where the converging flow accelerates the fluid, producing a thin, stable sheet with precise control over its characteristics.[57]

involves three key aspects: the flow system, the jet impingement platform, and the visualization methods.

The most widely used method for generating a liquid sheet involves two-jet impingement, where two liquid jets collide at an oblique angle, θ, producing a flat, mutually orthogonal sheet with thick rims at its edges, as shown in Figure 2.6(a). This initial sheet is referred to as the primary sheet. When the rims of the primary sheet collide, they form another perpendicular sheet, known as the secondary sheet. This process continues, creating a chain of liquid sheets, until the rims eventually coalesce, reforming into a jet.

The nozzle orientation has changed over the years, and instead of using two nozzle system, recently Belšak has produced the gas-focused liquid micro-sheets[56]. Recently, a new concept of using a single fluid convergent nozzle that laterally spreads out the sheet has been successfully studied by Ha and Santiago [57]. The microfluidic device used to generate the liquid sheet without the need of two jets colliding at an oblique angle is shown in the Figure 2.6(b).

All methods of liquid sheet jet generation leverage the interplay of fluid dynamics and surface tension to create thin liquid films but are suited for different scales and applications depending on the control and stability required for the sheets.

2.2.3 Factors Influencing The Liquid Sheet

For the generation of the liquid sheet, several key factors influence both the liquid sheet and the resulting chain structure. Many studies have been conducted to understand the effects of non-dimensional numbers that relate to the flow properties, liquid properties, and geometry. These parameters play a critical role in determining the shape, behavior, and stability of the liquid sheet. Below are some of the important parameters:

• **Reynolds Number** (**Re**): This non-dimensional number represents the ratio of inertial forces to viscous forces in the flow. It influences the transition between laminar and turbulent flow, thereby affecting the stability and breakup of the liquid sheet.

$$Re = \frac{\rho U L}{\mu} \tag{2.1}$$

The incoming flow may exhibit either laminar or turbulent characteristics depending on the flow rate. A higher Re signifies turbulent flow, while a lower Re indicates laminar flow. Turbulent incoming flow tends to destabilize the liquid sheet, as varying flow rates can introduce instabilities following the impingement [?]. In a fully developed laminar flow, a parabolic velocity profile emerges, with the axial velocity reaching twice the average velocity.

• **Weber Number** (**We**): This non-dimensional number is defined as the ratio of inertial forces to surface tension forces, the Weber number helps in assessing the tendency of the liquid sheet to break up into droplets. Higher Weber numbers indicate a higher likelihood of breakup. Viscosity acts as the damping mechanism for the sheet, the surface tension has a significant effect where the thickness of the LS is the minimum whereas liquid density influences the width of the sheet. Since the liquid sheet is very thin, surface tension becomes a critical factor at the sheet's edge. Lower surface tension can also cause the sheet to break up, leading to the formation of sprays. The Weber number is defined as:

$$We = \frac{\rho U^2 L}{\sigma} \tag{2.2}$$

• **Capillary Number** (**Ca**): The Capillary number represents the ratio of viscous forces to surface tension forces. It affects the shape and dynamics of the liquid sheet, especially in microfluidic applications.

$$Ca = \frac{\mu U}{\sigma} \tag{2.3}$$

In the context of a liquid sheet generated by two nozzles at an oblique angle, the Ca significantly influences the sheet's behavior. When the Capillary number is much smaller than 1 surface tension dominates over viscous forces. In this regime, the liquid sheet exhibits greater stability and is more likely to retain its shape. The dominance of surface tension helps to keep the liquid sheet intact, preventing it from breaking apart into smaller droplets. When the capillary number is much larger than 1, viscous forces dominate surface tension. In this case, the liquid sheet is less stable, and it is more prone to break up into smaller droplets due to the dominant effect of viscous shear forces. The capillary number also affects the thickness of the liquid sheet. A higher capillary number tends to result in a thinner sheet because viscous forces lead to more significant shear deformation, causing the sheet to spread out more. Ca_e have an influence in determining the size of the liquid sheet, where Ca_e denotes the capillary number at the exit of the nozzle . At low Re and Ca_e regimes, the size is reduced.

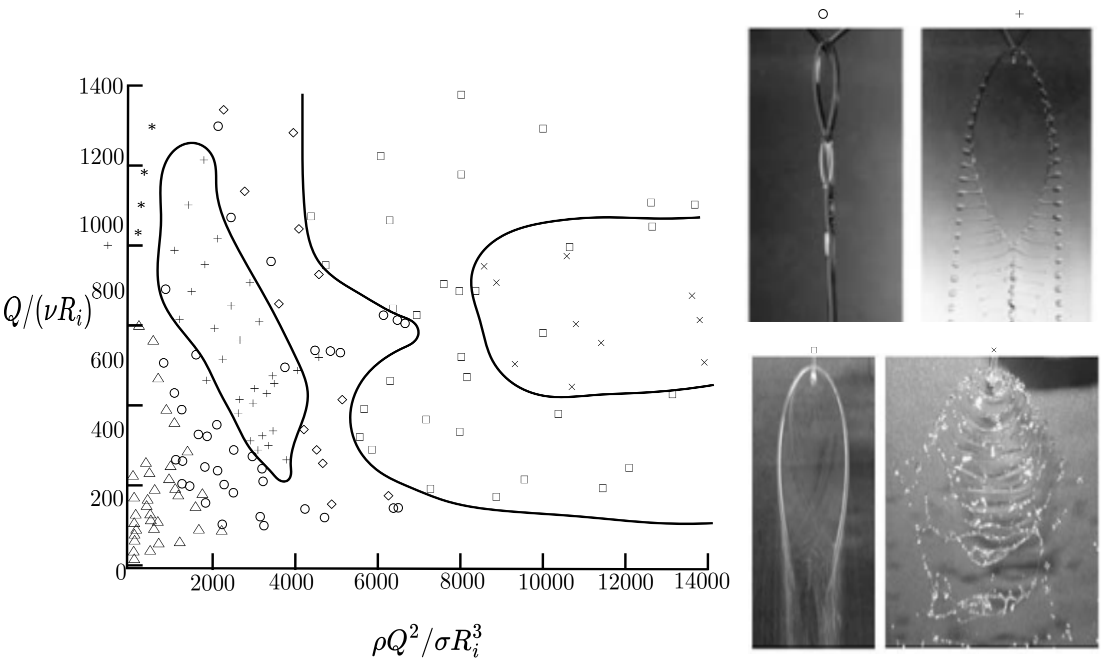
• **Collision Angle or Geometry of the System**: In methods like the oblique collision of jets, the angle at which they collide significantly impact the formation and stability of the liquid sheet. In microfluidic approaches, on the other hand, the nozzle geometry and confinement influence the sheet's thickness, extent, and stability.

2.2.4 Morphological Regimes of Liquid Sheet Jets

The study by Bush and Hasha presents various geometrical structures related to liquid-sheet jets. To classify the regimes of liquid-sheet jets effectively, the researchers redefined the expressions for the Reynolds number (2.1) and Weber number (2.2) using the volumetric flow rate Q and jet radius R_j. This reformulation allows for a more precise description of the interplay between inertial forces, curvature forces, and the transfer of kinetic energy through viscosity.[42] The non-dimensional quantities discussed above, then become:

$$\mathrm{Re} = \frac{Q}{\nu R_j} \qquad \text{(kinematic viscosity, } \nu = \frac{\mu}{\rho}\text{)} \tag{2.4}$$

$$We = \frac{\rho Q^2}{\sigma {R_j}^3} \tag{2.5}$$



(a) (b)

Fig. 2.7 Various regimes caused by the collision of the viscous jets: (a)Regimes of flow structure [42] b) Different flow regimes where,$\triangle$-oscillating streams,$\star$-Sheets with disintegrating rims,$\circ$-Fluid chains,+-fish-bone structure, $\diamondsuit$-Spluttering chains,$\square$-disintegrating sheets,$\times$-violent flapping [42]
.

$$Ca = \frac{\mu Q}{\sigma t w_o} \tag{2.6}$$

The formation of liquid-sheet jet structures is depicted as a function of the previously discussed dimensionless parameters in Figure 2.7(a), while Figure 2.7(b) provides images of these structures. The evolution of the liquid sheet structure has been tracked over a range of viscosity values, and despite variations, a consistent sequence of structural transitions is observed. At the lowest flow rates, the two jets merge into a single stream upon collision. Although the boundary between regimes is somewhat indistinct, increasing the flow rate leads to the development of stable fluid chains. A key distinguishing factor between the two structures is the difference in thickness between the sheet and the rims. In coalescing jets, the thickness is relatively uniform, whereas in liquid-sheet chains, the ratio of thickness between the sheet and the rims is much lower.

Instabilities in liquid sheets cause the initial link to shrink, eventually leading to the formation of droplets that diverge from the longitudinal axis. In an intermediate regime, known as fish-bone structures, elements of both stable and unstable behaviors are combined. These fish-bone structures exhibit temporal periodicity and can only be distinctly observed under strobe illumination due to their otherwise blurred appearance. To suppress these instabilities and regenerate stable liquid-sheet chains, a slight increase in flow rate can be employed. This adjustment helps stabilize the liquid sheet, ensuring controlled and consistent behavior, which is crucial for various applications.

These instabilities propagate through the final links of the chains as weak spluttering. This process eventually leads to the formation of holes within the sheet, causing it to disperse. However, the complete breakup of the liquid sheet and its disintegration into droplets is only signaled by the onset of violent sheet flapping.[42]

2.2.5 Geometrical Characteristics of Liquid Sheet Jets

A stable and controllable liquid sheet is crucial for numerous applications. Achieving this requires the ability to adjust its geometric properties, such as length, width, and thickness.

The thickness of the liquid sheet can be tailored through several design modifications:

1. **Impingement Angle**: Increasing the impingement angle enhances momentum transfer perpendicular to the sheet, effectively compressing it and reducing its thickness.[58]

2. **Working Fluids**: Utilizing different working fluids can also facilitate the creation of thinner sheets. For example, solvents with lower density can be employed to generate thinner liquid sheets.[56]

3. **Surface Tension**: Surface tension significantly influences the shape of the liquid sheet. By using liquids with lower surface tension, a larger frontal area with reduced thickness can be achieved.[59]

The effect of surface tension on the thickness of liquid sheet are as shown in Figure 2.8. Figure 2.8(a) illustrates that utilizing a liquid with higher surface tension (σ) results in a thicker liquid sheet, while lower surface tension produces a thinner sheet. Similarly, Figure 2.8(b) demonstrates the variation in the position at which the minimum thickness is attained. It is important to note that Taylor established that the liquid sheet thickness (h) is inversely proportional to the radial distance ($1/r$) and is also influenced by the collision angle (θ).[46] Subsequently, Hasson and Peck extended this analysis by correlating the thickness distribution with the azimuthal angle (ϕ) of the liquid sheet.[60] The relation is presented in Equation 2.7.

$$\frac{hr}{R_j^2} = \frac{\sin^3(\theta)}{(1 - \cos(\phi)\cos(\theta))^2} \tag{2.7}$$

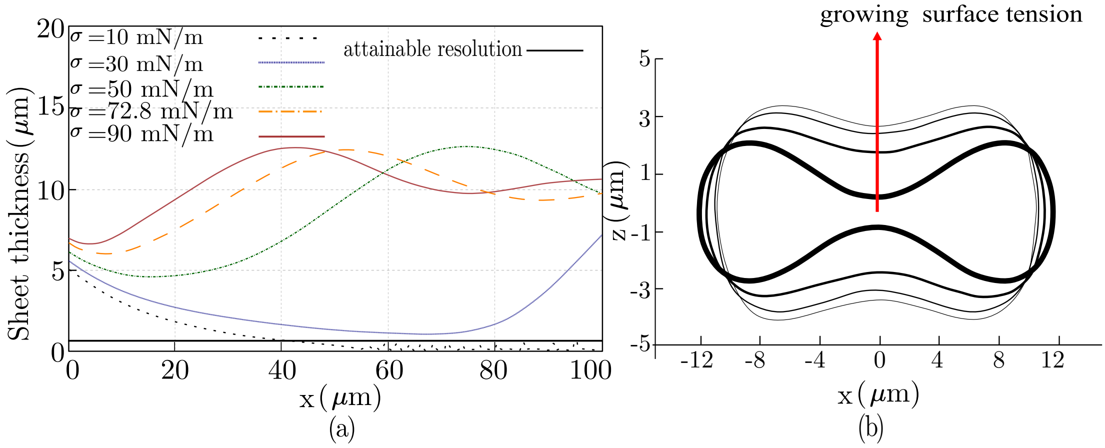


Fig. 2.8 Variation of sheet thickness with varying σ :(a)LS thickness for varying liquid surface tension coefficient [55](b)Contour depiction at the position of minimum thickness [55]

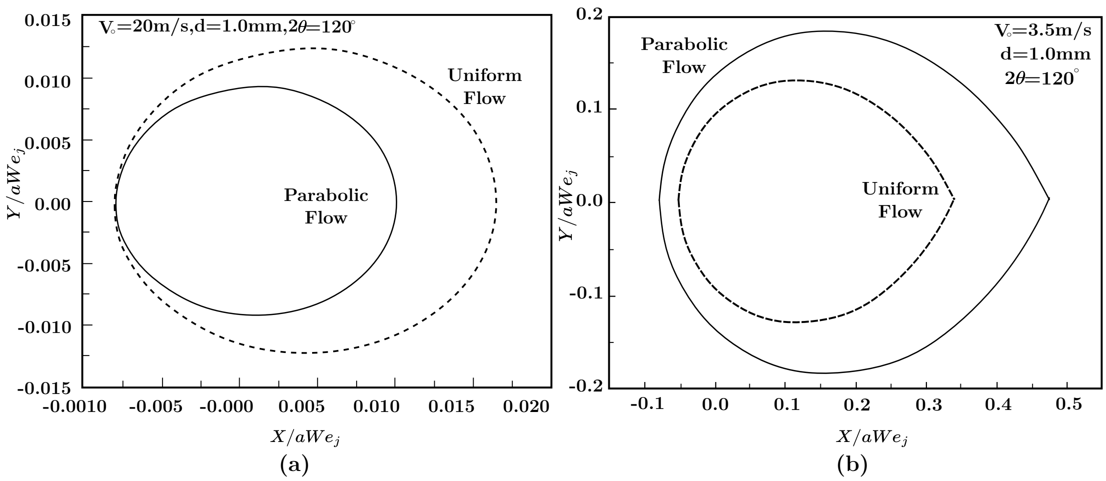


Fig. 2.9 Effect of velocity profile of jet on the LS shape: (a) Jet at low Re [60] (b) Jet at high Re [60]

It should also be noted that the Reynolds number (Re) is a parameter dependent on the velocity of the incoming jets. The Re of the jets has a direct impact on the size of the liquid sheet, as demonstrated in Figures 2.9a and 2.9b.[61]

Moreover, Ha and Santiago derived a scaled correlation for predicting the thickness (h), width (w_j), and length (l_j) of the generated liquid sheet (LS) based on linear regression. These correlations, outlined below, are dependent on the nozzle geometry:

$$\frac{h}{w_o} = 0.36\frac{t}{x}(1 + 1.5\alpha^2)[1 + cot(0.67\theta)] \tag{2.8}$$

$$\frac{l_j}{t} = 0.23We_eCa_e^{-0.1}(1 + 1.5\alpha^2)^{-0.5}[1 + cot(0.67\theta)]^{-0.5} \tag{2.9}$$

$$\frac{w_j}{t} = 0.074 We_e Ca_e^{-0.2}(1+1.5\alpha^2)^{-1}[1+cot(0.67\theta)]^{-1} \tag{2.10}$$

Here, α represents the aspect ratio, defined as $\alpha = t/w_o$, where t is the depth or thickness of the nozzle, and w_o is the nozzle exit width. Additionally, θ denotes the nozzle angle, and x is the distance from the point where the sheet is generated, measured along the centerline in the x-direction.

2.2.6 Disintegration Mechanism of LS

To achieve a stable liquid sheet for experimental studies, it is crucial to understand the forces and mechanisms that contribute to its instability. Early research has focused on deciphering the fragmentation and disintegration processes of liquid sheets. Although a complete theoretical framework is still lacking, certain forces have been identified as contributing to the growth of large-amplitude waves, ultimately leading to the sheet's destabilization.[62] Dombrowski's photographic analysis revealed that the primary sign of instability is the formation of threads and droplets detaching from the sheet's rims.[63] The sequence of events during the destabilization of a liquid sheet typically follows these steps:

1. Jet impingement.
2. Formation of a flapping sheet.
3. Evolution into cylindrical ligaments.
4. Eventual droplet formation.

The flapping behavior of liquid sheets formed by impinging jets exhibits a similar pattern to that observed in single liquid sheets.[64] However, when the study is extended to non-Newtonian fluids, researchers have identified six distinct instability patterns.[65]

1. Open rim without shedding droplets
2. Closed rim with shedding droplets
3. Open rim with shedding droplets
4. Rimless sheets
5. Bow-shaped ligaments
6. Fully developed spray

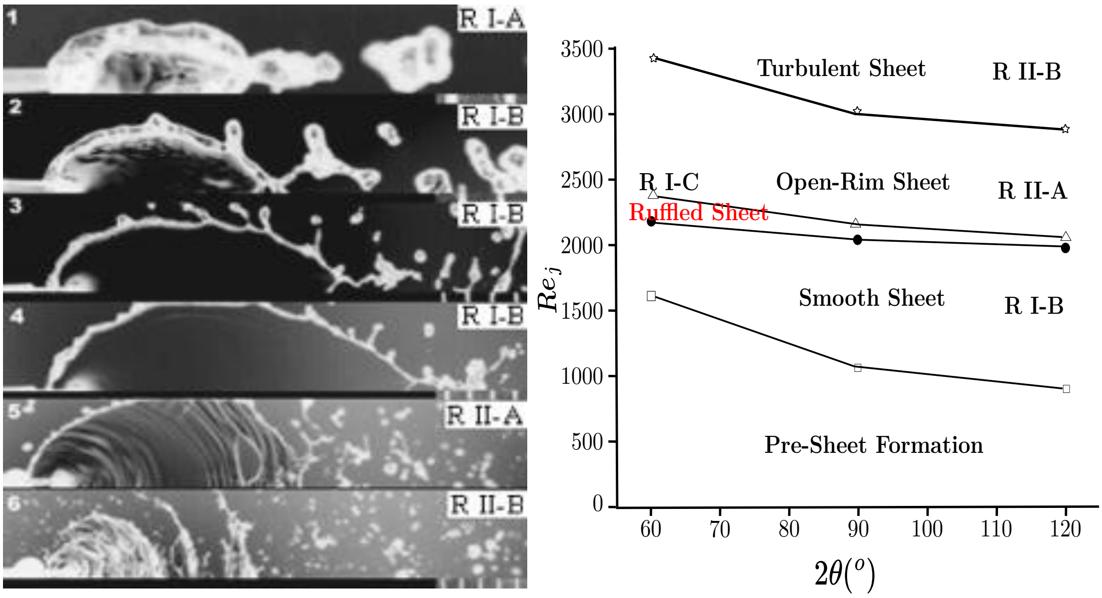
Understanding these instability patterns is essential for developing methods to maintain the stability of liquid sheets across various applications. Two principal mechanisms have been identified as contributing to the disintegration of liquid sheet jets:

1. **Aerodynamic Waves**: These waves are generated by interactions between the liquid sheet and the surrounding air, leading to its collapse. This phenomenon primarily affects the sheet's surface, causing waves that amplify and destabilize the sheet.

2. **Hydrodynamic or Impact Waves**: These waves originate from the internal fluid dynamics or from external impacts on the liquid sheet. They introduce disturbances within the sheet that propagate and ultimately result in its breakup.

These mechanisms allow for the classification of instabilities into two distinct breakup regimes. To effectively characterize these instabilities and distinguish between the different breakdown mechanisms, two critical dimensionless parameters are utilized: the Reynolds number (Re) and the Weber number (We).

To form a liquid sheet, the velocity of the colliding jets must exceed a specific threshold.[66] Below this limit, the jets simply bounce off each other without coalescing. When the velocity is sufficiently high, van der Waals forces at the collision point cause the jets to merge.[67] Small disturbances then lead to fragmentation, as illustrated in Figure 2.10a. In sheets with continuously closed rims, capillary phenomena occur. The breakdown of liquid sheets in the very low-velocity regime is shown in images 1 and 2 of Figure 2.10a. In this regime, the sheet becomes ruffled, and a single distorted jet forms at the link, breaking down according to Rayleigh instability. As velocity increases, momentum forces begin to dominate over surface tension. Initially, disturbances grow along the rims, detaching from the sheet in the form of droplets. With further increases in velocity, instabilities originate at the collision point. After the onset of Kelvin-Helmholtz instability, these disturbances, in the form of waves, travel along the sheet, ultimately causing its disintegration and resulting in droplet formation, as depicted in image 4 of Figure 2.10a. Figure 2.10b shows the different types of sheets formed at various angles and incoming jet Reynolds numbers (Re_j). The turbulent sheet shown in image 6 of Figure 2.10a can be associated with the data in Figure 2.10b, where the Re_j ranges from 3000 to 3500 for such breakup behavior.



(a) (b)

Fig. 2.10 Instability and breakup of liquid sheets: (a) Sheet breakup from jet impingement[65], (b) Breakup regime as a function of jet Reynolds number (Re_j)[65].

The destabilization of liquid sheets due to aerodynamic waves was modeled in the pioneering work of Dombrowski and Johns.[48] They derived equations that describe the diameter of the detaching ligaments and droplets from a liquid sheet, offering a theoretical framework to understand and predict these phenomena. The equation for the diameter of the detaching ligaments is as follows:

$$d_{\text{ligament}}(d_L),\ d_{\text{droplets}}(d_D) = f(\rho, \sigma, U, K, k, f, \mu)$$

$$d_L = 2\left(\frac{4}{3f}\right)^{1/3}\left(\frac{K^2\sigma^2}{\rho_g \rho U^2}\right)^{1/6}\left[1 + 2.6\mu\sqrt[3]{\frac{k\rho_g^3 U^8}{6f(\rho)^2\sigma^5}}\right]^{1/5} \tag{2.11}$$

$$d_D = \left(\frac{3\pi}{\sqrt{2}}\right)^{1/3} d_L\left[1 + \frac{3\mu}{\sqrt{\rho\sigma d_L}}\right]^{1/6} \tag{2.12}$$

In the study of liquid sheet breakup and droplet formation, various methods have been employed to model and predict droplet characteristics.[68]

1. **Instability Wave Amplification:**

   Weihs' linear stability theory posits that the diameter of droplets formed by Kelvin-Helmholtz (K-H) instability waves is roughly half the wavelength (k) of the most rapidly growing waves. This can be represented as:

$$d_0 = \frac{\pi}{k}$$

2. **Anti-symmetric Wave Theory at Lower We:**

   In low Weber number (We) regimes, where surface tension plays a dominant role, the anti-symmetric wave theory is utilized. This theory links droplet size to the thickness at the sheet's edge.[51]

3. **Empirical Correlations from Experimental Data:**

   Recent experiments have combined theoretical approaches to derive empirical correlations for breakup length (X_b) and droplet size (d_D) .[69]

   Breakup Length (X_b):

$$\frac{X_b}{d_0} = 5.451\, s^{-2/3} \left[We \cdot f(\theta)\right]^{-1/3}, \quad s = \frac{\rho_{\text{gas}}}{\rho_{\text{liquid}}} \tag{2.13}$$

   Droplet Size (d_D):

$$\frac{d_D}{d_0} = 1.144 s^{-1/6} [We \cdot f(\theta)]^{-1/3} \tag{2.14}$$

   Here,

$$f(\theta) = \frac{(1 - \cos^3\theta)^2}{\sin^3\theta} \tag{2.15}$$

Chapter 3

Experimental Methodology

Fluidic systems offer flexible control and stability, largely because they lack moving parts. In this work, a microfluidic system will be employed to generate a stable liquid sheet (LS) with minimal or no capillary waves. Unlike the conventional approach of using two jets impinging at an angle, a slit-type nozzle provides a simpler and more reproducible design, allowing for more precise control of the fluid. Additionally, this device can produce a uniform sheet thickness by reducing turbulence, resulting in a more stable liquid sheet.

This study aims to investigate the effects of injection velocity and nozzle structure on the formation of a planar liquid sheet. The device is designed to consistently reproduce the liquid sheet with minimal uncertainty. The use of a slit-type nozzle helps prevent harmonic perturbations that typically arise from jet collisions, significantly reducing rim destabilization. Producing a liquid sheet with minimal undulating movements is crucial to avoid rim fragmentation.[70] Overall, two microfluidic devices have been manufactured and tested, a stack-based model which consists of a three-layer assembly, and Silicon-based model based on advanced microfabrication techniques.

3.1 Stack-based model fabrication

The Stack-based liquid sheet device was designed as a three-layer assembly, as shown in Figure 3.1. A top glass wall, a Polyimide sheet containing the nozzle geometry, and a bottom glass wall with an inlet for flow from the liquid pump. The purpose of this design is to avoid the alignment issues associated with using two colliding jets. The base of the device is made from aluminum and features an O-ring to ensure even pressure distribution. The nozzle has a rectangular exit with a wide range of controllable aspect ratios, allowing for a variety of experimental configurations.

In the device design, careful attention must be given to creating a well-defined channel or nozzle geometry to promote a stable liquid sheet. Precise alignment of the micro-channels and inlets is crucial for successful LS generation. Additionally, polishing near the outlet is essential. According to the literature[57], surface treatments such as hydrophobic coatings are highly recommended to ensure the incoming flow spreads uniformly along the nozzle, further contributing to sheet stability. To achieve precise control over flow rates or pressure, microfluidic pumps or pressure controllers should be utilized. It is also important to select materials that are compatible with the liquids to ensure that they are not adsorbed or react with the fluids, as this could affect stability. Also, the use of a low viscosity fluid is highly recommended to prevent clogging of the device, which can occur due to high pumping pressures.[71] The Polymide sheet is cut using UV laser ablation, with the nozzle layer made from a commercially available Polyimide sheet (Kapton, Shagal Marketing Solutions Ltd, Israel), sandwiched between 1 cm thick Plexiglas. Since even a minor misalignment in these layers can cause instabilities in the resulting liquid sheet, special care is taken during assembly to minimize human error. Any protrusions caused by misalignment are polished using 800 to 2000 grit sandpapers after securing the assembly at the post. A commercially available spray coating water-repellent treatment (Rain-X, ITW Global Brands) is applied to the area near the nozzle exit to make it hydrophobic. This helps reduce dripping near the nozzle, which can lead to non-uniform jet behavior. The working liquid is pumped using an HPLC pump (Knauer 80 P) with appropriate tubing and fittings to ensure consistent flow.

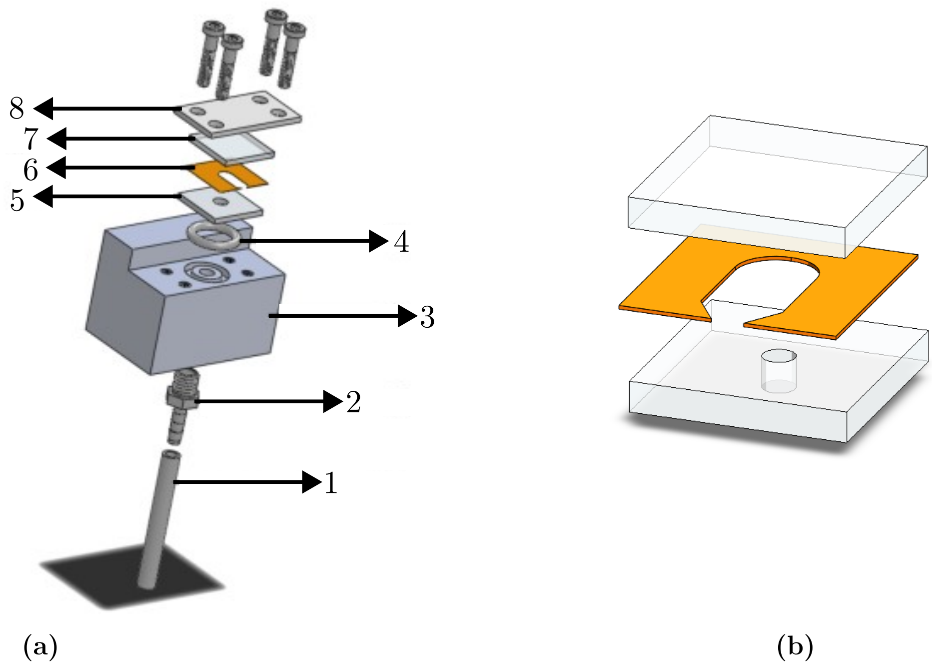


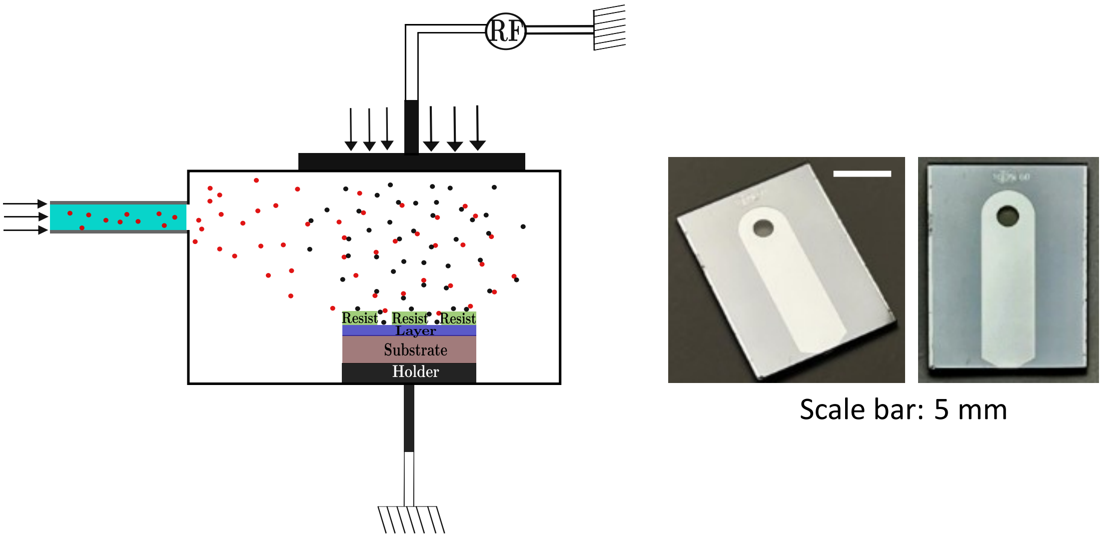
Fig. 3.1 (a)Stacked model with (1)-tubing (2)-inlet fitting (3)-Al body (4)-O-ring seal (5)- base plate (6)-polyimide nozzle (7)-top plate (8)-tigtening lid (b) Exploded view of different layers in the stacked model

3.2 Silicon-based model micro-fabrication

3.2.1 Dry Etching Technique

Dry etching is capable of producing structures[72] with a higher aspect ratio and is selected for nozzle fabrication due to its ability to achieve depths on the order of several micrometers while maintaining excellent sidewall smoothness, which is critical for nozzle performance. Although the process yields smooth surfaces, scalloping may occur near the nozzle exit, necessitating polishing to address these minor imperfections. The fabrication method employed is Deep Reactive Ion Etching (DRIE), a specialized form of the Reactive Ion Etching (RIE) technique.[73]

The primary technology employed in Deep Reactive Ion Etching (DRIE) is the Bosch process.[74] This process utilizes time multiplexing, consisting of alternating etching and polymer deposition phases within the plasma etching system, with each phase lasting several seconds. The initial step involves standard isotropic plasma technology, where ions in the plasma strike the wafer from a nearly vertical direction, as illustrated in 3.2(a). During the etching phase, the polymer is quickly sputtered away, but this occurs only on the horizontal surfaces, preserving the polymer on the sidewalls and enabling the profile to evolve in a highly anisotropic manner. Subsequently, a chemically inert passive layer is deposited, which serves to protect the entire substrate from further chemical attack and inhibits additional etching. Because the polymer dissolves only during the chemical portion of the etching process, it accumulates on the sidewalls, safeguarding them from etching. The etch rates achieved through this method are 3 to 4 times higher than those of traditional wet etching techniques. DRIE is capable of producing features with exceptional resolution and small feature sizes in thin film structures, as demonstrated in 3.2(b). This capability is attributed to its high aspect ratio, elevated etch rates, and excellent selectivity.



(a) (b)

Fig. 3.2 Fabrication process of the wafer nozzle model: (a) Deep Reactive Ion Etching (DRIE) technique applied to create high-aspect-ratio microstructures on the silicon wafer(b) Micro-scale nozzle geometry fabricated using DRIE, producing intricate, well-defined features essential for the nozzle's final design and functionality

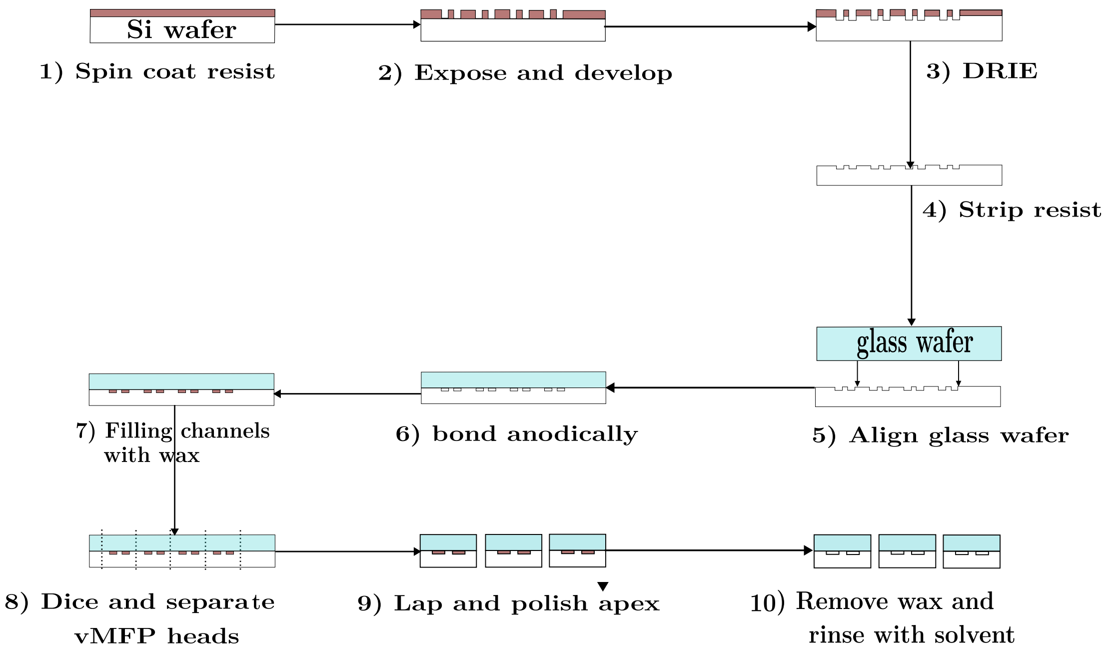


Fig. 3.3 Schematic representation of the steps involved in the Deep Reactive Ion Etching (DRIE) technique outlining the sequential processes, including photoresist application, pattern exposure, etching, and post-processing, which are essential for achieving high-aspect-ratio microstructures on anodically bonded glass wafers with silicon. [72]

3.2.2 Model Fabrication and Operation

While recent trends favor the use of glass-to-glass wafer nozzle chips for liquid sheet generation, the converging nozzle chips investigated in this study were fabricated using the method described by Kaigala et al.[73] This approach employed a silicon wafer with a thickness of 350 μm alongside a glass wafer with a thickness of 700μm, rather than utilizing two glass wafers. Figure 3.3 schematically illustrates each step of the process. CAD designs encompassing all geometric variations for the converging nozzles and inlet holes were employed to create the photolithography masks. To facilitate batch fabrication, multiple geometric variations are incorporated into each mask, arranged in a 2x5 configuration.

The fabrication of nozzle geometries involved a series of precise steps conducted at the Binnig and Rohrer Nanotechnology Center cleanroom facilities at ETH Zurich. Initially, 4-inch silicon wafers underwent plasma cleaning at 600 W for 3 minutes to eliminate organic contaminants. Subsequently, the wafers were treated with hexamethyldisilazane (HMDS) at 110^{o}C for 1 minute to enhance the adhesion of the photoresist layer. A 3.3 μm thick layer of AZ4533 photoresist was uniformly spin-coated onto the wafer at 4000 rpm for 40 seconds, with a ramp rate of 2000 rpm/s. The wafers were then baked at 110°C for 60 seconds to ensure optimal adhesion and stability of the photoresist. The desired pattern was transferred onto the photoresist using a mask aligner (MA6) through exposure, followed by development with an AZ developer (AZ400, diluted 1:4).

Deep reactive ion etching (DRIE) was performed with SF_6 as the etchant and C_4F8 as the passivation layer. Post-etching, acetone was employed to remove the remaining positive photoresist, and the wafers were dried using nitrogen. To eliminate any residual organic contaminants, the wafer underwent O2 plasma treatment for 5 minutes at 600 W. For the inlet holes, similar processes were repeated on the backside of the wafer. An anodic bonding process at 450°C and 1250 V was utilized to bond Borofloat 33 glass wafers (SCHOTT AG) to the processed silicon wafer. Using wax to fill the channels after anodic bonding is a practical approach to shield the delicate etched surfaces. This method ensures the structural integrity of the channels during processes like dicing, preventing potential fractures or surface damage. The dicing for obtaining individual nozzles were carried out in the Micro and Nano Fabrication Unit (MNFU) in Technion.Completing the dicing in the MNFU at Technion, followed by polishing, adds precision to the nozzle fabrication, especially at the outlet cross-section. This additional polishing step is essential for minimizing surface roughness, which can influence flow behavior at the nozzle exit,

As a result, this setup, with the anodic bonding of glass to an etched silicon wafer, creates a transparent window essential for visualizing the internal flow during PIV analysis. The anodic bond not only provides a strong, hermetic seal but also ensures that the transparent glass layer aligns precisely with the etched features on the silicon wafer, facilitating clear and accurate flow tracking with minimal optical distortion.

3.3 Measurement Techniques

3.3.1 High-Speed Shadowgraph Imaging

Shadowgraphy is an effective optical technique that visualizes flow patterns and structures by exploiting the principle of light extinction. In this method, a coherent light source illuminates the flow region of interest[75]. As the light traverses the medium, variations in the refractive index—caused by differences in fluid density, temperature, or particle presence—lead to changes in light transmission[76]. Some light is refracted, absorbed, or scattered, resulting in intensity differences that create shadow effects. These shadows provide detailed outlines and characteristics of the structures of the flow, revealing intricate flow patterns that may not be easily detectable with other methods. The non-intrusive nature of shadowgraphy ensures that the flow remains undisturbed, making it a valuable tool for qualitative analysis and a precursor for further quantitative investigations into fluid dynamics.

The deflected light rays are projected onto a recording plane, typically a screen or a digital sensor. The variations in light intensity create a shadow-like image, where darker regions correspond to areas with greater deflections and, consequently, more pronounced density gradients [77]. This shadow pattern is then captured using high-speed or high-resolution cameras, depending on the dynamics of the flow being investigated.

By analyzing these shadow images,information about the structures can be inferred. This technique is particularly advantageous in scenarios where significant density changes occur. Shadowgraphy's non-intrusive nature, combined with its simplicity and capability to capture real-time flow behavior, makes it an indispensable tool in experimental fluid dynamics. Furthermore, recent advancements in digital imaging and data processing have significantly improved the precision and clarity of shadowgraphy, rendering it especially effective for visualizing liquid sheets emerging from slit nozzles. These enhancements facilitate the detailed capture of the liquid sheet's thickness variations, surface undulations, and instabilities as it exits the nozzle. Clear, high-resolution images of the liquid sheet's formation, breakup, and potential fragmentation are provided by shadowgraphy, offering critical insights into flow behavior. This level of precision is considered invaluable for optimizing the design of slit nozzles, particularly in applications such as atomization, spray, and fluid dispensing, where uniformity and stability of the liquid sheet are essential.

3.3.2 Particle Image Velocimetry (PIV)

The Particle Image Velocimetry (PIV) technique is employed to quantitatively measure the instantaneous velocity field across numerous points within a fluid. This method provides comprehensive, two-dimensional flow field data without interfering with the flow itself; that is, it is a non-intrusive technique. The PIV technique can be roughly divided into two steps,[78] visualization and image analysis as can be seen in the Figure 3.4.[79]

In this technique, small seeding particles are introduced into the fluid flow to serve as tracers as shown in Figure 3.4. The selection of these particles is critical, as they must accurately follow the fluid dynamics without altering the flow behavior. To achieve this, the flow system used for this technique must provide a uniform flow devoid of irregularities. Ideally, particles should be as small as feasible to closely follow the fluid motion. However, they should not be excessively small, as this could lead to inadequate light scattering and, consequently, weak image contrast, impairing the accuracy of the measurements.[80] Fluorescent polymer microspheres are commonly utilized due to their bright coloration and precise optical properties, making them ideal for high-resolution flow visualization in

PIV experiments. These microspheres are specifically designed with densities and sizes that enable them to faithfully track the flow, ensuring accurate and reliable measurements [81].

The choice of illumination method and its configuration can significantly impact the quality of PIV measurements. Therefore, careful consideration and optimization are essential for successful experiments. In general, there are two types of illumination methods: Continuous-wave (CW) laser and pulsed-laser. While pulsed lasers are commonly used in traditional PIV techniques, continuous laser illumination is utilized in micro-PIV. In micro-scale flows, where particle displacement is minimal, continuous-wave (CW) lasers can provide adequate illumination for capturing particle motion [82].

The laser beam is transformed into a thin sheet using optical components such as cylindrical lenses and beam expanders. This laser sheet illuminates a two-dimensional slice of the flow field. Furthermore, continuous-wave (CW) lasers emit a constant light, eliminating the need for synchronization with the camera shutter [83].

To acquire the spatial images I_1 and I_2 at time t_i and t_{i+1} respectively, which provide information about the displacement of the particles, as shown in Figure 3.4, high-speed cameras with high sensitivity are essential to capture the faint light scattered by very small particles. This is crucial for imaging in microchannels where illumination is limited [84].

After acquiring these images, image processing is performed to improve particle detection, reduce noise, and increase the accuracy of velocity vector calculations. This process includes a background subtraction technique used to remove static background noise and enhance the contrast of particle images against the background, which is critical in microfluidic environments.[85] Images can be transformed into the frequency domain using Fast Fourier Transform (FFT) to analyze the frequency components. This transformation helps identify periodic structures or noise in the images. FFT allows for the identification and removal of unwanted frequency components (noise) in the image. By applying a filter in the frequency domain, noise can be reduced, enhancing the clarity of particle images [86].

The filtered images in the frequency domain are converted back to the spatial domain using an inverse Fourier transform [87]. These images are then subjected to cross-correlation to determine the displacement of particles between the images by dividing the images into smaller interrogation windows and calculating the cross-correlation for each window to identify the peak correlation value (indicating the displacement) [86]. From the detected peaks, the displacement vectors are obtained, which are used to calculate the velocity vectors for each interrogation window. This velocity field is utilized for further post-processing.

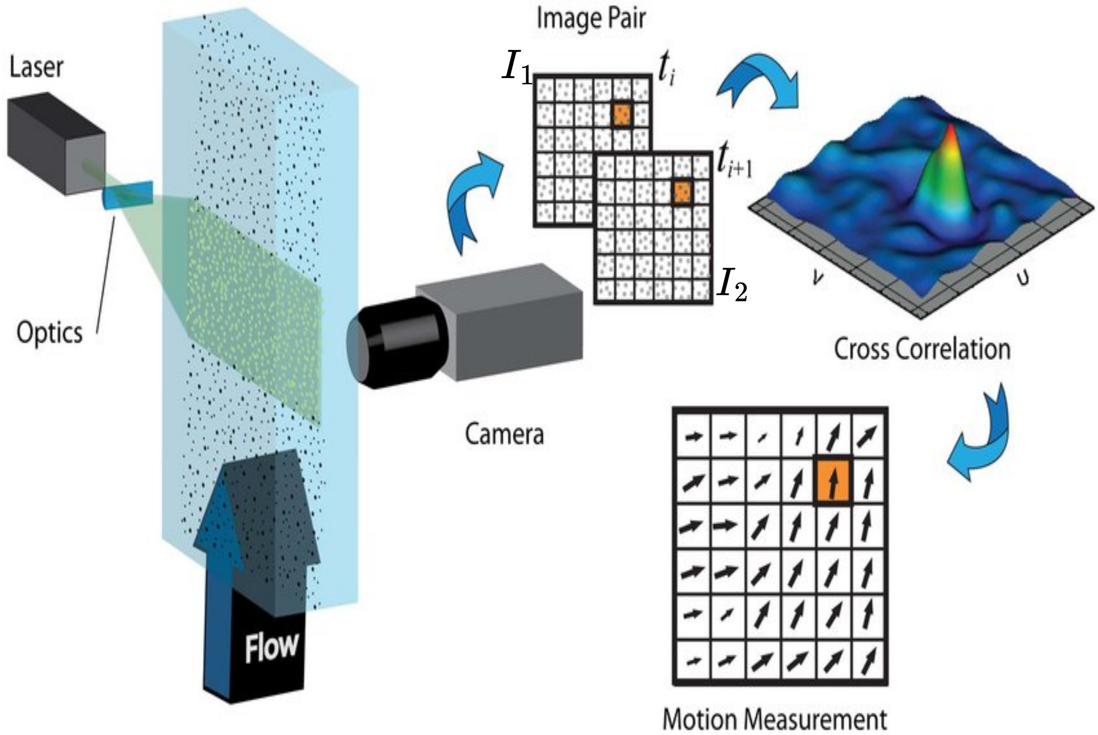


Fig. 3.4 A step-by-step representation of the processes involved in PIV, including particle seeding, laser sheet illumination, image capture by high-speed cameras, and cross-correlation analysis for velocity field computation [79]

For conducting the PIV experiment at the ThermoFluids and Interface Lab at Technion, dyed aqueous fluorescent tracer particles with a diameter of 2μm are used. These particles are diluted with water and injected into the nozzle via a syringe using the New Era Pump Model 100. A continuous wave (CW) laser with 10 W is employed to excite the particles, and a FASTCAM UX100 Mini camera is used to capture images of the particles at different times. The exposure time is maintained according to the requirements for each experiment conducted at varied flow rates, specifically 20μL/min, 40 μL/ min, 80 μL/min.

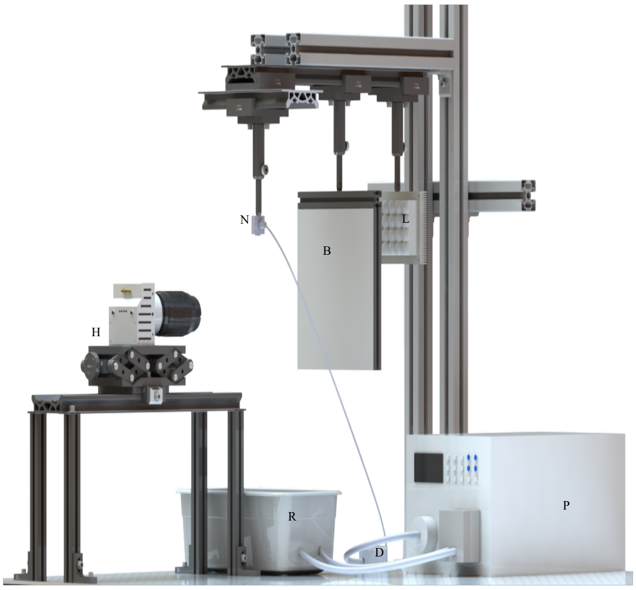


Fig. 3.5 A detailed arrangement illustrating the nozzle system, fluid reservoirs, pressure regulators, and optical components used to visualize and analyze the liquid sheet formation, with precise control over flow rates

3.4 Experimental Setup

The experimental apparatus used to generate liquid sheets using water is shown in Figure 3.5. The setup comprises a microfluidic system designed to deliver precise fluid flow under ambient conditions. The system consists of the following key components:

1. **High-Pressure Dosing Unit (80P, KNAUER) [P]**: This unit is designed to operate under ambient conditions, ensuring the consistent delivery of fluid to the microfluidic model.Given that pressures can escalate to nearly 6-8 bars during spray formation at higher flow rates, utilizing a high-performance liquid chromatography (HPLC) pump is essential. This pump is capable of handling such high pressures, ensuring reliable and consistent fluid delivery while preventing potential system failures or inconsistencies in spray generation. It pumps distilled water with the properties shown in the Table 3.1.

2. **Pulse Dampener (ZN60, LeadFluid) [D]**: Positioned between the pump and the microfluidic model, this component is essential for eliminating

pressure fluctuations. The dampener ensures a steady, pulse-free flow, which is critical for maintaining experimental accuracy and consistency in the fluid dynamics observed within the microfluidic model.

3. **Microfluidic Model [N]**: The fluid enters the model via rubber tubing, navigating through a converging channel after a 90-degree turn, as shown in Figure 3.6. This configuration minimizes turbulence and optimizes flow focusing, which is crucial for achieving desired fluidic behaviors within the model. Upon exiting the nozzle, the fluid is discharged vertically downward into the environment. The nozzle's design facilitates the formation of liquid sheets, characterized by thin, continuous layers of fluid as it transitions from the nozzle to the surrounding environment.

4. **Reservoir [R]**: The discharged fluid is collected in a reservoir, completing the fluid circuit. This component captures the fluid post-discharge, allowing for subsequent analysis or recycling within the system.

5. **Illumination Source [L]**: Backlight imaging methods are applied to acquire shadowgraph images of the produced sheets. Therefore, a continuous illumination source (MultiLED QX, GSVitec) is located behind the tested model. Its intensity is set at 10% to prevent overexposure in the recorded images.

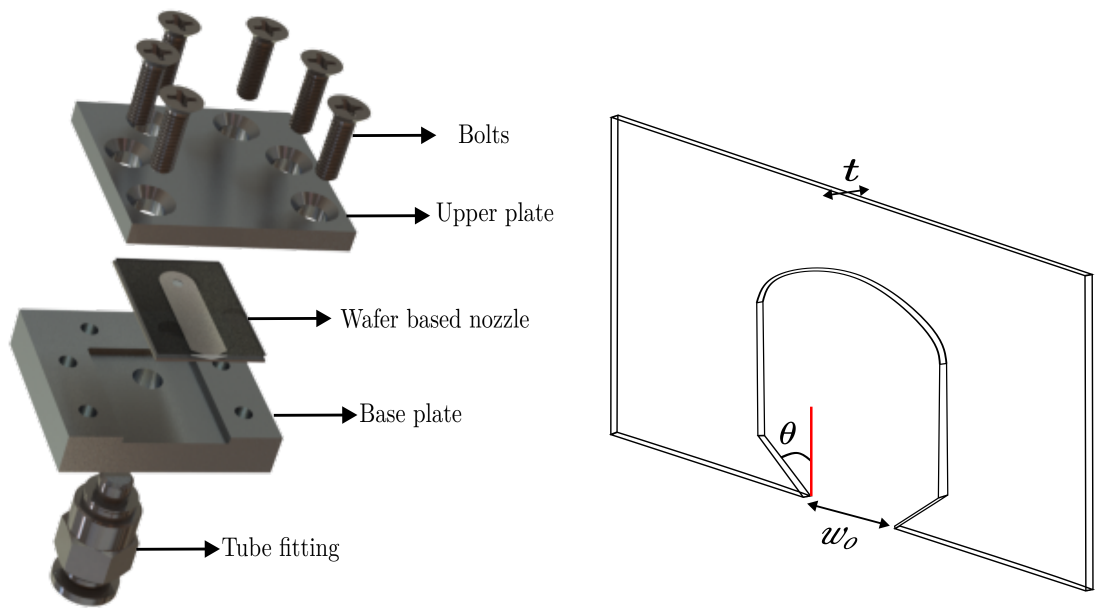
6. **Background Plane [B]**: A light-diffusion paper foil is attached to a hanging frame positioned between the light source and the model to create a uniformly illuminated background.

7. **High-Speed Camera [H]**: A high-speed camera (FASTCAM SA-Z, Photron) with a F2.8/100 mm lens (Samyang Telephoto Macro) is used to sufficiently capture the entire field of view and acquire snapshots of the liquid sheets. The camera is installed on a moving platform with potential 3-D positioning adjustments to ensure its alignment with the microfluidic device and the light source. During the experiments, images are acquired at 100 kHz allowing the capture of rapid dynamics and transient phenomena such as the wave formation and breakup, since a higher framerate helps to minimize the motion blur ensuring the fast moving features are clearly resolved. An exposure time of 8.93 μs is chosen to strike a balance between capturing sufficient light for a clear image of the liquid sheet structure and also to minimize the motion blur ensuring that there is no overlapping of the frames and smearing from the motion.
   Using a magnification of 70 μm/pixel strategically balances field of view and resolution, allowing for the comprehensive capture of the entire liquid sheet

structure and droplet formation. This level of detail minimizes pixelation, ensuring that both the sheet dynamics and droplet sizes are clearly resolved in the captured images. Although finer magnification enhances resolution and facilitates the observation of smaller droplets with greater clarity, it may come at the expense of broader structural information.

Fluid Property	**Value**
Density	998 Kg/m^3
Surface tension	0.072 N/m
Dynamic viscosity	0.001 Pa·s

Table 3.1 Properties of working fluid



(a) (b)

Fig. 3.6 Nozzle configuration for liquid sheet (LS) generation: (a) Exploded view of the nozzle assembly, illustrating the individual components for liquid sheet formation. (b) Schematic diagram of the nozzle, emphasizing key dimensions essential to the performance of liquid sheet generation

Chapter 4

Numerical Method

4.1 Introduction

This chapter presents the numerical methodologies used to analyze the velocity field inside the microfluidic nozzle, providing insights into the flow behavior prior to the generation of the liquid sheet. The CFD simulations are performed using ANSYS Fluent 2022 R1, that utilizes the Finite-Volume Method (FVM) for discretizing and solving the governing equations. Due to the proprietary nature of the software, the numerical procedures are outlined succinctly in this document, with the reader being referred to the ANSYS User's Manual for an exhaustive treatment [88].

First, the governing fluid dynamics equations, including continuity and momentum equations, are derived from mass and momentum conservation principles and form the foundation of the flow analysis. ANSYS Fluent employs an FVM approach, discretizing the computational domain into a finite number of control volumes. The governing equations are integrated over these control volumes, and the fluxes at the control volume faces are approximated using interpolation schemes, ensuring the conservation of mass and momentum across the domain. The numerical scheme is designed to balance accuracy with computational efficiency, employing methods such as higher-order upwind schemes to minimize numerical diffusion and ensure stability, particularly in high-gradient regions of the flow.

Next, the implementation of boundary conditions within the numerical framework is discussed. These conditions are essential for accurately simulating real-world scenarios and are applied using algorithms that represent physical constraints like no-slip walls, inlet, and outlet flows. The boundary conditions are customized to meet the simulation's specific needs, ensuring the numerical solution aligns with the physical realities of the problem.

4.2 Governing Equations

This section presents the conservation equations for mass and momentum used to simulate steady-state, laminar flow in a converging nozzle designed to generate a liquid sheet. The fluid dynamics, governed by the Navier-Stokes equations, capture the principles of mass and momentum conservation, essential for understanding flow behavior in confined geometries. The equations are expressed as follows:

1. Continuity equation (mass conservation): For an incompressible fluid, where the density ρ is constant, the continuity equation, which ensures the conservation of mass, is given by:

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{4.1}$$

   Here, u_i represents the velocity components in the i-th spatial direction, and x_i denotes the corresponding spatial coordinates. This equation implies that the divergence of the velocity field is zero, ensuring that the volume of fluid entering any differential control volume is exactly balanced by the volume exiting, with no net accumulation. This constraint is crucial in modeling incompressible flow, where any change in velocity must be accompanied by a corresponding change in flow direction to maintain constant density.

2. Momentum Equations: The momentum conservation in steady-state, incompressible laminar flow is described by the Navier-Stokes equations, which, in the absence of external body forces (e.g., gravity) and under steady-state conditions, reduce to:

$$\rho u_j \frac{\partial u_i}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \mu \frac{\partial^2 u_i}{\partial x_j^2} \tag{4.2}$$

   In this equation: ρ is the fluid density, which remains constant, u_i and u_j are the velocity components in the i-th and j-th directions, respectively, p denotes the pressure field, μ is the dynamic viscosity of the fluid, and $\frac{\partial^2 u_i}{\partial x_j^2}$ represents the viscous diffusion term, accounting for the internal frictional forces due to the fluid's viscosity.

   The momentum equation expresses a balance between the inertial forces (represented by the convective term $\rho u_j \frac{\partial u_i}{\partial x_j}$), pressure forces (represented by the gradient $\frac{\partial p}{\partial x_i}$), and viscous forces (represented by the Laplacian $\mu \frac{\partial^2 u_i}{\partial x_j^2}$). In a laminar flow regime, the viscous forces dominate, leading to a smooth, well-ordered flow without turbulence.

These equations provide a complete framework for analyzing steady-state, incompressible laminar flow in confined geometries, like the microfluidic convergent nozzle used in this study. The continuity equation ensures mass conservation, while the momentum equations govern force distribution, enabling accurate flow simulation.

4.3 Solver and software

For all numerical investigations of the converging nozzle designed to analyze the internal flow and generate a liquid sheet the commercial CFD software ANSYS Fluent version 2022 R1 is employed. This solver utilizes a second-order spatial discretization scheme based on an upwind flux method, ensuring high accuracy in capturing the flow dynamics within the nozzle. ANSYS Fluent is capable of providing steady-state solutions, leveraging its robust numerical framework to handle the steady-state nature of the incompressible flow analysis effectively. The implicit method employed for solving the governing equations allows for efficient handling of the steady-state conditions without the constraints associated with explicit time-marching methods.

The computational domain was discretized using ANSYS Meshing, which allowed for the creation of structured hexahedral meshes. These meshes are particularly advantageous for their accuracy and convergence properties, especially when the geometry permits such grids. The systematic refinement capabilities of hexahedral grids further enhance the reliability of the simulation results. The numerical results were visualized using the open-source software Paraview 3.10.1, with the data exported in the CGNS format. Paraview's flexibility, especially its ability to handle large datasets and be extended with MATLAB scripts, facilitated efficient post-processing, including automated procedures for analyzing and visualizing the flow characteristics within the nozzle.

4.4 Computational domain and boundary conditions

The computational domain for the numerical investigation has been specifically tailored to analyze the internal flow within a converging nozzle designed to generate a liquid sheet under steady-state, single-phase, incompressible flow conditions. The domain is created using SolidWorks v2024, and its schematic representation is provided in Figure 4.1. The model is optimized to accurately represent the nozzle's internal flow dynamics while minimizing computational resource requirements.

Instead of modeling the entire experimental setup, the simulation focuses solely on the nozzle's internal flow inside the nozzle.

Despite the nozzle's small thickness i.e; t=60μm, the following reasons explain why Hele-Shaw instabilities are not observed inside the nozzle:

1. For Hele-Shaw flow to be effectively observed, the thickness of the gap t should be much smaller than its width w (i.e., $t \ll w$)

   In the geometry used in this simulation, the gap between the plates is not extremely small compared to the flow's characteristic length scale i.e; $t = 0.067w_o$ depicting that the flow is not quasi-two-dimensional.

2. As Hele-Shaw flow operates at lower Re, lesser than 1, but our case has higher Re $\approx 66\,(>> 1)$ even for the flowrate of 20μL/min, the condition is not met.

3. The flow is not primarily driven by pressure gradient but is governed by inertia.

The geometry is designed to be identical to the actual wafer-based nozzle dimensions and the physical dimensions of the computational domain are shown in the Table 4.1.

Dimension		**Value**
Outlet width	w_o	900 μm
Inlet width	w_i	4.95 mm
Thickness	t	60 μm
Angle	θ	60^o

Table 4.1 Dimensions of the geometry

The computational boundary conditions are configured to match the physical setup of the converging nozzle. The boundary conditions at the inlet are defined with a specified velocity magnitude to simulate the inflow conditions accurately. Due to the complexities associated with the exact inlet conditions, a detailed simulation of the complete nozzle system may be conducted to determine the appropriate inlet profiles. The exit boundary condition is set to a zero gauge pressure outlet to simulate the flow exiting the nozzle, while the walls are set to no-slip conditions. These configurations ensure that the numerical model effectively captures the flow characteristics within the converging nozzle prior to the formation of the liquid sheet.

Figure 4.1 also illustrates the boundary conditions applied in the current simulations.

1. Inlet Boundary Condition: For steady-state, incompressible, laminar flow, the inflow boundary condition requires the specification of the velocity profile normal to the boundary. This includes defining the magnitude and direction of the velocity vector at the inlet. Since heat transfer is not considered in this scenario, the temperature at the inflow is assumed to be constant and does not require specification.

2. Outlet Boundary Condition: At the subsonic outflow boundary, the condition is specified by setting a reference pressure. This ensures that the flow exits the domain without introducing additional pressure gradients that could affect the stability and accuracy of the simulation. The reference pressure is chosen to be consistent with the ambient pressure or the pressure at the domain exit.

3. Wall Boundary Condition: For wall boundaries, the normal component of the velocity must be zero, i.e., $u_n = 0$, to enforce the no-slip condition. This condition ensures that the fluid velocity at the wall matches the wall's surface velocity, which is zero in this laminar flow scenario. Since heat transfer is not considered, there are no thermal boundary conditions applied at the wall.

4.5 Numerical Grid

In this study, a structured mesh was generated using ANSYS Meshing to discretize the 3D geometry of the convergent nozzle, as shown in Figure 4.1. The grid size is adjusted to balance accuracy and computational efficiency. The mesh consists of approximately 6.4×10^4 elements, with a refined grid structure to capture the flow dynamics inside the nozzle. Particular attention is given to the regions of high velocity gradient, such as the nozzle throat and the boundary layers near the walls, where the grid is refined to ensure accurate resolution of flow and the formation of the liquid sheet. The refinement strategy focuses on accurately capturing the near-wall flow behavior.

4.6 Numerical Accuracy

A mesh sensitivity analysis was also conducted to assess the numerical error for the case of flow inside the convergent nozzle. The chosen method involves continuous refinement of the numerical mesh and assesses discretization errors. The Grid Convergence Index (GCI), recommended by the Fluids Engineering Division of ASME, is used for quantifying these errors.

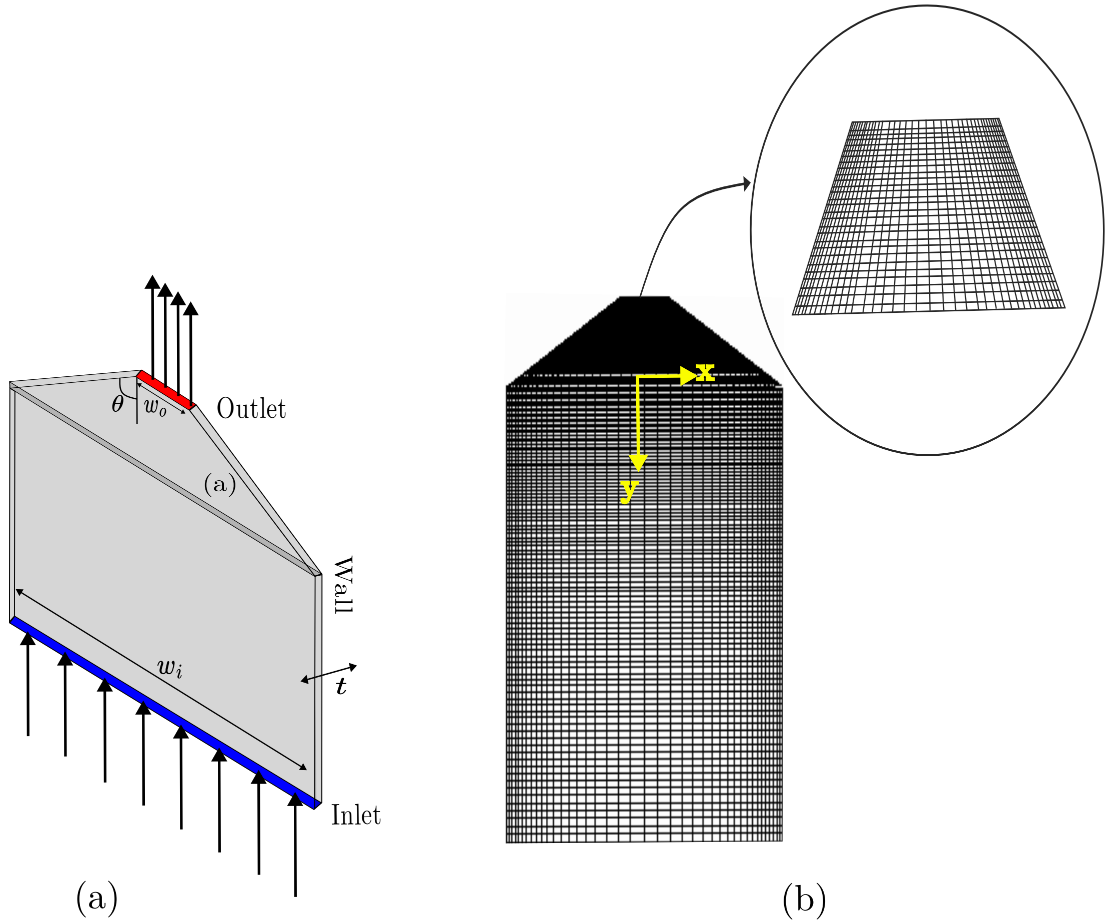


Fig. 4.1 Geometry and mesh of the nozzle: (a) Computational domain showing the detailed structure of the converging nozzle used for the simulations (b) Mesh generation illustrating the distribution and refinement of the mesh within the nozzle to ensure accurate capture of flow characteristics

This allows the calculation of error bands and demonstrates grid independence for the simulation. However, it is important to note that the GCI only quantifies discretization errors and does not account for modeling errors, such as those related to boundary conditions or flow assumptions specific to laminar, incompressible, steady-state conditions.

Grid refinement is conducted systematically using a recommended refinement factor of about 1.65, which corresponds to an increase in grid resolution by approximately doubling the number of elements between successive meshes. Given the use of structured grid elements for discretization, the refinement is applied uniformly across all three spatial directions. To ensure the accuracy of the simulation, only geometrically similar cells are used throughout the domain. The distance from the channel wall to the first grid cell is also adjusted according to the refinement factor, ensuring that the near-wall resolution is consistent with the overall grid refinement strategy.

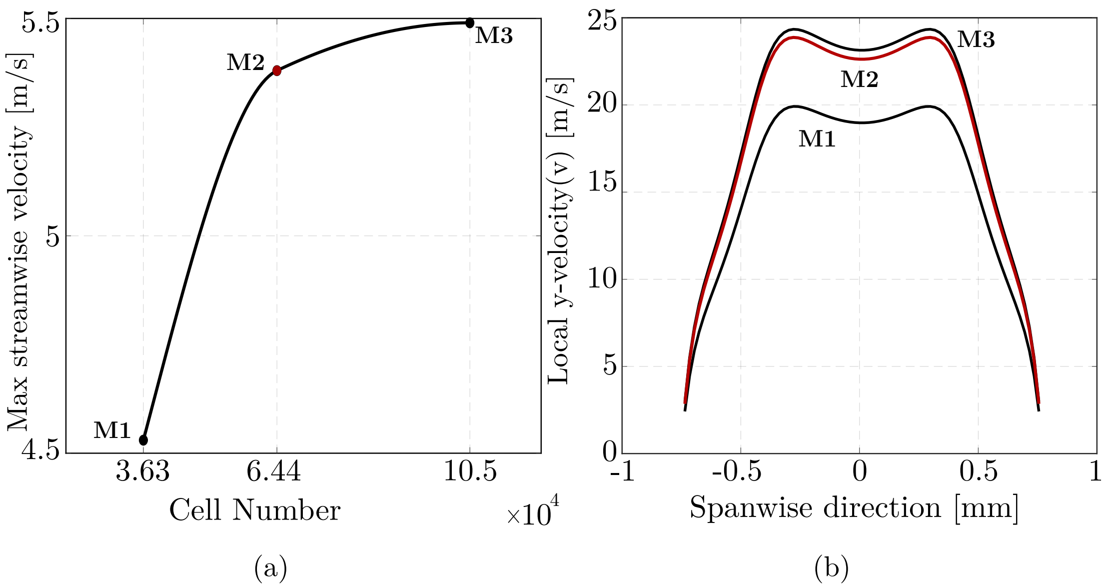


Fig. 4.2 (a) Grid independence test demonstrating solution stability across different mesh densities (b) Velocity variation analysis for multiple mesh resolutions to ensure accuracy of flow predictions

For the Grid Convergence Index (GCI) calculation, the velocity variation in the spanwise direction is evaluated. The respective values in the velocity calculation are shown in Table 4.2. The discretization uncertainty was estimated to be only 2% (for the finest mesh M3) when compared to the mesh M2 as shown in Figure 4.2. So, this meshing parameters (M2) are applied to all grids used in the study.

$$\Delta(\%) = \frac{m_{i+1} - m_i}{m_{i+1}} * 100\% \tag{4.3}$$

Grid	**Grid cells**	**Velocity (m/s)**	**Δ (%)**
M1 (coarse)	36300	19.9	-
M2 (intermediate)	64400	23.87	16.1
M3 (fine)	105000	24.34	1.9

Table 4.2 Grid parameters used for the mesh sensitivity analysis of the calculated velocity magnitude at the centerline of the convergent nozzle for a flow rate of 80 μL/min

4.7 Numerical validation

This section describes the numerical results obtained for the converging channel. The numerical model is used to calculate the flow characteristics within the channel. The analysis is divided into several parts, focusing on the influence of different parameters, such as Reynolds number (flow rate), nozzle convergence angle. Although many factors can affect the flow distribution, this section specifically investigates these key parameters.

The computational parameters for the baseline configuration, including the inlet width, channel convergence angle, and outlet width, are assumed to be consistent across different flow rate variations. This simplifying assumption allows for an examination of how the channel's convergence affects the flow characteristics.

A comparison between the CFD simulations and Particle Image Velocimetry (PIV) experimental results to analyze the internal flow dynamics of a nozzle designed to produce a liquid sheet. The CFD simulations provide detailed insights into the velocity and pressure distributions within the nozzle, enabling a thorough understanding of the flow characteristics (Table 4.3). Concurrently, the PIV technique offers high-resolution experimental data, capturing the intricate flow behaviors with precision. By systematically juxtaposing these methodologies, the accuracy is evaluated and reliability of the CFD models in predicting the flow patterns observed in the experiments. This comparison elucidates key similarities and discrepancies, offering valuable insights into the fluid dynamics of the nozzle and informing potential refinements in simulation approaches.

Both CFD and PIV contours converge strikingly, offering a coherent and detailed depiction of the flow dynamics within the converging nozzle. The velocity gradient, evident from the contours, is pronounced, with a differential of up to 4.0 mm/s for the flowrate of 80 μ L/min between the centerline and the periphery of the nozzle, indicative of significant shear effects and enhanced fluid acceleration.

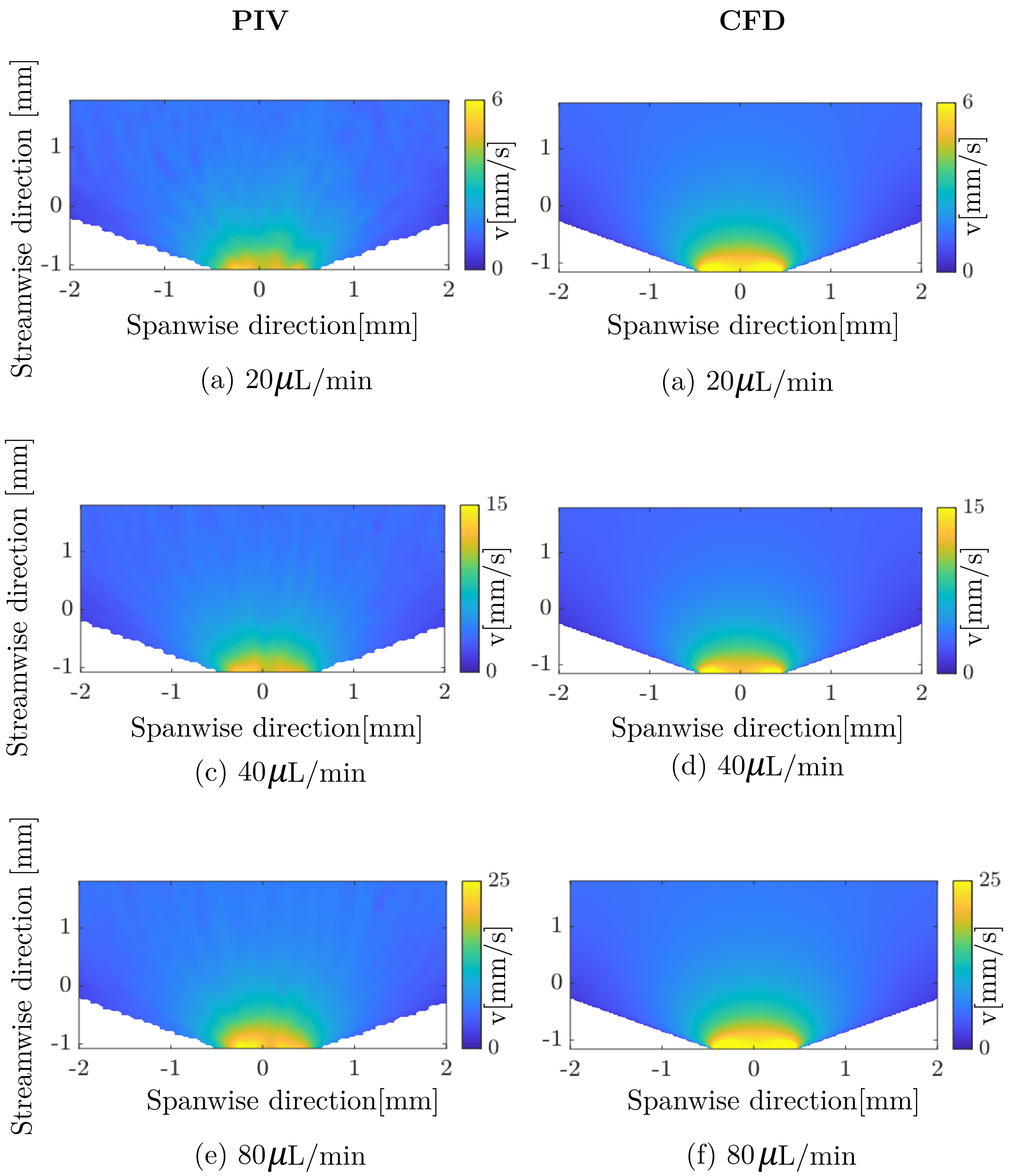


Fig. 4.3 Contours for PIV measurements: (a) 20μL/min, (c) 40μL/min, (e) 80μL/min and CFD simulations: (b) 20μL/min, (d) 40μL/min (f) 80μL/min

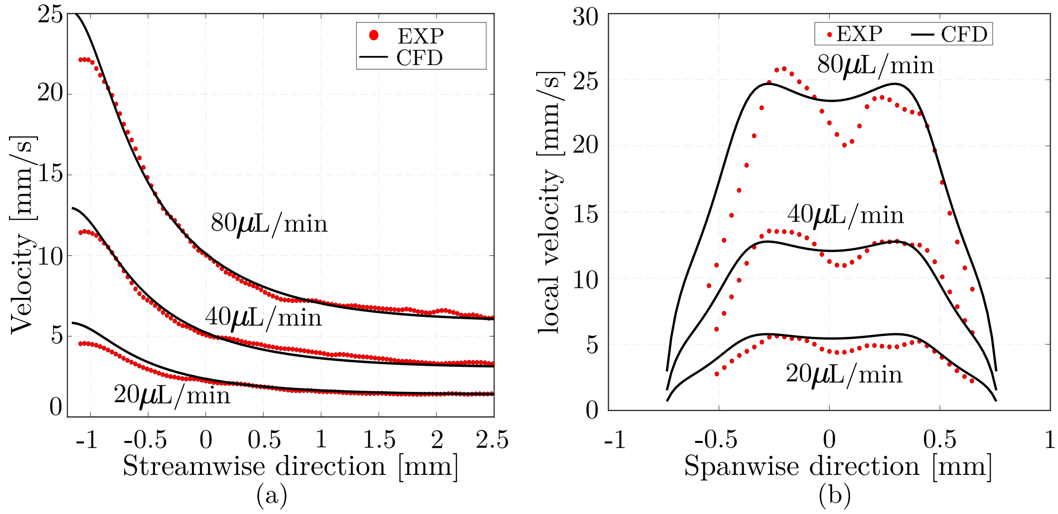


Fig. 4.4 (a) Velocity magnitude for PIV vs CFD (b) y-velocity component along the length of the nozzle at the exit for PIV vs CFD

The close alignment between the PIV and CFD results underscores the accuracy of the numerical simulations in capturing the complex flow phenomena observed experimentally. Both sets of contours demonstrate substantial rotational flow near the nozzle exit. This rotational component induces pronounced asymmetries and localized disturbances within the internal flow, which are critical to understanding the nozzle's performance.

This congruence in contour data enhances our comprehension of how elevated flow rates impact internal flow characteristics, providing crucial insights into the effects of velocity distribution.

Flow rate (μL/min)	**CFD** (mm/s)	**PIV** (mm/s)	**Error** (%)
80	23.4	21.0	17.0
40	12.1	11.3	6.6
20	5.4	4.8	10.0

Table 4.3 Error for velocity observed at the centerline at the exit for varying flow rates

Chapter 5

Results

This section presents the findings from a comprehensive investigation into the generation, characterization, and breakup of liquid sheets within a nozzle. The study addresses several key elements, including the formation of a stable liquid sheet, an analysis of its breakup dynamics, and precise measurements of its length and width. The results are organized into the following categories:

1. **CFD derived internal nozzle flow:** Numerical simulations were performed to visualize the internal flow within the nozzle. This investigation was conducted for varying flow rates (Q), thicknesses (t), convergent angles (θ), and outlet widths (w_o) of the nozzle to understand how these parameters affect the flow behavior and influence the stability of the liquid exiting the nozzle.

2. **Effect of flowrate (Q):** The findings from our investigation into the effects of flow rate variation on the formation and characteristics of a liquid sheet are presented here. Understanding how flow rate influences liquid sheet behavior is crucial. This section systematically explores the dynamics involved in generating a stable liquid sheet and elucidates the mechanisms leading to its breakup. The study focuses on the impact of flow rate on key parameters such as sheet thickness, breakup length, and stability, all of which play a pivotal role in determining the performance and efficiency of systems that employ liquid sheets.

3. **Effect of the converging angle (θ):** This section examines the effects of the nozzle converging angle on the formation and characteristics of a liquid sheet. The converging angle is a critical parameter that influences the fluid dynamics as the liquid exits the nozzle, affecting key attributes such as initial sheet velocity, thickness, stability, and breakup length.

4. **Effect of outlet width (w_o):** The outlet width of a nozzle plays a significant role in determining the behavior and characteristics of a liquid sheet, making it a crucial parameter in sheet generation. The area and aspect ratio of the nozzle exit influence several key factors, including the sheet's stability, breakup dynamics, and spray pattern.

5. **Breakup regime classification:** Analyzing the breakup regimes offers critical insights into the behavior of liquid jets under various conditions. This analysis is essential for achieving optimized nozzle designs, as it helps to understand and control factors that influence jet stability and breakup.

6. **Influence of Polishing:** The surface finish of a nozzle, particularly the degree of polishing, plays a significant role in influencing the liquid dynamics as the fluid exits the nozzle. Polishing impacts key factors such as flow smoothness, velocity profile, and the stability of the resulting jet. This section focuses on investigating these effects in detail.

5.1 CFD derived internal nozzle flow

5.1.1 Effect of Flowrate

The different inflow rates directly affects the breakup regime and the droplet dynamics of the liquid sheet. So, it is important to look into the flow behavior within the nozzle. Three different flow rates i.e; 20, 40 and 80 ml/min are taken for the geometry specification mentioned in Table 4.1.

At a flow rate of 20 mL/min , the simulations delineated a distinct flow regime characterized by a subtle interplay between laminar behavior and localized vorticity near the nozzle exit. The velocity field at the exit was predominantly uniform, with a peak velocity of approximately 6.4 m/s at the centerline as shown in the Figure 5.1(a), precisely where the convergent geometry begins to exert its influence. This indicates that the flow remained largely laminar, with negligible rotational disturbances—a critical factor in maintaining the integrity of the liquid sheet.

The Figure 5.1(b) reveals a meticulously controlled velocity profile near the convergent area, underscoring the system's ability to sustain a smooth and steady acceleration through the nozzle.The minimal vorticity detected near the exit signifies a delicate balance within the flow dynamics, where the convergence-induced deceleration does not precipitate significant turbulence captured in contours in Figure 5.2.

This controlled flow regime is indicative of a well-engineered system, where the laminar flow ensures minimal shear stress and negligible turbulence, thereby optimizing the fluid's behavior at the exit. The findings suggest that the nozzle design effectively mitigates the potential for rotational instabilities.

For a flow rate of 40 mL/min, CFD analysis demonstrated a marked shift in flow behavior compared to the 20 mL/min case as can be observed in Figure 5.1(a) which is the case for the flow near the converging part as can be seen in 5.1(b). At this higher flow rate, the peak velocity increased significantly, reaching approximately 12.27 m/s at the centerline , more than doubling the peak observed at the lower flow rate. This escalation in velocity indicates a transition to a more dynamic flow regime, where the convergent geometry intensifies the fluid's momentum.

In comparison to the 20 mL/min flow, which exhibited a relatively gentle velocity gradient, the 40 mL/min case shows a steeper gradient, indicating a more pronounced acceleration near the nozzle centerline. This steeper gradient results from the higher flow rate's capacity to generate a stronger pressure differential and more rapid fluid acceleration. Consequently, the flow at 40 mL/min becomes more dynamic and concentrated.

The increase in flow rate also enhances rotational effects near the nozzle exit which can be seen in the contour (Figure 5.2). At higher velocities, the fluid's inertia amplifies any inherent rotational tendencies, leading to more pronounced swirl and asymmetry in the liquid sheet after it exits the nozzle. This phenomenon occurs because the higher flow rate introduces greater shear forces and centrifugal effects within the nozzle, which intensify as the fluid accelerates.

While the increased flow rate at 40 mL/min improves the acceleration and uniformity of the liquid sheet, it also introduces complexities associated with rotational flow. These rotational effects can lead to asymmetries and potential instability in the liquid sheet, in contrast to the more stable, laminar flow observed at 20 mL/min. Thus, the nozzle design must be optimized to balance the benefits of higher flow rates with the need to manage rotational instabilities effectively.

At a flow rate of 80 mL/min, the CFD analysis reveals a pronounced shift in flow characteristics compared to the 20 mL/min and 40 mL/min cases. The peak velocity at the centerline escalates to approximately 23.5 m/s, a substantial increase relative to the 6.4 m/s observed at 20 mL/min and 12.27 m/s at 40 mL/min. This increase in peak velocity signifies a dramatic enhancement in the fluid's kinetic energy and momentum as the flow rate rises.

The velocity gradient within the nozzle also becomes significantly steeper at 80 mL/min. This steep gradient, characterized by a more pronounced difference in velocity between the centerline and the nozzle walls, reflects a heightened shear force exerted on the fluid. As the flow rate increases, the acceleration of the fluid becomes more concentrated near the centerline, leading to more substantial shear effects across the cross-section of the nozzle.

In terms of rotational dynamics, the higher flow rate of 80 mL/min introduces a more complex flow regime compared to the 20 mL/min and 40 mL/min cases (based on the contours). The interaction between increased flow rate and rotational forces creates a more dynamic and potentially unstable flow pattern, compared to the more stable and less rotationally influenced flows observed at lower flow rates.

The transition from 20 mL/min to 80 mL/min can generate Kelvin-Helmholtz instabilities, characterized by wave-like disturbances along the shear layer. These instabilities can grow and eventually lead to vortices that disrupt the smooth flow of the liquid sheet.

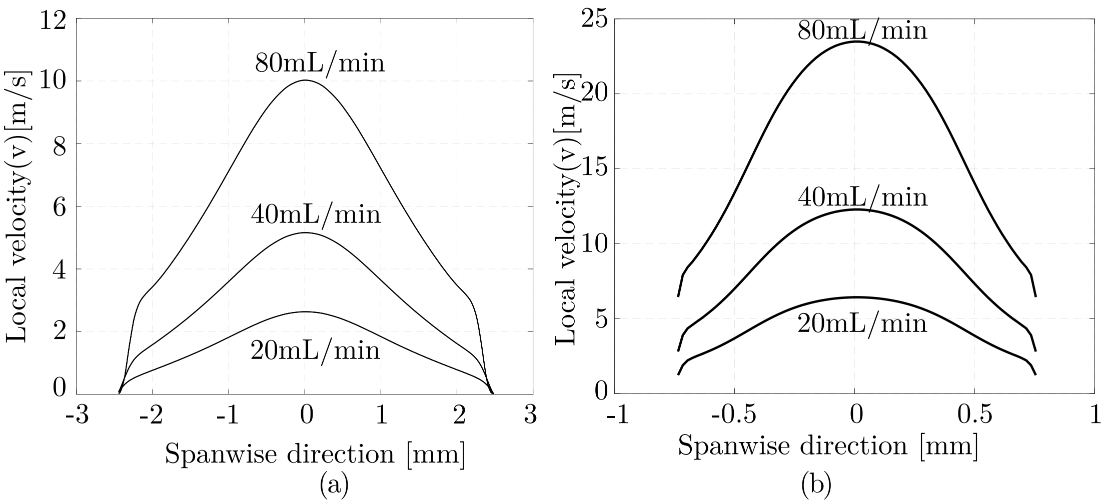


Fig. 5.1 y-component velocity (v) (a) near the exit (b) At the convergent region for a) 20 ml/min(b) 40 ml/min (c) 80 ml/min showcasing the influence of varying flow rates on the velocity distribution within the system

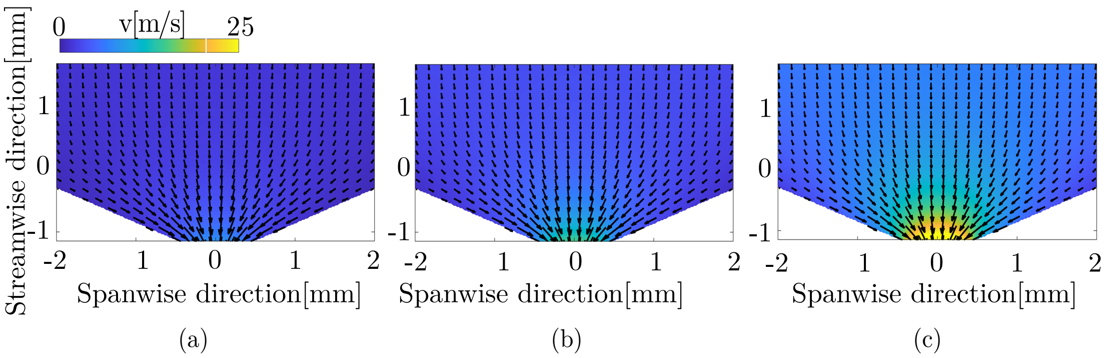


Fig. 5.2 y-Component velocity contours at varying flow rates: (a) 20 ml/min, (b) 40 ml/min, and (c) 80 ml/min. The consistent color bar across all contour plots facilitates a comparative analysis of velocity magnitudes, highlighting the variations in fluid dynamics as the flow rate increases

5.1.2 Effect of Converging Angle

A numerical simulation has been conducted on a geometry with a reduced convergent angle, while maintaining all other geometric parameters constant. The study investigates the impact of the decreased convergent angle on flow behavior at varying flow rates of 20, 40, and 80 ml/min. The plots are intentionally presented in dimensional form, with varying convergent angles, to visualize not only the profile of changes in flow behavior but also the magnitude of these variations under actual conditions.The flow exhibits self-similarity near the nozzle exit, indicating consistent flow patterns despite changes in flow rate.

At a flow rate of 20 ml/min, the variation in local velocity across the spanwise direction is minimal, reflecting a relatively uniform flow distribution. In contrast, at the higher flow rate of 80 ml/min, the spanwise velocity differences become more pronounced, highlighting the influence of increased flow rate on flow dynamics. The reduced convergent angle results in a consistently higher velocity at each point compared to the geometry with a larger angle of convergence, demonstrating the significant effect of the nozzle's geometric configuration on flow acceleration. This behavior is clearly illustrated in Figure 5.3, which compares the velocity profiles at different flow rates and convergence angles.

This analysis highlights the pivotal role of geometric refinement in optimizing fluid flow through convergent nozzle, particularly in influencing the characteristics of the exiting fluid. The findings emphasize how even subtle changes in nozzle geometry can lead to accelerated flow, resulting in a higher exit velocity and more streamlined discharge.

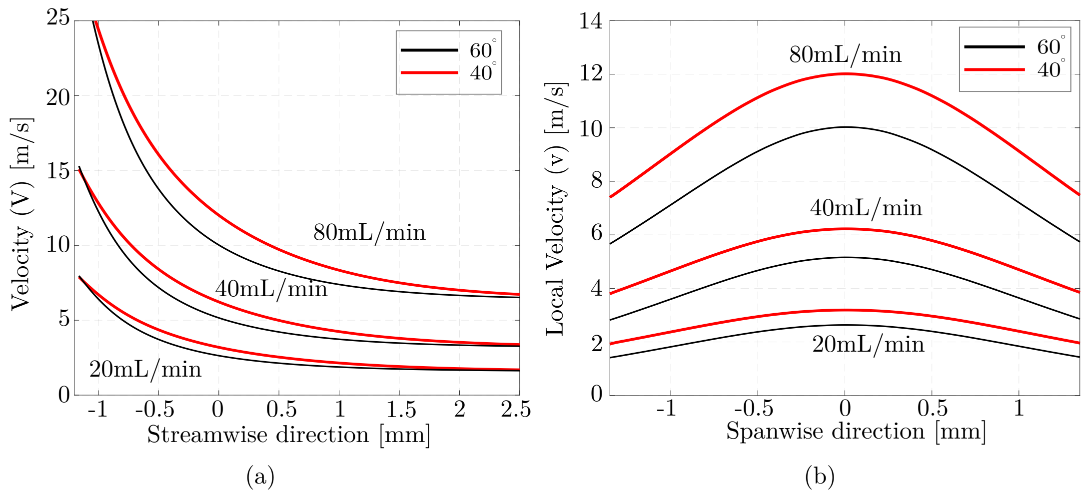


Fig. 5.3 Velocity profile due to outlet variation (a) streamwise direction (b) near the exit illustrating how adjustments in the nozzle angle influence the flow profile, providing insights into the relationship between nozzle orientation and fluid dynamics.

Furthermore, the velocity magnitude reaches the same value at the exit along the streamwise direction for both the convergent angle as the outlet cross-section for both the case is the same which is evident in Figure 5.3(a) Though, there are visible differences in the streamwise flow, near the convergent part of the nozzle.

The spanwise local velocity component are self-similar for both the angles with 60° having a higher value compared to the one with the reduced angle since the narrower convergent angle makes the fluid to gain acceleration as presented in 5.3(b).

5.1.3 Effect of Outlet Width

When the outlet width is reduced, while maintaining the same convergence angle and other geometric parameters, a noticeable increase in velocity is observed compared to the configuration with a larger outlet. This phenomenon is dictated by the principle of mass conservation, which requires the fluid velocity to increase as the cross-sectional area decreases to maintain a constant volumetric flow rate. Specifically, the continuity equation $Q = A \cdot v$, where Q is the volumetric flow rate, A is the area, and v is the velocity, governs this behavior. As the outlet area is reduced, the velocity increases to ensure that the same amount of fluid passes through the smaller cross-sectional region (Figure 5.4).

For the y-component of the velocity profile (Figure 5.4(a)), the smaller outlet leads to a visibly sharper increase in velocity magnitude, especially at the onset of convergence at $x = 0$ mm. Due to the higher exit velocities caused by the smaller cross-sectional area, the fluid is subjected to increased shear forces, which in turn promote the atomization of the liquid as it exits the nozzle. This atomization leads to the development of a spray, where the fluid breaks up into smaller droplets due to instabilities driven by the interaction between the high-velocity flow and the surrounding ambient medium. The higher velocities associated with the smaller outlet intensify this effect, producing a finer and more dispersed spray pattern.

In contrast, for the larger outlet (900 μm), the velocity decreases due to the expanded cross-sectional area, leading to a more distributed and less intense flow. The velocity profiles in this case exhibit more gradual gradients, indicating a lower rate of momentum transfer across the flow domain.

This comparison underscores the critical impact of outlet geometry on flow dynamics, particularly in terms of velocity distribution and flow focusing. The findings highlight how manipulating the outlet area can be used to control exit velocities and adjust the overall flow characteristics.

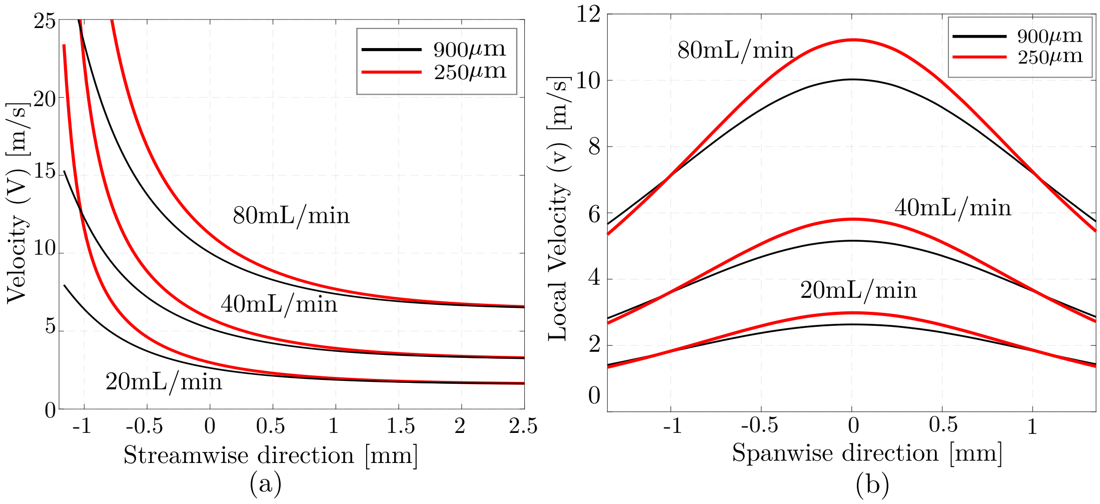


Fig. 5.4 Velocity profile variations due to changes in outlet dimensions: (a) Streamwise direction and (b) near the exit for 20 ml/min, 40 ml/min, and 80 ml/min demonstrating that as the outlet dimension decreases, the flow velocity increases, highlighting the relationship between outlet geometry and fluid dynamics.

For the decreased outlet area i.e; 250 μm, the spanwise velocity near the exit has a higher local velocity, eg: for the case of 80ml/min, the local velocity for w_o =250 μm is 11 m/s which is greater than the larger outlet width.(Figure 5.4b)

It should be noted that the smaller outlet width of the nozzle, at a higher inflow velocity creates a larger vorticity around the corners of the exit, this phenomena can lead to the formation of unstable sheet.

5.2 Effect of flowrate (Q):

Figure 5.5 presents the progressive expansion of the liquid sheet as the flow rate increases. Within the Reynolds number (Re) range of 1380 to 4608, the increased momentum significantly transforms the sheet's morphology. The presence of a small converging angle intensifies the jet's momentum in the gravitational direction, thereby facilitating the formation of a complex sheet chain [89]. Initially, this chain consists of seven interconnected links that diminish in size before eventually coalescing into a cylindrical jet stream.

The initial link of the chain is characterized by a markedly thin sheet, while subsequent links exhibit minimal variations in thickness between the rims and the central sheet. Despite the increased flow rate, as demonstrated in Figure 5.5(b), which causes an enlargement of both primary and secondary links, the chain still experiences disintegration at the third link and maintains a jet-like breakup pattern. At elevated flow rates, the liquid sheets undergo fragmentation, resulting in a finer spray composed of smaller droplets. This droplet formation occurs at the second link, oriented perpendicularly to the imaging plane, while a triangular spray pattern emerges at the apex of the primary sheet, as illustrated in 5.5(c) and Figure 5.5(d).

The formation of unstable liquid sheets becomes prominent when the Reynolds number (Re) exceeds 3225, marking a transition to a regime where fluid dynamics are dominated by instability growth. As Re increases, the momentum of the fluid amplifies the growth of these instabilities, a phenomenon well-documented by Sarchami et al.[90] In comparing figures 5.5(d) and 5.5(e), the latter image reveals more pronounced rim instabilities, which manifest as discontinuities along the sheet rims. These instabilities propagate through the sheet, preventing the rims from coalescing again at the sheet's apex, thereby altering the breakup dynamics.

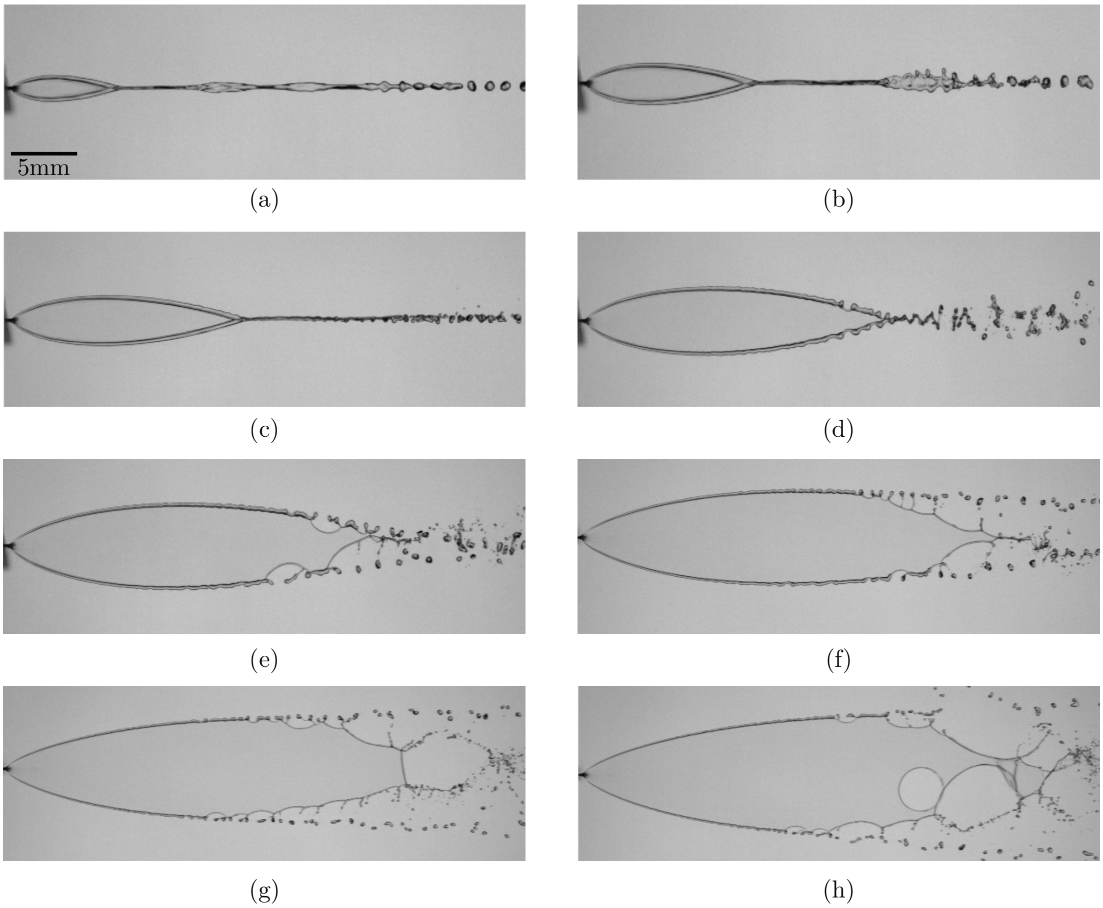


Fig. 5.5 LS variation for t=100 μm, w_o =750 μm & θ =40° for the flowrates of (a) 30, (b) 40, (c) 50, (d) 60, (e) 70, (f) 80, (g) 90 & (h) 100 ml/min

Initially, the liquid sheet disintegrates into very thin ligaments, which are subsequently fragmented into fine droplets. This process is illustrated in Figures 5.5(f) through 5.5(h), where droplets of various diameters are observed. However, this fragmentation comes at the cost of high pressure experienced by the pumping unit.

This fragmentation process is a critical aspect of atomization, affecting the distribution and size of droplets within the spray. The finer-droplet spray, originating from the thin ligaments, is interspersed with a larger droplet spray produced by the breakup of the thicker sheet rims, highlighting the complex interaction between sheet thickness and breakup dynamics.

As the flow rate increases from 80 to 100 ml/min, the increased momentum leads to the earlier breakup of the rims, occurring closer to the nozzle's outlet. This shift is indicative of a transition in the atomization regime, where the enhanced kinetic energy overcomes surface tension forces more readily. The shift in breakup direction is also notable; rim droplets, initially moving inward toward the sheet's longitudinal axis, begin to move outward, forming positive spray angles. This change in trajectory results in a wider spray angle and potentially more efficient coverage in applications such as spray cooling or coating.

Higher flow rates accentuate the amplification of instabilities, with the emergence of holes and aerodynamic waves on the sheet's surface. These surface perturbations are captured in Figures 5.5(g) and 5.5(h) and are characteristic of increased aerodynamic interaction and surface tension forces competing with inertial forces. This interaction ultimately leads to the fragmentation of ligaments and their breakdown into smaller droplets, a process described by N. Dombrowski and Johns [48] as crucial in the formation of fine sprays in high-speed liquid sheet disintegration.

The primary sheet's dimensions are significantly affected by variations in flow rate. Within its stable range, a gradual increase in both the length and width of the sheet is observed. Under identical geometrical conditions, the size of the sheet exhibits a quadratic dependence on the flow rate, expressed by a factor involving the Weber number (We) and the capillary number (Ca), where the exponent m differs for length and width, as reported by Hoffman et al. [91] This relationship highlights the complex interaction between inertial forces, represented by the Weber number, and surface tension effects, represented by the capillary number.

As flow rates surpass 70 ml/min, the width of the sheet continues to increase, whereas changes in the sheet's length become progressively smaller. This observation suggests a transition in the sheet dynamics, where the dominant forces affecting the breakup evolve with increasing flow rates. Notably, the breakup length appears to stabilize, as depicted in Figures 5.5(g)and 5.5(h). This stabilization indicates that the breakup length reaches a maximum before decreasing.[69][92]

The stabilization and subsequent reduction of the breakup length highlight a critical transition in the balance between stabilizing surface tension and destabilizing inertial forces. Initially, increasing the flow rate extends the breakup length by enhancing sheet stability. However, beyond a certain threshold, the increased inertial forces foster the development of instabilities that lead to an earlier onset of breakup. This behavior aligns with theoretical models of liquid sheet breakup dynamics, where the interplay between these forces dictates the breakup length and sheet morphology.

The insights gained from understanding the dependency of sheet size on flow rate provide a detailed understanding of the physical mechanisms governing liquid sheet stability and breakup, contributing to the broader field of fluid dynamics and interfacial phenomena.

The process applied for determining the geometry of the generated LS is as follows:

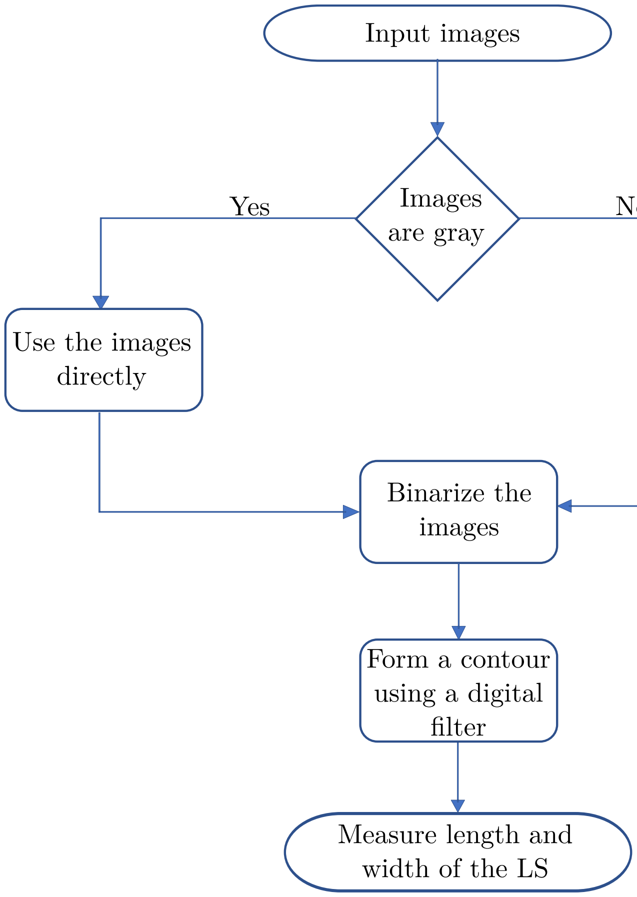


Fig. 5.6 Flowchart for obtaining the dimensions of the liquid sheet

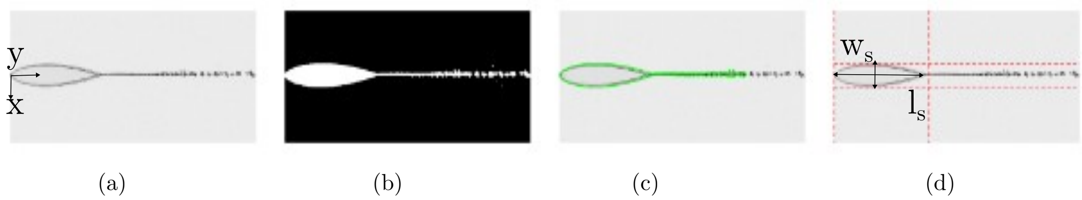


Fig. 5.7 Workflow applied on obtained shadowgraphy image where (a)grayscale shadowgraphy image, (b) binarized image,(c)edge contour and (d)dimensions of the liquid sheet

Based on the flowchart shown in 5.6, it is seen that to quantitatively analyze the dimensions of the liquid sheet, the captured shadowgraphy images such as shown in Figure 5.7(a) were first converted to grayscale to simplify processing. A thresholding technique, was then applied to binarize the images, effectively isolating the liquid sheet from the background as it can be observed in Figure 5.7(b). Edge detection algorithms were subsequently utilized to delineate the boundaries of the sheet structure accurately as can be seen in Figure 5.7(c). The contours of the liquid sheet were traced using MATLAB's 'bwboundaries' function, allowing for precise mapping of its edges. By carefully analyzing these contours, the primary geometric characteristics of the sheet, including its length and width, were calculated as shown in Figure 5.7(d) providing critical data for further dimensional and stability assessments.

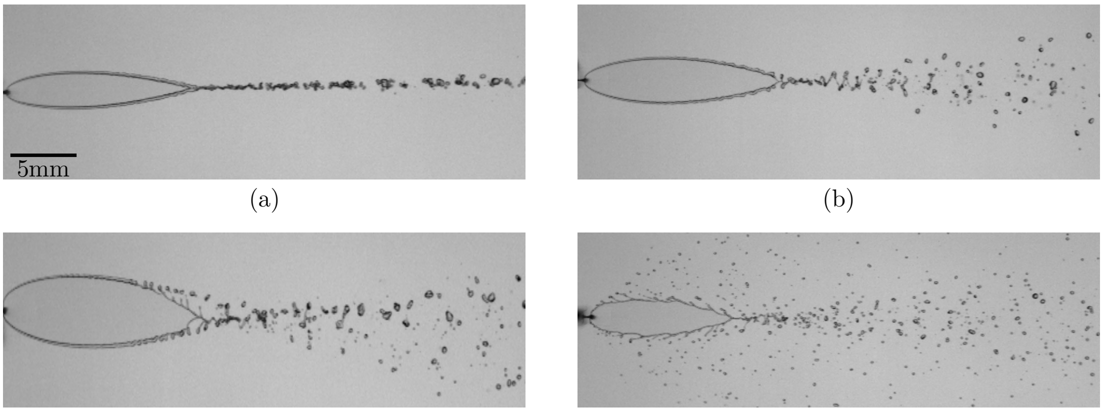
Based on 5.6, the quantitative analysis of the liquid sheet dimensions began by converting the captured shadowgraphy images to grayscale, streamlining the subsequent processing steps. To isolate the liquid sheet structure from the background, thresholding techniques were applied, resulting in a binarized image. Edge detection algorithms were then employed to precisely delineate the boundaries of the sheet. Utilizing MATLAB's 'bwboundaries' function, the contours of the sheet structure were mapped, facilitating accurate dimensional analysis. Through a detailed examination of these contours, the primary geometric characteristics of the sheet, specifically its length and width, were calculated. This analysis yielded essential quantitative data, instrumental for assessing the dimensional stability of the liquid sheet.

5.3 Effect of the converging angle (θ):

Figure 5.8 illustrates how the nozzle angle affects the stability and behavior of the liquid sheet. In Figure 5.8(a), a stable water sheet is produced with minimal disturbances. The initial breakup of the primary sheet into a spray occurs at the second link, resulting in droplets moving perpendicular to the imaging plane. This indicates a relatively stable regime where the sheet maintains coherence before transitioning into spray form.

As the nozzle angle is varied, depicted in Figure 5.8(b), the stability of the sheet begins to decline. Small-scale disturbances become evident near the apex of the sheet, suggesting the onset of instability. These disturbances can be attributed to changes in the distribution of forces across the sheet due to the altered nozzle angle, which affects the sheet's ability to maintain its structural integrity.

Despite these disturbances, stable sheets are still formed, as evidenced by the continued presence of coherent structures. The rims of the sheet exhibit thinning, and gaps are observed, but the sheet demonstrates a notable resilience to these perturbations. However, the process of spray generation has become more visible within the imaging plane, indicating a shift in the breakup dynamics where droplets are more directly observable along the sheet's trajectory.



(a) (b) (c) (d)

Fig. 5.8 LS variation for t=50μm, w_o =750μm at the flow rate of 30ml/min for angles (a)θ =40°, (b)θ =50°, (c)θ =60°, (d) θ =70°

Additionally, there is a slight increase in the sheet's size within this range of nozzle angles. This suggests that the nozzle angle influences not only the stability but also the dimensions of the sheet. The angle likely impacts the initial momentum distribution and the balance between inertial and surface tension forces, thereby affecting the sheet's spread and thickness.

The observations in Figure 5.8 underscore the sensitivity of liquid sheet dynamics to geometrical parameters such as the nozzle angle. By altering the angle, the flow characteristics, including velocity distribution and pressure gradients, are modified, leading to changes in sheet morphology and breakup behavior.

Although increasing the nozzle angle to $\theta = 60°$ results in wider and longer sheets in this nozzle model, the sheets generated are unstable. The increase in angle amplifies disturbances that destabilize the continuous rims more frequently, as neighboring rim non-uniformities come into closer proximity, leading to interactions that promote disintegration. This rim and sheet breakup results in sprays with droplets of varying sizes, highlighting the complex dynamics involved.

The inward motion of the disintegrated rims contributes to the formation of a large-droplet spray, which bounds the small-droplet spray originating from the breakup of the sheet itself. The onset of higher amplitude instabilities near the nozzle's aperture is a key driver of the increased spray formation. These instabilities arise from the enhanced aerodynamic interactions as the fluid exits the nozzle, where the increased angle alters the velocity profile and pressure distribution across the sheet.

Figure 5.8(d) prominently illustrates how increasing the angle further accelerates disintegration. The disturbances completely destabilize the continuous rims, resulting in the formation of a thinner, rimless section. Such flow patterns have been previously observed in laminar jet collisions at low velocities.[?] In this context, the edges of the sheet, unable to maintain the curved trajectory imposed by the nozzle's geometry, fragment into thin ligaments that eventually detach. This fragmentation process is driven by a balance of forces where the inertia of the flow overcomes the cohesive surface tension forces, leading to rapid droplet formation.

As these ligaments break apart, droplet formation occurs almost instantaneously. The resulting spray downstream of the water sheet occupies a larger area due to its outward expansion, which is influenced by the increasing converging nozzle angle. This outward expansion is a result of the nozzle's geometry, which modifies the flow's radial distribution, enhancing the spread of the spray.

The increased nozzle angle intensifies the aerodynamic shear forces acting on the liquid sheet, promoting a rapid onset of instabilities that lead to disintegration. The balance between inertial, surface tension, and aerodynamic forces becomes critical in this regime, dictating the breakup patterns and droplet sizes. The spray's expanded area reflects the altered dynamics of sheet breakup, where the increased nozzle angle creates conditions that favor the rapid development of instabilities and subsequent fragmentation.

The interplay between nozzle geometry and liquid sheet behavior is essential for predicting and controlling the characteristics of sprays, particularly for the applications where precise droplet size distribution is critical. The insights gained from these observations help us with a deeper understanding of fluid dynamics in complex flow systems.

In fluid dynamics, the breakup dynamics of liquid sheets are significantly influenced by the Weber number (We), which represents the relative importance of inertial forces compared to surface tension forces, and the impingement angle (θ). The breakup length and width of liquid sheets exhibit distinct scaling behaviors with variations in these parameters. Initially, as the impingement angle increases, the liquid sheet experiences an increase in both length and width due to enhanced kinetic energy distribution along the sheet's surface.

The experimental data reveal that the maximum length of the liquid sheet occurs at an impingement angle of $\theta = 50°$. At this angle, the inertial forces optimally counterbalance the surface tension forces, allowing the sheet to stretch to its fullest extent. However, the width of the sheet continues to expand with increasing angle, reaching its maximum at $\theta = 60°$. This phenomenon suggests that the lateral spreading of the liquid is influenced more strongly by the impingement angle than the axial stretching at higher angles.

This behavior is corroborated by observations of low Reynolds number sheets, specifically at $Re = 87$ and $We = 558$ which is similar to the study carried out by Yang et al.[53] When the impingement angle is further increased to $\theta = 70°$, the liquid sheet begins to contract. This contraction is likely due to the destabilization of the sheet, as the excessive angle leads to increased fragmentation and droplet formation, reducing the effective length and width. Moreover, it is reported that the maxima for sheet length and width occur at approximately $\theta = 51.5°$ and $\theta = 58.5°$, respectively. [66]

5.4 Effect of outlet width (w_o):

Figure 5.9 illustrates the effects of varying the nozzle outlet geometry on the dynamic behavior of water sheets. The images indicate a progressive reduction in sheet dimensions as the outlet width increases, leading to a decrease in both the length and width of the primary water sheet. This reduction can be attributed to the lower aspect ratio of the nozzle resulting from the increased outlet width, which alters the fluid dynamics and consequently diminishes the size of the water sheet.

At the lower end of the parameter spectrum, where the nozzle outlet width is minimal, the water sheet exhibits pronounced instability. This instability manifests as premature fragmentation before the rims of the sheet have coalesced, indicating a failure to achieve a stable sheet structure. In this regime, the sheet flapping behavior becomes more pronounced, and significant aerodynamic wave disturbances develop. These disturbances are reminiscent of Kelvin-Helmholtz instabilities, which are characterized by the shearing of fluid layers and the subsequent formation of vortical structures.[93]

As the amplitude of these aerodynamic waves increases, it eventually approaches the radius of the ligaments within the sheet. When this critical amplitude-ligament radius ratio is reached, the sheet undergoes a breakup event, characterized by the detachment of the ligaments from the sheet. This phenomenon aligns with the theoretical framework proposed by Dombrowski and Johns, where wave-induced perturbations cause the ligaments to exceed their structural integrity limits, leading to fragmentation.[48] The intricate interaction between aerodynamic forces and fluid mechanics highlights the pivotal role of nozzle geometry in governing the stability and integrity of water sheets. This tells about the essential role of nozzle aspect ratio in ensuring the stability and coherence of water sheets.

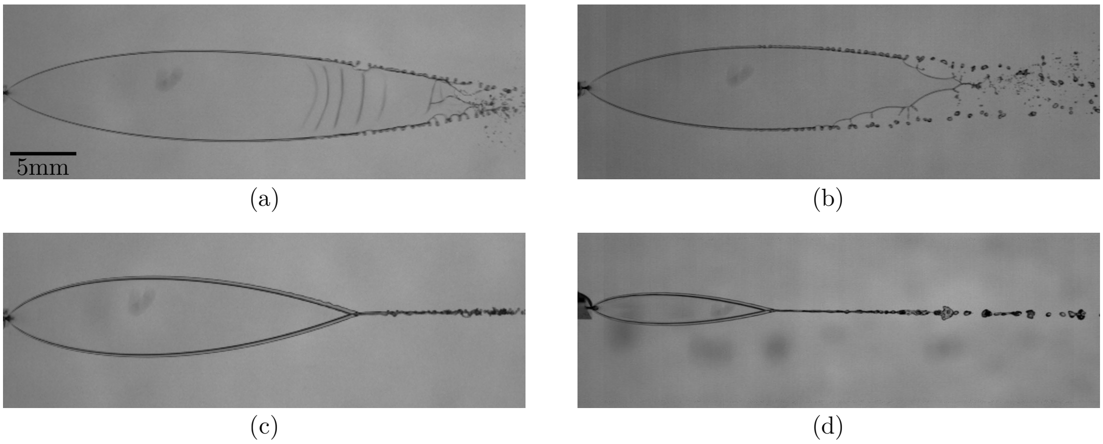


Fig. 5.9 LS variation for thickness t=150 μm & angle θ=50° at 60 ml/min for (a) 250μm, (b) 500μm, (c) 750μm, (d) 1000μm

This highlights the significant impact of nozzle aspect ratio on the stability and structural integrity of water sheets.

At elevated velocities, the wave amplification on the surface of a liquid sheet intensifies, ultimately leading to sheet breakup. However, when the outlet width (w_o) is increased while maintaining constant geometric and flow parameters, the nozzle's outlet aspect ratio increases, resulting in a reduction of exit velocity from 22.9 m/s to 6.7 m/s. This decrease in velocity leads to a reduction in wave amplitude after reaching its peak value, indicating that an increase in w_o effectively dampens disturbances within the fluid flow. Consequently, the liquid sheets become smaller yet exhibit enhanced stability due to the diminished wave activity.

Occasionally, minor instabilities with small amplitudes can manifest along the rims of the liquid sheets, but they generally do not cause sheet breakup. Instead, the presence of these instabilities suggests that the sheet structure can absorb or dissipate energy without disintegration. Additionally, a thin, two-dimensional spray is observed downstream of the primary liquid sheet at $w_o = 500\,\mu$m and $w_o = 750\,\mu$m. This spray, oriented perpendicular to the imaging plane, indicates a change in the breakup dynamics.

The alteration in breakup behavior is attributed to the interaction between wave growth, velocity, and the distance from the nozzle. As the aspect ratio increases, the breakup location becomes more sensitive to these factors, affecting the spatial development of the spray. Specifically, the increase in the nozzle's outlet aspect ratio leads to a reduction in exit velocity due to the conservation of mass flow rate and the spreading of the fluid stream over a larger exit area. This reduction in exit velocity mitigates the growth of capillary waves on the surface of the liquid sheet, resulting in a more stable sheet. The decreased wave amplitude reduces the likelihood of wave-induced breakup, allowing the liquid sheet to maintain its integrity over a longer distance.

Furthermore, the formation of a thin, two-dimensional spray downstream of the primary liquid sheet suggests a transition in the breakup mechanism. At higher aspect ratios, the breakup of the liquid sheet is influenced more by the aerodynamic forces acting on the sheet rather than by the intrinsic instabilities of the fluid flow. This shift in the breakup mechanism indicates that the nozzle's geometric configuration plays a significant role in determining the stability and breakup behavior of the liquid sheet.

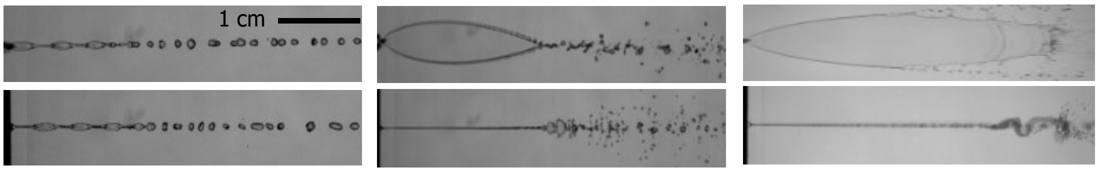
Further increasing the outlet width to $w_o = 1000\,\mu$m results in a notable reduction in the size of the primary sheet. The observed progressive decrease in the length and width of the sheets with increasing outlet widths corroborates Equations 5.1 and 5.2 (same as Eqn. 2.10), as w_o is a parameter in both the Weber number (We) and the non-dimensional parameter α.[57].

$$\frac{l_s}{t} = 0.36We_e\,Ca_e^{-0.1}(1+1.5\alpha^2)^{-0.5}[1+cot(0.67\theta]^{-0.5} \tag{5.1}$$

$$\frac{w_s}{t} = 0.074We_e\,Ca_e^{-0.2}(1+1.5\alpha^2)^{-1}[1+cot(0.67\theta)]^{-1} \tag{5.2}$$

Moreover, as shorter and narrower water sheets are generated, their rims tend to become thicker and more pronounced. This thickening is attributed to the redistribution of fluid mass along the edges of the sheet, which enhances the sheet's stability to a certain extent. However, these thicker rims are also more susceptible to disintegration via the Rayleigh-Plateau instability, leading to the formation of larger droplets. This mode of breakup, characterized by a low-velocity jet, is consistent with the predictions of the Rayleigh-Plateau instability theory, which describes the breakup of a cylindrical jet into droplets due to surface tension forces.[94]

The synergy between the nozzle's geometric parameters and the resulting fluid sheet behavior is crucial for understanding and optimizing the breakup processes. Specifically, as the outlet width w_o increases, the Weber number (We), which is a dimensionless quantity representing the ratio of inertial forces to surface tension forces, decreases. This reduction in We indicates a decrease in the relative influence of inertial forces, leading to a more stable sheet with reduced wave growth. The parameter α, which is related to the aspect ratio of the nozzle, further influences the stability and breakup dynamics of the sheet.



(a) (b) (c)

Fig. 5.10 Different morphology of liquid sheets obtained at varying Re, top image showing the front view and the bottom image shows the side view:(a)Stable liquid chain, (b)Stable liquid sheet with spray formation and (c) Unstable liquid sheet with spray formation

5.5 Breakup Regime Classification

Liquid sheet formations have been rigorously analyzed and classified according to their geometric configurations and the specific instability mechanisms that precipitate their disintegration.[42] In this study, we delineate the flow phenomena into three principal regimes:

1. **Stable Liquid Chains:** These formations consist of a sequence of interconnected liquid sheets that are subject to capillary instabilities as can be seen in 5.10(a). The stability of each link in the chain is influenced by its aspect ratio, defined as the ratio of the length to the width of the liquid sheet. Links with larger aspect ratios are more susceptible to instabilities due to enhanced capillary effects, while shorter, broader links exhibit greater stability. Over time, these structures break apart into discrete droplets via a jet-induced breakup process.As the liquid exits the nozzle, inertial forces associated with the jet flow overcome the cohesive forces of surface tension, leading to the fragmentation of the interconnected sheets.The droplets formed from the breakup of stable liquid chains exhibit a range of sizes, which are influenced by factors such as the length and width of the liquid sheets, the intensity of the capillary instabilities, and the jet velocity.

2. **Stable Liquid Sheets:** Figure 5.10(b) shows table liquid sheets represent a distinct and highly coherent regime, characterized by their sustained structural integrity and uniformity. Unlike their unstable counterparts, stable liquid sheets exhibit unique physical properties and breakup dynamics, maintaining their coherence over extended distances. This section provides an in-depth analysis of the physics and defining features of stable liquid sheets, highlighting their stability and the conditions that govern their behavior.The primary instability mechanism for stable liquid sheets is the formation of capillary waves. These waves are driven by surface tension forces that cause small perturbations on the sheet's surface to grow, leading to eventual breakup. Since, the liquid sheet interacts with air (environment),

Kelvin-Helmholtz instabilities might develop. These instabilities arise from velocity differences between the liquid and air, contributing to the growth of perturbations and droplet formation.

The breakup of stable liquid sheets typically results in the formation of uniform-sized droplets. The regularity of droplet size is a direct consequence of the uniform thickness and consistent breakup pattern of the sheet.

3. **Unstable Liquid Sheets:** Similar to stable liquid sheets, these formations also possess fewer than two stable links but display more complex breakup dynamics. The inherent instability of these sheets leads to a two-scale atomization process: A salient feature of these sheets is that their rims do not converge at the link apex as shown in Figure 5.10(c). This non-convergence is a critical factor that distinguishes them from other types of liquid sheet formations. The lack of convergence at the apex results in an inherent instability within the sheet, setting the stage for distinct atomization dynamics.

   The atomization pattern can be categorized as follows:

   (a) **Coarse Droplet Distribution:** Due to the non-convergent nature of the rims, a coarse droplet distribution is generated. This coarse spray originates from the rims of the liquid sheets, where the liquid is subjected to higher perturbations and disturbances. The droplets formed in this region are typically larger in size, reflecting the increased instability at the sheet's edges.

   (b) **Fine Droplet Spray:** Concurrently, the internal breakup of the sheet, driven by its intrinsic instability and perturbations, leads to the formation of a finer droplet spray. This finer spray results from the internal fragmentation processes within the liquid sheet, where capillary forces and dynamic instabilities induce the breakup into smaller droplets.

   Unstable liquid sheets exhibit apex regions, where the sheet reaches its maximum height before undergoing breakup. The primary apex is the initial peak formed closest to the nozzle, while subsequent apexes may form further downstream. The breakup of unstable liquid sheets typically occurs between the first and second apexes. This region between the apexes is characterized by increased instabilities and perturbations that lead to the disintegration of the sheet into droplets.

   In cases where the second link of the liquid sheet only partially forms, the breakup dynamics are notably affected. The incomplete formation of the second link leads to an early onset of instabilities, causing the sheet to break up before a fully coherent second link can establish. Moreover, the spray generated from the breakup of such sheets is often narrow and oriented perpendicularly to the imaging plane. This specific orientation results from the directionality of the instabilities and the partial formation of the second link.

As the breakup point approaches the apex of the primary sheet, the wetting area of the resultant spray increases. This expansion is due to the dispersion of droplets over a wider area, influenced by the breakup dynamics and fluid properties. The characteristics of the spray, including droplet size distribution and spread, are directly impacted by the location of the breakup point relative to the apex. Closer breakup points result in a more concentrated spray, while points further from the apex lead to broader dispersion.

5.5.1 Regime Distribution

Figures 5.12 present a detailed analysis of the regime distributions mapped against nozzle angles and Reynolds numbers for various nominal thickness values of the nozzles. This section provides a scientific and quantitative evaluation of the effects of nozzle geometry on the stability and breakup dynamics of liquid sheets.

1. **Influence of Nozzle Angle** (θ)**:** The increasing nozzle angle (θ) accelerates the onset of breakup due to changes in fluid dynamics at the nozzle exit. This effect is particularly pronounced with greater nozzle thickness, as illustrated more clearly in Figures 5.12(b) and (c). The flow transitions to liquid sheet with spray at a comparatively lower number for a thinner nozzle as can be observed from map in Figure5.12(a).

   A larger nozzle angle alters the velocity profile and shear forces at the liquid sheet's surface, promoting the growth of perturbations that lead to breakup. The growth rate of surface perturbations is governed by the Weber number (We). Higher nozzle angles result in increased exit velocities and shear forces, enhancing the (We) and thus the rate at which perturbations grow, leading to earlier breakup.

2. **Influence of Nominal Thickness** (t)**:** As the nominal thickness (t) of the nozzles increases, the regime regions for liquid chains and liquid sheets extend into larger Reynolds number (Re) ranges. The Reynolds number Re characterizes the flow regime. Thicker nozzles produce more substantial flow inertia, allowing stable liquid sheet formations over broader flow conditions due to increased resistance to capillary and shear instabilities.

The quantification of Re for different regime transitions are observed as mentioned below: For stable liquid sheets, the minimum Reynolds number required to obtain a stable liquid sheet is $Re = 1360$ for a nozzle with t=50μm and $\theta = 40^o$. This threshold signifies the critical flow conditions necessary to sustain a coherent liquid sheet without transitioning into a disintegrated spray.

While, for the unstable liquid sheets, the threshold Reynolds number for the onset of unstable liquid sheets is $Re = 2266$ for the same nozzle parameters t=50μm and ($\theta = 40^o$). This value represents the critical point where inertial forces overcome surface tension, leading to sheet instability and breakup.

The results align well with the findings of Li and Ashgriz [66], who also identified threshold Reynolds numbers for regime transitions. This agreement underscores the robustness of the current study methodology and the general applicability of the observed phenomena.

Differences in threshold Reynolds numbers can be attributed to variations in experimental setups, nozzle geometries, and fluid properties. Li and Ashgriz's work focused on specific breakup mechanisms and may have used different flow conditions, leading to slight discrepancies.

The challenge lies in accurately identifying the transition points between these flow regimes, which can significantly impact the downstream processes. Traditional experimental methods, while effective, can be time-consuming and may not readily adapt to real-time monitoring needs. To overcome these limitations, a binary classification model that leverages advanced machine learning techniques to analyze flow images captured during nozzle operation is proposed.

This approach involves the extraction of key features from the images, which are then used to train a machine learning model capable of distinguishing between the different flow states. By implementing this model, a reliable and efficient method for predicting liquid sheet stability.

The flow regime is classified into two primary categories: sheet (class 1) and liquid chain (class 0) for simplicity in the binary classification framework as can be seen in the Figure 5.11. Assuming that the classification model produces a continuous output value, a threshold function f(θ,Re) is defined as shown in Equation 5.3, which is used to determine the probability or score associated with the positive class (liquid sheet). This output value is typically a probability between 0 and 1.

$$f(\theta, \mathrm{Re}) = P_1 + P_2\theta + P_3\mathrm{Re} \tag{5.3}$$

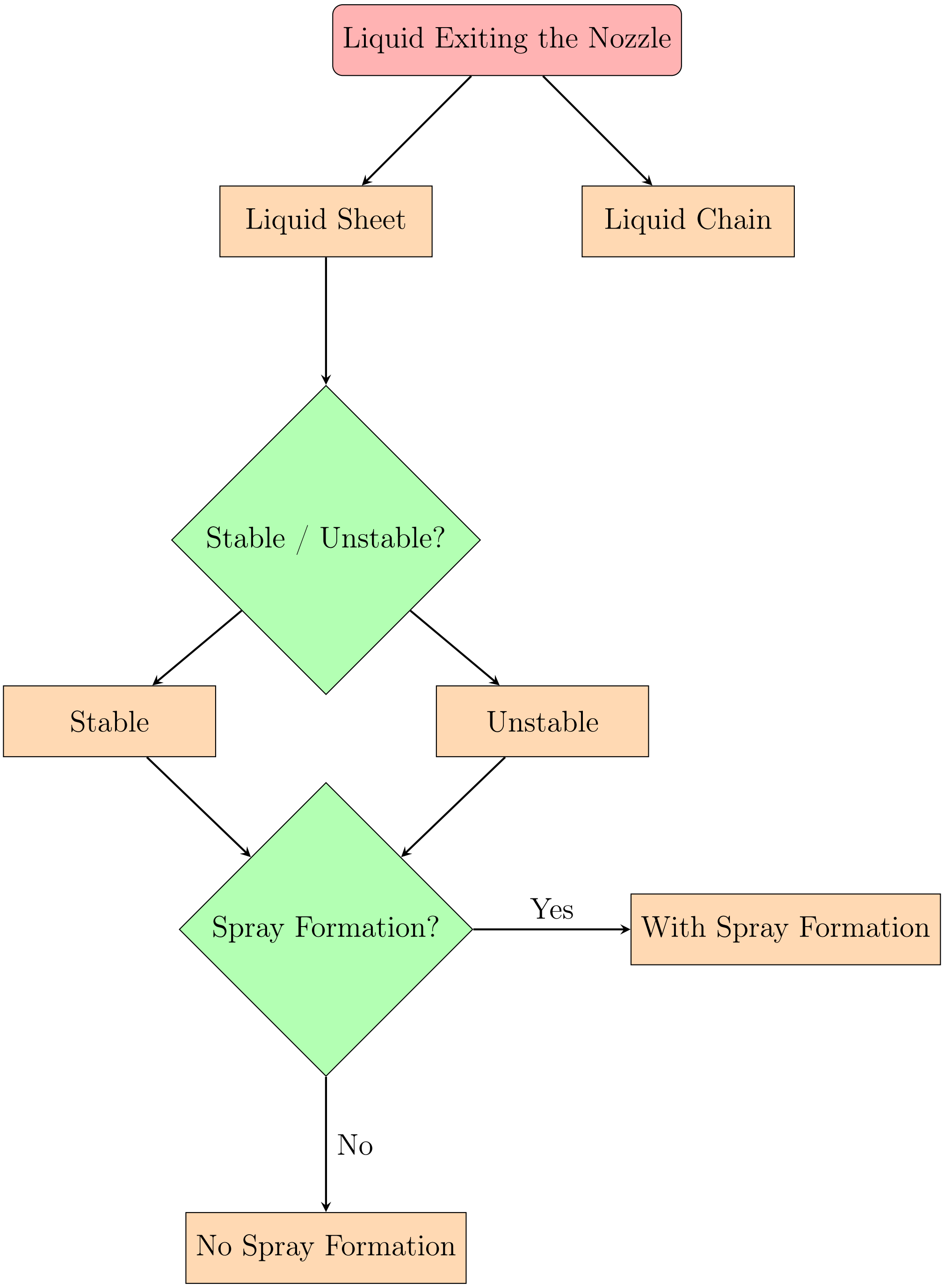


Fig. 5.11 Classification of exiting flow structures: This classification categorizes liquid sheets into various types, illustrating the binary nature of the classification problem, which enables the generation of comparative plots across different flow regimes.

The coefficients P_1, P_2, and P_3 are crucial in defining the shape and behavior of the threshold function.

$$f(\theta, \mathrm{Re}) = \begin{cases} \text{Liquid Sheet} & \text{if } f \geq 0 \\ \text{Liquid Chain} & \text{if } f < 0 \end{cases} \tag{5.4}$$

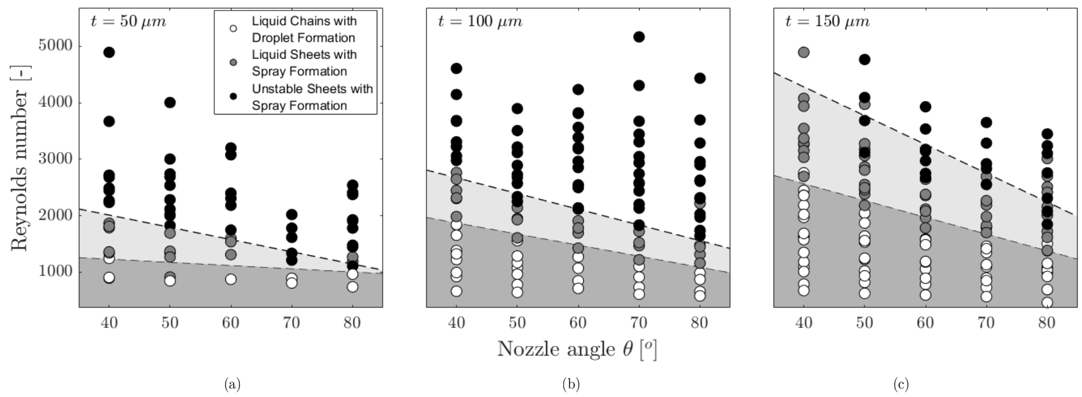


Fig. 5.12 Liquid sheet breakup regime maps for varying nozzle thicknesses: (a) 50 μm, (b) 100 μm, and (c) 150 μm. This figure illustrates the different flow regimes based on the classification problem, demonstrating how nozzle thickness influences liquid sheet behavior and breakup characteristics.

5.6 Influence of Polishing

The identification of outlier data points within the experimental results is primarily linked to fabrication-induced defects, which are a consequence of imperfections during critical stages of the device manufacturing process, such as dicing and polishing. These fabrication flaws, while often subtle, can have profound effects on the overall device performance and experimental accuracy, particularly in the context of high-precision flow testing. As demonstrated by Crissman et al. [95], even minor irregularities introduced during these stages can result in significant deviations from expected performance, thereby skewing the results and leading to misleading interpretations of the underlying physics.

Figure 5.13 presents a comprehensive assessment of both the qualitative and quantitative impacts that dicing and polishing have on the outlet surface of the device. In this case, the dicing process was conducted with cutting lines placed in close proximity to the outlet planes, leading to minimal margins available for subsequent polishing. This reduced polishing margin significantly limits the capacity for surface refinement, a factor that has been observed to exacerbate edge imperfections. When comparing Figures the critical importance of an optimized polishing protocol becomes apparent. Polishing is not only essential for removing surface irregularities but also for restoring the outlet geometry, which directly affects the fluid dynamics during testing.

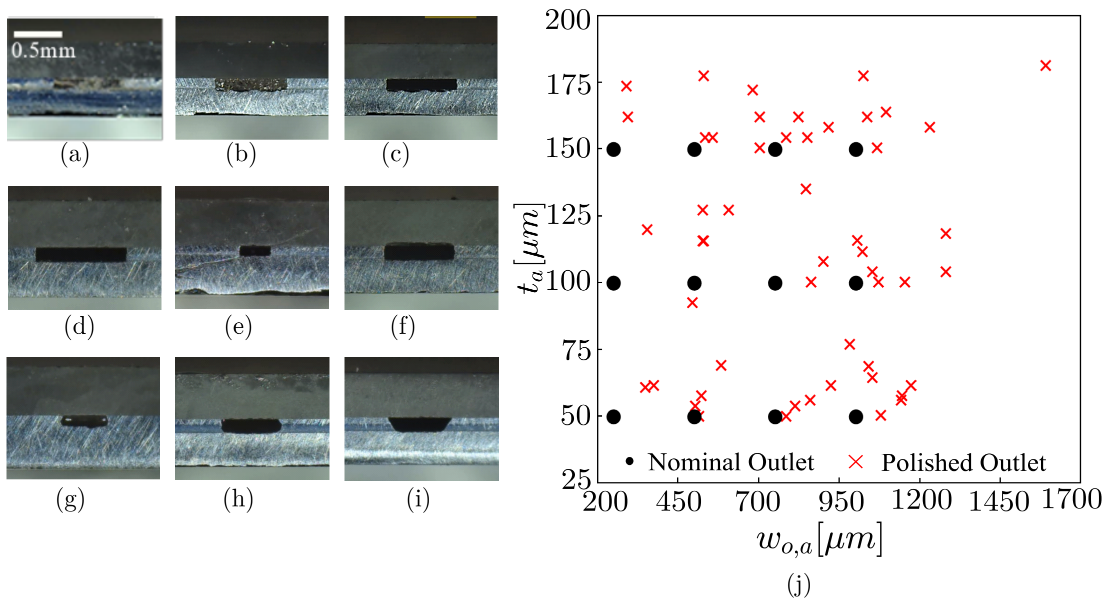


Fig. 5.13 Outlet morphology morphology (a) before, (b) after polishing & (c) after wax removal - Outlet flaws and deviations from (d) the ideal rectangular shape (e) dicing crack interfering with the outlet, (f) glass wafer edge chamfering, (g) high edge roughness, (h) elliptic & (i) trapezoid outlet shapes. Scale bar: 0.5 mm. - (j) Design point shift due to outlet polishing.

In Figure 5.13(a) the presence of residual wax is particularly notable. The wax serves as an optical contrast agent that reveals the outlet profile more clearly, but also underscores the challenges posed by dicing. The edges of the outlet surfaces are often susceptible to unevenness due to two key factors: the mechanical wear of the dicing tool and the heterogeneous nature of the wafer material itself as can be seen in Figure 5.13(b)and 5.13(c). Tool wear, an often-overlooked variable, leads to a gradual degradation in the sharpness of the cuts, producing non-uniform edges that can introduce turbulence or flow perturbations during fluid testing. Additionally, variations in wafer properties, such as differences in hardness or crystalline structure, can exacerbate this uneven cutting behavior, leading to inconsistent surface characteristics across different wafers or even within the same device.

The detrimental impact of these protruding edges is particularly pronounced in precision flow testing like the case of generating a sub-micron thickness liquid sheet, where surface irregularities can distort flow patterns, generate unwanted secondary flows, or introduce localized disturbances. These factors collectively contribute to measurement inaccuracies by making it challenging to define consistent dimension. Furthermore, as surface roughness can influence boundary layer formation and transition to turbulence, even minor imperfections can lead to disproportionately large deviations in flow behavior, where laminar flow regimes are highly sensitive to surface conditions.

As a result, the polishing process emerges as a critical step, not merely for aesthetic or surface smoothness but for ensuring the functional integrity of the device during testing. A meticulous polishing protocol is necessary to mitigate the adverse effects of dicing and to produce a surface geometry that supports accurate, repeatable measurements. Ha et al. [57] have demonstrated that without sufficient polishing, residual edge defects can persist, leading to inconsistent experimental results that undermine the validity of the study. Therefore, to maintain the fidelity of fluid dynamic testing and ensure the reproducibility of results, the implementation of stringent polishing techniques is imperative.

5.6.1 Various Irregularities Observed After Dicing

Polishing reveals defects arising from both the dicing and etching processes, as deviations from the ideal outlet geometry become more pronounced after surface refinement. Ideally, the outlet should conform to a precise, rectangular shape, as shown in Figure 5.13(d), which serves as the reference for a perfectly formed outlet. However, the fabrication process, particularly dicing, can introduce structural imperfections that compromise this ideal geometry.

Figure 5.13(e) highlights the effects of dicing tool wear, which can cause fragmentation of the silicon wafer. In this instance, cracks have propagated through the wafer, extending all the way to the nozzle outlet. Such cracks present a significant challenge, as they weaken the structural integrity of the outlet and can lead to fluid leakage or flow distortion during testing. Additionally, Figure 5.13(f) shows a chamfered edge on the glass wafer, which is another common consequence of dicing tool wear. Chamfering, while subtle, can alter the exit flow profile, introducing deviations in the boundary layer development and subsequently affecting flow uniformity.

In addition to dicing, etching processes contribute to the formation of outlet defects. During etching, the nozzle channel surface often deviates from being uniformly flat due to inconsistencies in material removal across the channel's thickness. These variations in surface flatness can exacerbate flow instabilities by promoting localized areas of increased roughness. For instance, Figure 5.13(g) illustrates localized roughness along the outlet edge, which likely arises from non-uniform etching. This roughness can induce secondary flow disturbances, increasing turbulence and potentially skewing experimental measurements.

Moreover, etching-induced geometric distortions further compromise the outlet's structural integrity. In Figures 5.13(h) , the outline of the outlet has deviated to an elliptical shape, while in Figure 5.13(i) , the outlet exhibits a trapezoidal geometry. These deviations from the desired rectangular shape can significantly influence fluid dynamics, particularly in microfluidic devices where precise outlet geometry is critical for maintaining laminar flow. The elliptical and trapezoidal shapes, for example, can cause asymmetry in the flow field, altering the velocity profile and introducing undesirable shear stresses that could impact both the accuracy and repeatability of experimental data.

The defects introduced by both dicing and etching processes emphasize the necessity of stringent quality control measures during fabrication. As K. S. Chen et al. [96] have demonstrated, the interaction between these defects and the fluid flow can lead to significant performance degradation, underscoring the importance of optimizing each step in the fabrication process to minimize imperfections.

5.6.2 Challenges After Polishing

Polishing, while effective in smoothing the chip's aperture and eliminating surface irregularities, introduces its own set of challenges by altering the outlet plane and consequently changing the outlet's dimensional profile. As shown in Figure 5.13(j), there is a clear dimensional discrepancy between the original and the polished outlet design points. The figure highlights the extent to which polishing modifies the outlet dimensions, deviating from the nominal specifications initially set during the design phase.

Each of the nominal design points corresponds to multiple models with varying nozzle angles, intended to produce a consistent set of outlet geometries. However, the polishing process introduces non-uniform material removal across different regions of the outlet, leading to unpredictable deviations from the original design. As demonstrated by the comparison in Figure 5.13(j), significant discrepancies are observed between the initial, unpolished state and the final, polished outlet geometries. These differences are not trivial; they effectively transform the outlet into a new design point, distinct from the original specifications. Such dimensional variations can alter critical performance parameters, including flow rate, velocity profile, and even pressure drop across the nozzle.

Polishing thus has the unintended consequence of multiplying the outlet design points. In practical terms, this means that each polished nozzle chip is characterized by a unique aspect ratio, dependent on the specific polishing conditions and material properties. As a result, it becomes nearly impossible to establish coherent trends between different models that share the same nominal outlet dimensions. Despite starting from identical design specifications, the final polished outlets exhibit enough variation that comparisons between models are rendered inconsistent.

The alteration of the outlet dimensions also complicates the evaluation of performance trends, as the modified geometry introduces deviations in flow behavior that are not accounted for in the initial design calculations. This lack of coherence between models with the same nominal design highlights the difficulty in ensuring reproducibility in experimental testing. The unique aspect ratio that results from polishing-induced changes in outlet geometry can introduce variabilities in the boundary layer formation, velocity distribution, and flow stability within the microfluidic system.

5.6.3 Discrepancies Between Experimental Observations and Theoretical Predictions

The significant alterations in the outlet dimensions also become evident when calculating the primary dimensions of the liquid sheets. These length and width calculations are performed using custom-developed MATLAB scripts. By applying the scaling analysis proposed by Ha et al. [57], which predicts the sheet length $l_{c,n}/t_n$ and width $w_{c,n}/t_n$ based on the nominal outlet dimensions, we obtain the graphs shown in Figures 5.14 and 5.15 .The calculations are presented in a dimensionless format, normalized by the nozzle thickness to allow for a direct comparison across different outlet geometries.

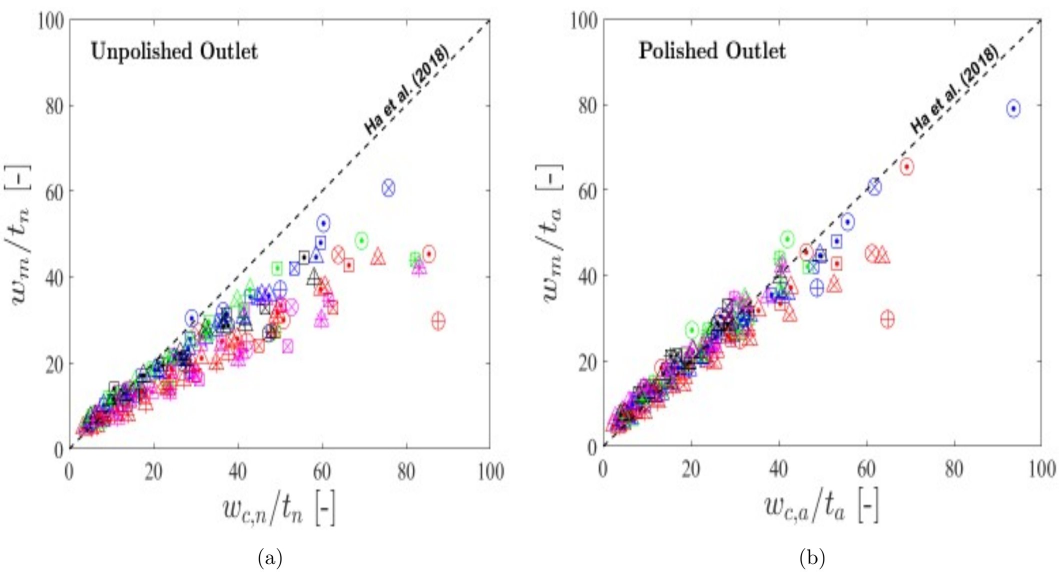


Fig. 5.14 Width measurement comparison with scaling models (a) before polishing -nominal outlet dimensions and (b) after polishing -actual outlet dimensions.

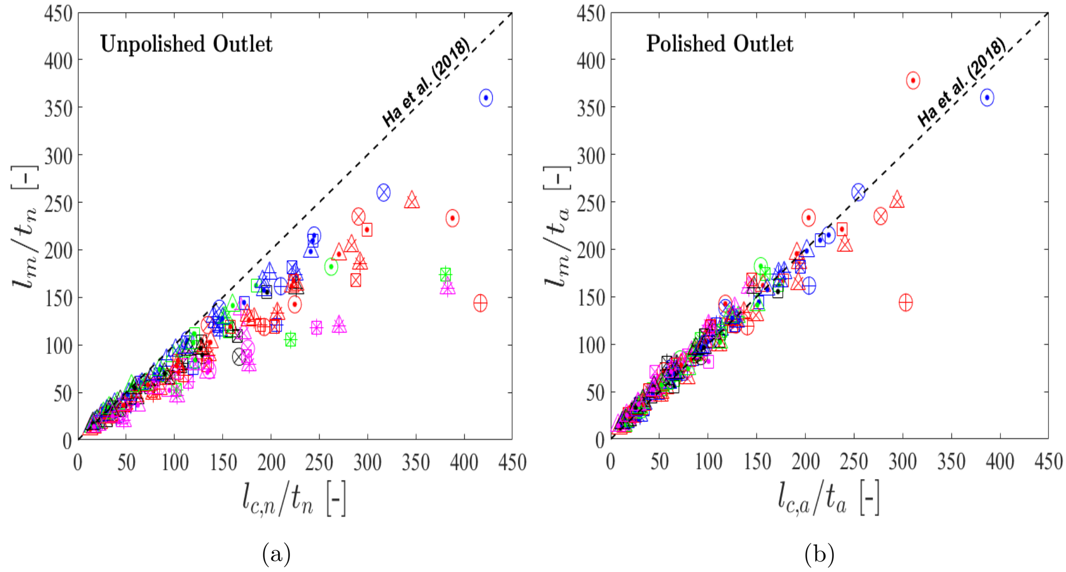


Fig. 5.15 Length measurement comparison with literature scaling models (a) before polishing -nominal outlet dimensions- and (b) after polishing -actual outlet dimensions

The predicted sheet dimensions are juxtaposed against experimental data. It is important to note that the experimental data points correspond exclusively to those tests that produced stable liquid structures, such as continuous chains and sheets. These stable structures provide a robust basis for comparison against the theoretical models. However, a systematic underestimation of the sheet dimensions is observed when using the nominal outlet dimensions for scaling. Specifically, both the measured lengths and widths are consistently smaller than the predicted values across the dataset, indicating a discrepancy between the theoretical scaling and the experimental outcomes.

This deviation from the predicted dimensions is particularly pronounced for larger sheets, where the nominal outlet dimensions appear to overestimate the actual size of the liquid structures. Limited agreement is observed for the smallest sheets, where the experimental data comes closer to the predicted values, albeit with some residual variance. These findings highlight the limitations of applying nominal outlet dimensions in scaling models, particularly when fabrication-induced geometric variations are not accounted for.

To address these discrepancies, the sheet dimensions are recalculated using the actual measured outlet dimensions rather than the nominal ones. These recalculations yield the modified graphs in Figures 5.14(b) and 5.15(b) where a much closer alignment between the predicted and measured dimensions is observed. The improved agreement across the entire dataset demonstrates that the incorporation of the actual outlet dimensions corrects the scaling analysis, bringing the predictions into better conformity with the experimental data.

The impact of this recalibration underscores the importance of using precise, experimentally measured outlet dimensions in theoretical modeling. The substitution of actual dimensions compensates for the geometric irregularities introduced during the fabrication process, which, if left unaccounted for, can skew theoretical predictions. This adjustment not only enhances the accuracy of the size predictions but also confirms the validity of the scaling models .

Figure A.5 provides an extensive legend detailing all the symbols used across Figures 5.14 and 5.15 ensuring clarity in data interpretation.

Chapter 6

Conclusions

This thesis provides an exhaustive and highly detailed exploration of the generation and dynamics of a liquid sheet formed using a microfluidic device, specifically designed with a slit-type converging nozzle. By integrating experimental observations with sophisticated numerical simulations, this research delves into the internal flow dynamics of the nozzle. It also offers a comprehensive analysis of the liquid sheet's behavior, from its initial formation at the nozzle exit to its eventual breakup into droplets during spray formation.

The research highlights the pivotal role played by critical parameters—such as the geometry of the nozzle, flow rate, and nozzle aspect ratio—in influencing the liquid sheet's fundamental characteristics, including its geometry, stability, and breakup behavior. By systematically varying these parameters, the study uncovered the significant impact they have on the liquid sheet's formation, such as altering the sheet's length, width and the onset of instability. These changes directly affect the liquid sheet's evolution, as the study demonstrated how modifications to these parameters could delay or hasten the sheet's breakup into droplets, a process governed by fluid instabilities. The liquid sheet's stability, which is a central concern in controlling the transition from a continuous sheet to a spray, was shown to be highly sensitive to the interactions between the internal flow dynamics and external disturbances.

The thesis incorporates detailed numerical simulations to supplement the experimental results obtained through PIV, using advanced computational fluid dynamics (CFD) techniques to model the internal flow within the nozzle. These simulations provide a nuanced understanding of the intricate flow patterns and velocity distributions that emerge within the converging nozzle, capturing the detailed effects of geometric variations and flow conditions. By correlating these internal flow dynamics with the subsequent formation of the liquid sheet, the simulations reveal the underlying mechanisms through which the nozzle's design and operational parameters influence the sheet's stability. The research helps us to understand that disturbances within the nozzle's flow field propagate to the liquid sheet, initiating instability modes that eventually lead to sheet breakup. The analysis of these instability mechanisms—such as the onset of Kelvin-Helmholtz instabilities and their amplification along the liquid sheet—was central to understanding the spray formation process.

Beyond the fluid dynamics, this work also contributes significantly to the microfabrication aspect of microfluidic devices. The development of the slit-type converging nozzle was achieved through the use of Deep Reactive Ion Etching (DRIE), a highly precise fabrication technique essential for producing the intricate geometric features required for accurate liquid sheet formation. The DRIE process, characterized by its ability to create deep, high-aspect-ratio etches in silicon, was critical in realizing the desired nozzle shape, which directly influenced the flow behavior and the liquid sheet's dynamics. The thesis offers an in-depth, step-by-step account of the DRIE process, from the design phase to the final fabrication, providing critical insights encountered during the development of the microfluidic nozzle. This detailed fabrication workflow serves as a valuable reference for future researchers aiming to replicate or optimize similar microfluidic devices, particularly those intended for controlled liquid sheet generation.

In addition to the focus on fabrication, the thesis provides a framework for assessing the liquid sheet's breakup dynamics, linking the instability growth to the droplet size distribution observed during spray formation. This framework integrates both the experimental and numerical aspects of the research, offering a comprehensive model that can predict the breakup behavior based on key operational parameters. By identifying the conditions under which the liquid sheet transitions from stable to unstable, and ultimately to droplet formation, the thesis provides crucial insights into the control mechanisms that can be employed in practical applications.

In conclusion, this particular work represents a significant advancement in the understanding of liquid sheet generation and its dynamics, providing a holistic view that spans both experimental and numerical domains, as well as the intricacies of microfluidic device fabrication. The comprehensive analysis of the effects of nozzle geometry, flow rate, and aspect ratio on liquid sheet behavior opens new avenues for optimizing microfluidic systems for a wide range of applications. Moreover, the methodological rigor demonstrated in both the experimental design and the fabrication processes offers a strong foundation for future work, making this thesis a critical reference for researchers and engineers engaged in the study of fluid dynamics and microfluidic technologies. The results and innovations presented here are poised to inform and inspire ongoing research in fields that require precise, efficient control of fluid dynamics at the microscale.

Appendix A

Experiments with Polyimide Nozzle

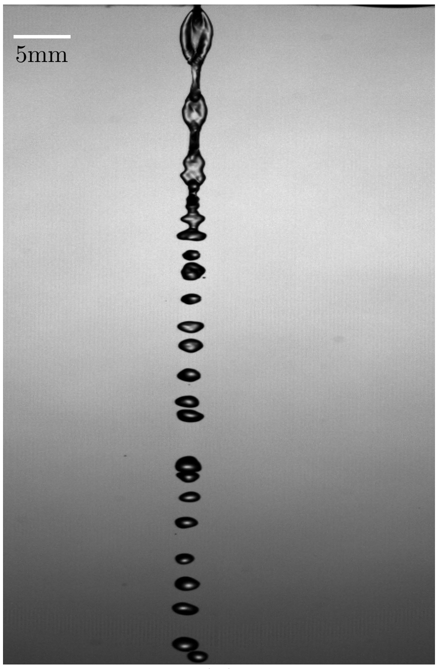
A.1 Uncertainty in Measurements for Stacked Model

In the experiments carried out using the stacked model, the liquid sheet shape is considerably unstable and there is a prominent effect of instabilities causing the sheet to undergo rough breakup. Furthermore, polishing of the nozzle at the exit of the nozzle causes sheet to become deviated from the axis of the nozzle resulting to the outgoing fluid to be significantly tilted.

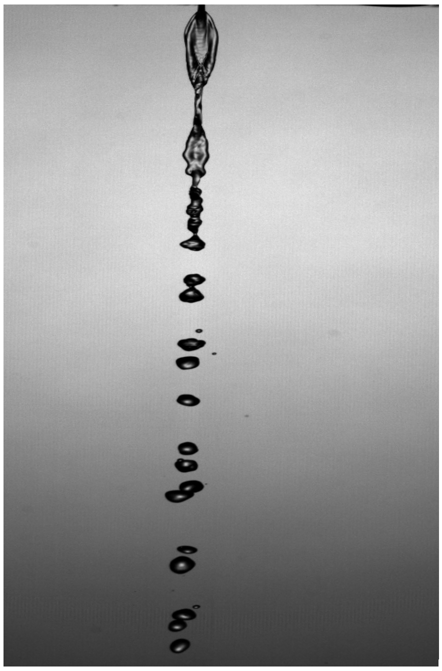
The experiments are carried out using water for outlet widths (w_o) $1000\mu m$ and $500\mu m$, angles (θ) $40, 60, 80$ deg and thickness (t) of $50, 127\mu m$. The fluid properties of water are referred from Santiago[57] and are given as follows:

Fluid	**Water**
density (kg/m^3)	1000
viscosity (Pas)	$1.2e-03$
surface tension (N/m)	0.072

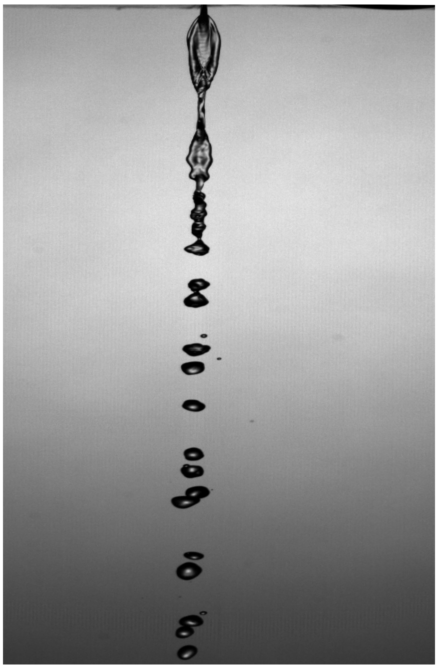
Table A.1 Working Fluid Properties



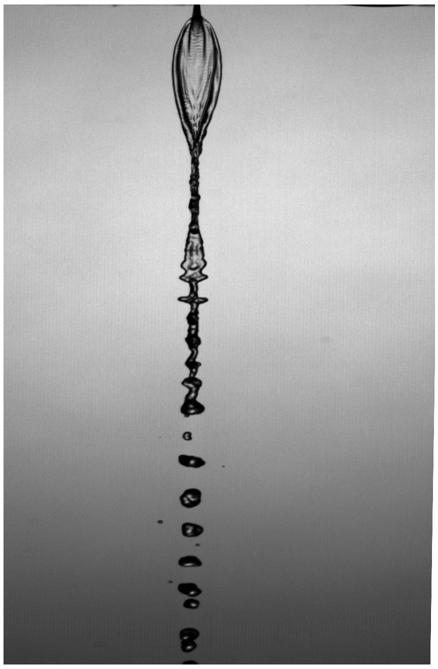
(a)



(b)



(c)



(d)

Fig. A.1 Images of LS for $t = 127\mu m$, $w_o = 1000\mu m$ and $\theta = 40^o$ for flowrate (a)$q = 30\,ml/min$ (b) $q = 40\,ml/min$ (c) $q = 50\,ml/min$ (d) $q = 60\,ml/min$

The differences in lengths are taken based on the calculations carried out by Santiago[57], by taking in consideration the eqns (11) and (12):

$$(l_j/t) = 0.23We_eCa_e^{-0.1}(1+1.5\alpha^2)^{-0.5}(1+cot(0.67\theta))^{-0.5} \qquad \text{(A.1)}$$

$$(w_j/t) = 0.074 We_e Ca_e^{-0.2}(1+1.5\alpha^2)^{-1}(1+cot(0.67\theta))^{-1} \tag{A.2}$$

Here,

Re_e is the Reynolds number at the nozzle exit given as:

$$(Re_e) = \rho Q/\mu w_o \tag{A.3}$$

Ca_e is the capillary number at the nozzle exit given as:

$$(Ca_e) = \mu Q/\sigma t w_o \tag{A.4}$$

We_e is the weber number at the nozzle exit given as:

$$(We_e) = \rho Q^2/\sigma t^2 w_o \tag{A.5}$$

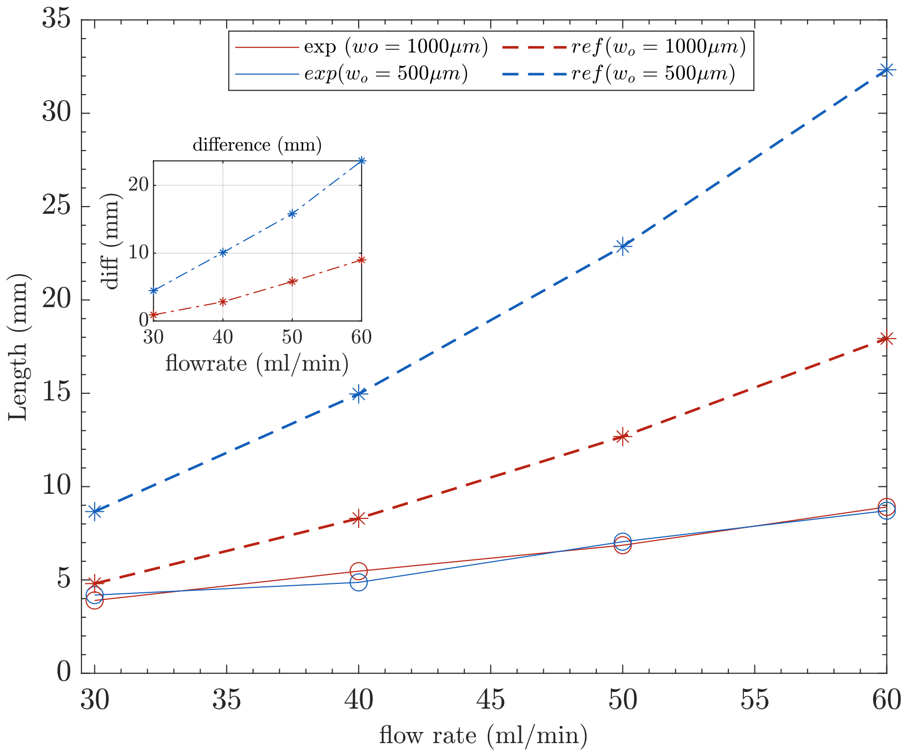


Fig. A.2 Length vs flowrate for $t = 127\mu m$ and $\theta = 40^o$

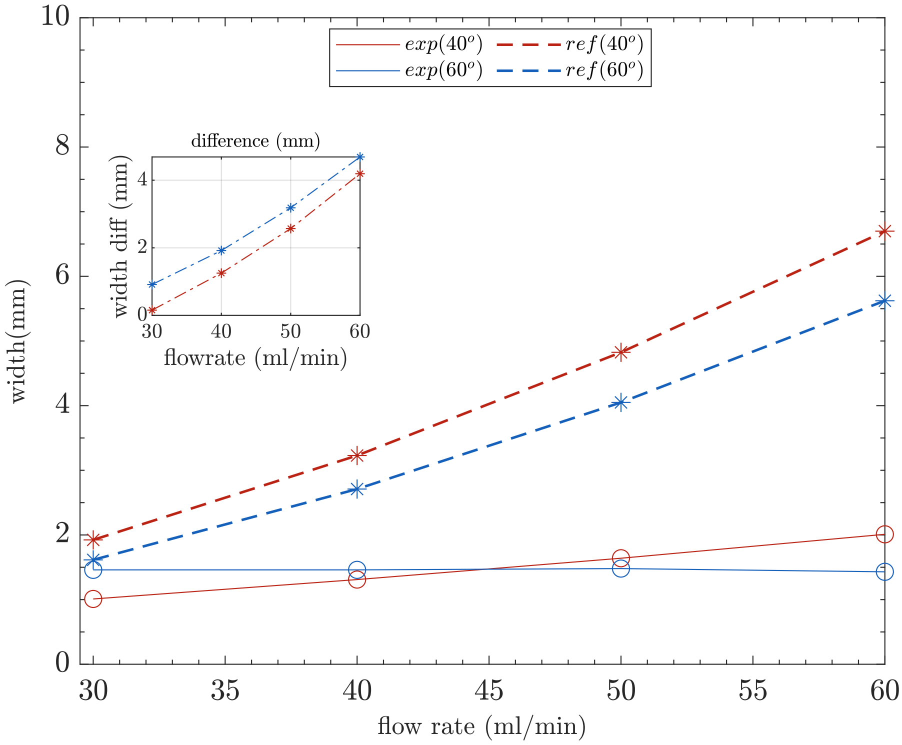
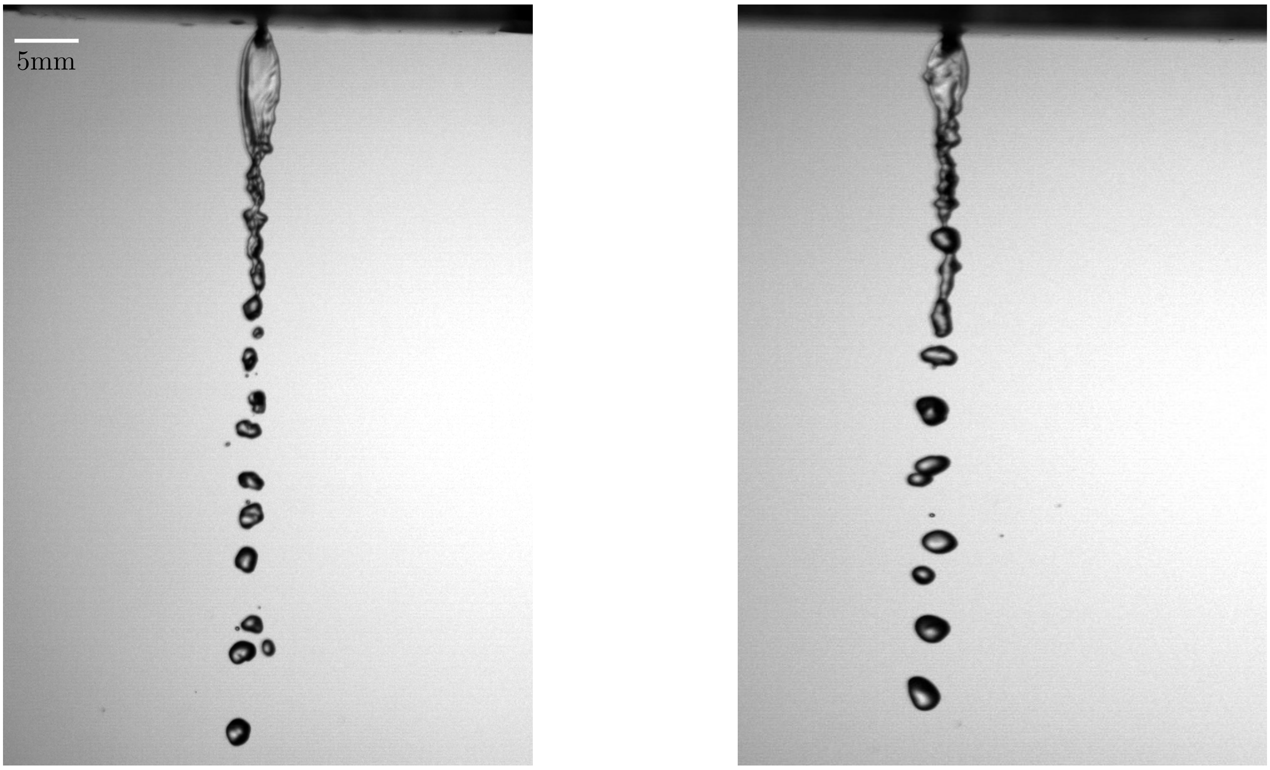


Fig. A.3 width vs flowrate for $t = 127\mu m$, $w_o = 500\mu m$ and $\theta = 40^o, 60^o$

The major difficulty faced while using the stacked model is the lack of reproducibility of the LS generated. The following figure shows the variation of the LS that is observed for the same geometry $t = 127\mu m, w_o = 1000\mu m, \theta = 60^o$ and flow rate $(q = 30ml/min)$.



(a) (b)

Fig. A.4 LS variation for $t = 127\mu m$, $w_o = 100\mu m$ and $\theta = 30^o$

Figure A.4.(b) above has a visibly shorter breakup length than A.4.(a). Also,the A.4.(b) is more unstable though both the experiments were carried out in the same fashion with equal care.

As this uncertainty can lead to variation in the measurements of various morphological elements, this method of generating a liquid sheet can not be relied upon.

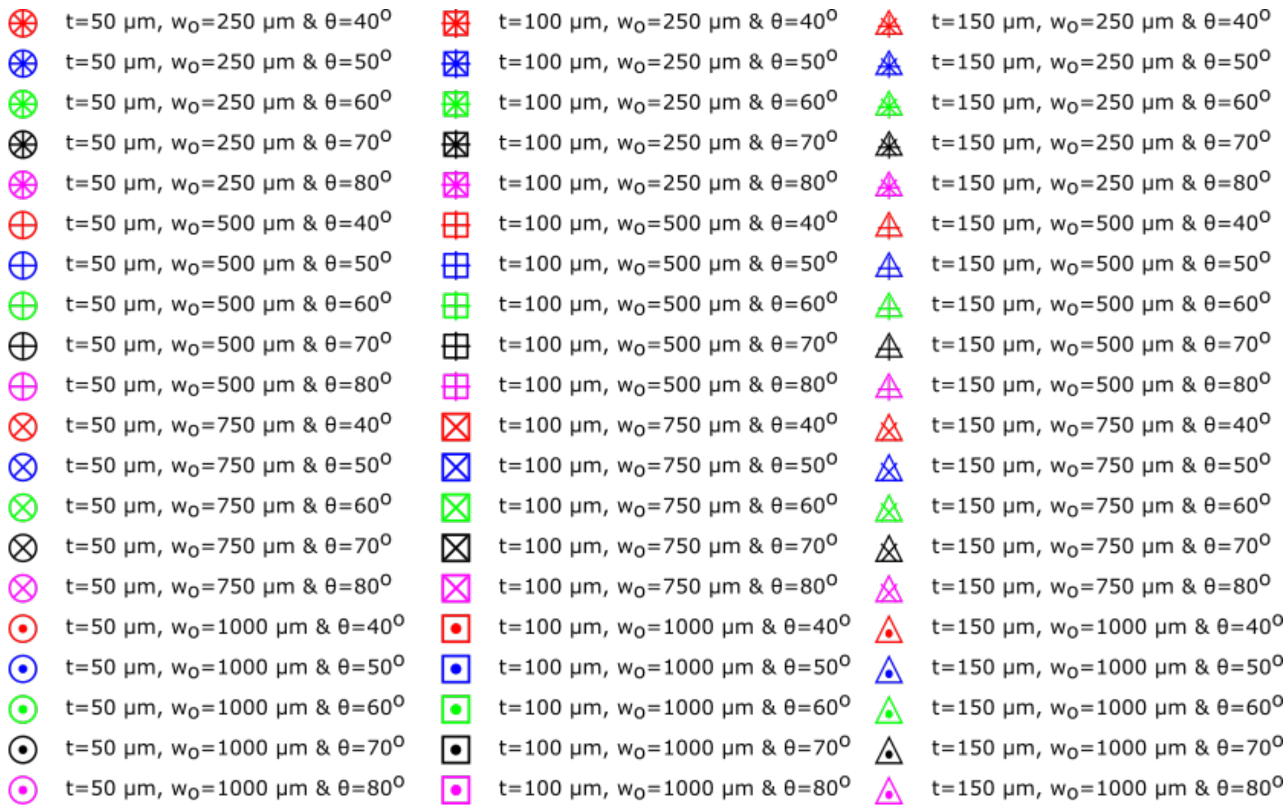


Fig. A.5 Markers used in predicting the length and width

References

[1] Ivett A. Leyva. The relentless pursuit of hypersonic flight. *Physics Today*, 70(11):30–36, 11 2017. ISSN 0031-9228. doi: 10.1063/PT.3.3762. URL https://doi.org/10.1063/PT.3.3762.

[2] Ya-Zheng Yang, Jia-Ling Yang, and Dai-Ning Fang. Research progress on thermal protection materials and structures of hypersonic vehicles. *Applied Mathematics and Mechanics*, 29(1):51–60, January 2008.

[3] Vivek Ahuja, Roy J. Hartfield, and Andrew Shelton. Optimization of hypersonic aircraft using genetic algorithms. *Applied Mathematics and Computation*, 242:423–434, 2014. ISSN 0096-3003. doi: https://doi.org/10.1016/j.amc.2014.05.120. URL https://www.sciencedirect.com/science/article/pii/S0096300314008200.

[4] H.J. Gladden. *Hypersonic Engine Component Experiments in High Heat Flux, Supersonic Flow Environment*. NASA technical memorandum. NASA, Lewis Research Center, 1993. URL https://books.google.co.il/books?id=UKY3AQAAMAAJ.

[5] NASA. X-43a hyper-x, 2024. URL https://www.nasa.gov/reference/x-43a/. NASA technical memorandum.

[6] J. Pletzer, D. Hauglustaine, Y. Cohen, P. Jöckel, and V. Grewe. The climate impact of hydrogen-powered hypersonic transport. *Atmospheric Chemistry and Physics*, 22(21):14323–14354, 2022. doi: 10.5194/acp-22-14323-2022. URL https://acp.copernicus.org/articles/22/14323/2022/. Published in Atmospheric Chemistry and Physics.

[7] Destinus D. The first hypersonic drone to be introduced. https://www.destinus.com/page/destinus-d, 2024. Accessed: 2024-11-02.

[8] Wikipedia contributors. Stratosphere — Wikipedia, the free encyclopedia, 2024. URL https://en.wikipedia.org/w/index.php?title=Stratosphere&oldid=1232893136. [Online; accessed 14-July-2024].

[9] W.A. McConarty, F.M. Anthony, Bell Aerospace Company, and Langley Research Center. *Design and Evaluation of Active Cooling Systems for Mach 6 Cruise Vehicle Wings*. NASA contractor report. National Aeronautics and Space Administration, 1971. URL https://books.google.co.il/books?id=QicuAQAAIAAJ.

[10] Javier Urzay. The physical characteristics of hypersonic flows, 2020. URL https://api.semanticscholar.org/CorpusID:221817491.

[11] Charles R. Smith. Aerodynamic heating in hypersonic flows. *Physics Today*, 74(11):66–67, 11 2021. ISSN 0031-9228. doi: 10.1063/PT.3.4888. URL https://doi.org/10.1063/PT.3.4888.

[12] Olivia Hultgren. Faster than the speed of sound. https://cse.umn.edu/aem/feature-stories/faster-speed-sound, -.

[13] Haydn Kasen, Wadley. *Thermal Management at Hypersonic Leading Edges," University of Virginia.* School of Engineering and Applied Science. Online Archive of University of Virginia Scholarship, 2013. URL https://libraetd.lib.virginia.edu/public_view/6d56zw87f.

[14] David Glass. Physical challenges and limitations confronting the use of uhtcs on hypersonic vehicles. In *17th AIAA International Space Planes and Hypersonic Systems and Technologies Conference 2011*, 04 2011. ISBN 978-1-60086-942-6. doi: 10.2514/6.2011-2304.

[15] Steven P. Schneider. Hypersonic laminar–turbulent transition on circular cones and scramjet forebodies. *Progress in Aerospace Sciences*, 40(1):1–50, 2004. ISSN 0376-0421. doi: https://doi.org/10.1016/j.paerosci.2003.11.001. URL https://www.sciencedirect.com/science/article/pii/S037604210300112X.

[16] M. HOLDEN, J. MOSELLE, A. WIETING, and C. GLASS. *Studies of aerothermal loads generated in regions of shock/shock interaction in hypersonic flow*. Aerospace Research Central, 1988. doi: 10.2514/6.1988-477. URL https://arc.aiaa.org/doi/abs/10.2514/6.1988-477.

[17] David Glass, Ray Dirling, Harold Croop, Tim Fry, and Geoffrey Frank. *Materials Development for Hypersonic Flight Vehicles*. Aerospace Research Central, 2006. doi: 10.2514/6.2006-8122. URL https://arc.aiaa.org/doi/abs/10.2514/6.2006-8122.

[18] Yiding Zhu, Wenkai Zhu, Dingwei Gu, Cunbiao Lee, and C. R. Smith. Hypersonic transition over a heated wall. *Physics of Fluids*, 33(10):101706, 10 2021. ISSN 1070-6631. doi: 10.1063/5.0068157. URL https://doi.org/10.1063/5.0068157.

[19] Wilson Santos and Mark Lewis. Aerodynamic heating performance of power law leading edges in rarefied hypersonic flow. In *36th AIAA Thermophysics Conference*, page 3894, 2003.

[20] Hemateja Ari, B. Teja, A. Kumar, and Rakesh Sg. Influence of nose radius of blunt cones on drag in supersonic and hypersonic flows. *IOP Conference Series: Materials Science and Engineering*, 225:012045, 08 2017. doi: 10.1088/1757-899X/225/1/012045.

[21] Jian-Jun Gou, Yue Chang, Zheng-Wei Yan, Bing Chen, and Chun-Lin Gong. The design of thermal management system for hypersonic launch vehicles based on active cooling networks. *Applied Thermal Engineering*, 159:113938, 2019. ISSN 1359-4311. doi: https://doi.org/10.1016/j.applthermaleng.2019.113938. URL https://www.sciencedirect.com/science/article/pii/S135943111930715X.

[22] Vinh Tung Le, Ngoc San Ha, and Nam Seo Goo. Advanced sandwich structures for thermal protection systems in hypersonic vehicles: A review. *Composites Part B: Engineering*, 226:109301, 2021. ISSN 1359-8368. doi: https://doi.org/10.1016/j.compositesb.2021.109301. URL https://www.sciencedirect.com/science/article/pii/S1359836821006752.

[23] David Glass. *Ceramic Matrix Composite (CMC) Thermal Protection Systems (TPS) and Hot Structures for Hypersonic Vehicles*. doi: 10.2514/6.2008-2682. URL https://arc.aiaa.org/doi/abs/10.2514/6.2008-2682.

[24] Jean-François Justin, Aurélie Julian-Jankowiak, Vincent Guérineau, Virginie Mathivet, and Antoine Debarre. Ultra-high temperature ceramics developments for hypersonic applications. *CEAS Aeronautical Journal*, 11(3): 651–664, September 2020.

[25] L. STEG and H. LEW. Chapter 32 - hypersonic ablation. In WILBUR C. NELSON, editor, *The High Temperature Aspects of Hypersonic Flow*, volume 68 of *AGARDograph*, pages 629–680. Elsevier, 1964. doi: https://doi.org/10.1016/B978-1-4831-9828-6.50037-0. URL https://www.sciencedirect.com/science/article/pii/B9781483198286500370.

[26] E H Hirschel and C Weiland. Design of hypersonic flight vehicles: some lessons from the past and future challenges. *CEAS Space Journal*, 1(1):3–22, September 2011.

[27] John V Becker. New approaches to hypersonic aircraft. In *INTERNATIONAL COUNCIL OF THE AERONAUTICAL SCIENCES, CONGRESS, 7TH*, number ICAS PAPER 70-16. INTERNATIONAL COUNCIL OF THE AERONAUTICAL SCIENCES,, 1970.

[28] RG Helenbrook and Frank M Anthony. Design of a convective cooling system for a mach 6 hypersonic transport airframe. Technical report, NASA, 1971.

[29] Zhang Silong, Xin Gang Li, Jingying Zuo, Qin Jiang, Kunlin Cheng, Yu Feng, and Wen Bao. Research progress on active thermal protection for hypersonic vehicles. *Progress in Aerospace Sciences*, 119:100646, 2020. URL https://api.semanticscholar.org/CorpusID:224967030.

[30] Weichen Li, Jieliang Zhao, Xiangbing Wu, Lulu Liang, Wenzhong Wang, and Shaoze Yan. Structure design and heat transfer performance analysis of a novel composite phase change active cooling channel wall for hypersonic aircraft. *Micromachines*, 15(5), 2024. ISSN 2072-666X. doi: 10.3390/mi15050623. URL https://www.mdpi.com/2072-666X/15/5/623.

[31] HS Ifti, T Hermann, M McGilvray, L Larrimbe, R Hedgecock, and L Vandeperre. Flow characterization of porous ultra-high-temperature ceramics for transpiration cooling. *AIAA JOURNAL*, 60(5):3286–3297, 2022.

[32] Wilson Santos. *Leading Edge Thickness Impact on Drag and Lift in Hypersonic Wedge Flow*. aerospace Research Central, 2012.

[33] STEPHEN E. GRENLESKI and FREDERICK S. BILLIG. Investigation of an actively cooled leading edge for hypersonic ramjet engines. *Journal of Aircraft*, 5(4):359–366, 1968. doi: 10.2514/3.43951. URL https://doi.org/10.2514/3.43951.

[34] Chaokai Yuan, Jinping Li, Zonglin Jiang, and Hongru Yu. Experimental investigation of liquid film cooling in hypersonic flow. *Physics of Fluids*, 31(4):046101, 04 2019. ISSN 1070-6631. doi: 10.1063/1.5088024. URL https://doi.org/10.1063/1.5088024.

[35] Xiao Wei Zhu and Jing Quan Zhao. Study on helium impingement cooling for a sharp leading edge subject to aerodynamic heating. *Applied Thermal Engineering*, 107:253–263, 2016. ISSN 1359-4311. doi: https://doi.org/10.1016/j.applthermaleng.2016.06.013. URL https://www.sciencedirect.com/science/article/pii/S1359431116308973.

[36] Manojprabakar Sargunaraj, Marcel Otto, Ladislav Vesely, Erik Fernandez, Jayanta Kapat, and Valerio Viti. Use of supercritical co 2 impingement cooling for a hypersonic leading edge. In *AIAA SCITECH 2022 Forum*, 01 2022. doi: 10.2514/6.2022-0266.

[37] Arvid Jahedi. *Experimental and Numerical Investigation of the Quenching Process on Rotary Hollow Cylinder by Multiple Impinging Jets*. PhD thesis, University of Gävle, 02 2021.

[38] J. Yang, L. C. Chow, and M. R. Pais. Nucleate Boiling Heat Transfer in Spray Cooling. *Journal of Heat Transfer*, 118(3):668–671, 08 1996.

[39] Ji-Xiang Wang, Wei Guo, Kai Xiong, and Sheng-Nan Wang. Review of aerospace-oriented spray cooling technology. *Progress in Aerospace Sciences*, 116:100635, 2020. ISSN 0376-0421. doi: https://doi.org/10.1016/j.paerosci.2020.100635. URL https://www.sciencedirect.com/science/article/pii/S0376042120300476.

[40] Jungho Kim. Spray cooling heat transfer: The state of the art. *International Journal of Heat and Fluid Flow*, 28(4):753 – 767, 2007. doi: 10.1016/j.ijheatfluidflow.2006.09.003. URL https://www.scopus.com/inward/record.uri?eid=2-s2.0-34447319643&doi=10.1016%2fj.ijheatfluidflow.2006.09.003&partnerID=40&md5=745c98de0552be40dc15fabf1e765013.

[41] Wen-Long Cheng, Wei-Wei Zhang, Hua Chen, and Lei Hu. Spray cooling and flash evaporation cooling: The current development and application. *Renewable and Sustainable Energy Reviews*, 55:614–628, 03 2016. doi: 10.1016/j.rser.2015.11.014.

[42] John Bush and ALEXANDER HASHA. On the collision of laminar jets: fluid chains and fishbones. *Journal of Fluid Mechanics*, 511:285 – 310, 07 2004. doi: 10.1017/S002211200400967X.

[43] Christophe Clanet. Waterbells and liquid sheets. *Annual Review of Fluid Mechanics*, 39(Volume 39, 2007):469–496, 2007. ISSN 1545-4479. doi: https://doi.org/10.1146/annurev.fluid.39.050905.110317. URL https://www.annualreviews.org/content/journals/10.1146/annurev.fluid.39.050905.110317.

[44] Savart F. Wemoire sur la constitution des veines liquids lancees par des orifices circulaires en mince paroi. *Ann. de chim.*, 53:337–386, 1833.

[45] H B Squire. Investigation of the instability of a moving liquid film. *British Journal of Applied Physics*, 4(6):167, jun 1953. doi: 10.1088/0508-3443/4/6/302. URL https://dx.doi.org/10.1088/0508-3443/4/6/302.

[46] Geoffrey Taylor. Formation of thin flat sheets of water. *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*, 259(1296):1–17, 1960. doi: 10.1098/rspa.1960.0207. URL https://royalsocietypublishing.org/doi/abs/10.1098/rspa.1960.0207.

[47] Y. J. Choo and B. S. Kang. The effect of jet velocity profile on the characteristics of thickness and velocity of the liquid sheet formed by two impinging jets. *Physics of Fluids*, 19(11):112101, 11 2007. ISSN 1070-6631. doi: 10.1063/1.2795780. URL https://doi.org/10.1063/1.2795780.

[48] N. Dombrowski and W.R. Johns. The aerodynamic instability and disintegration of viscous liquid sheets. *Chemical Engineering Science*, 18(3):203–214, 1963. ISSN 0009-2509. doi: https://doi.org/10.1016/0009-2509(63)85005-8. URL https://www.sciencedirect.com/science/article/pii/0009250963850058.

[49] E. VILLERMAUX and C. CLANET. Life of a flapping liquid sheet. *Journal of Fluid Mechanics*, 462:341–363, 2002. doi: 10.1017/S0022112002008376.

[50] Erik Miller, Beau Gibson, Erik McWilliams, and Jonathan P. Rothstein. Collision of viscoelastic jets and the formation of fluid webs. *Applied Physics Letters*, 87(1):014101, 06 2005. ISSN 0003-6951. doi: 10.1063/1.1984099. URL https://doi.org/10.1063/1.1984099.

[51] E. A. Ibrahim and A. J. Przekwas. Impinging jets atomization. *Physics of Fluids A: Fluid Dynamics*, 3(12):2981–2987, 12 1991. ISSN 0899-8213. doi: 10.1063/1.857840. URL https://doi.org/10.1063/1.857840.

[52] Jake D Koralek, Jongjin B Kim, Petr Brůža, Chandra B Curry, Zhijiang Chen, Hans A Bechtel, Amy A Cordones, Philipp Sperling, Sven Toleikis, Jan F Kern, Stefan P Moeller, Siegfried H Glenzer, and Daniel P DePonte. Generation and characterization of ultrathin free-flowing liquid sheets. *Nature Communications*, 9(1):1353, April 2018.

[53] Li-jun Yang, Fei Zhao, Qing-fei Fu, and Kun-da Cui. Liquid sheet formed by impingement of two viscous jets. *Journal of Propulsion and Power*, 30(4):1016–1026, 2014. doi: 10.2514/1.B35105. URL https://doi.org/10.2514/1.B35105.

[54] Philipp Erni and Amal Elabbadi. Free impinging jet microreactors: Controlling reactive flows via surface tension and fluid viscoelasticity. *Langmuir : the ACS journal of surfaces and colloids*, 29, 06 2013. doi: 10.1021/la401017z.

[55] David E. Hertzog, Benjamin Ivorra, Bijan Mohammadi, Olgica Bakajin, and Juan G. Santiago. Optimization of a microfluidic mixer for studying protein folding kinetics. *Analytical Chemistry*, 78(13):4299–4306, 2006. doi: 10.1021/ac051903j. URL https://doi.org/10.1021/ac051903j. PMID: 16808436.

[56] Grega Belšak, Sasa Bajt, and Bozidar Sarler. Computational modeling and simulation of gas focused liquid micro-sheets. *International Journal of Multiphase Flow*, 140:103666, 04 2021. doi: 10.1016/j.ijmultiphaseflow.2021.103666.

[57] Byunghang Ha, Daniel P. DePonte, and Juan G. Santiago. Device design and flow scaling for liquid sheet jets. *Phys. Rev. Fluids*, 3:114202, Nov 2018. doi: 10.1103/PhysRevFluids.3.114202. URL https://link.aps.org/doi/10.1103/PhysRevFluids.3.114202.

[58] Y J Choo and B S Kang. Parametric study on impinging-jet liquid sheet thickness distribution using an interferometric method. *Experiments in Fluids*, 31(1):56–62, July 2001.

[59] Samuel Menzi, Gregor Knopp, Andre Al Haddad, Sven Augustin, Camelia Borca, Dardan Gashi, Thomas Huthwelker, Daniel James, Jiaye Jin, Georgios Pamfilidis, Kirsten Schnorr, Zhibin Sun, Reto Wetter, Qiang Zhang, and Claudio Cirelli. Generation and simple characterization of flat, liquid jets. *Review of Scientific Instruments*, 91(10):105109, 10 2020. ISSN 0034-6748. doi: 10.1063/5.0007228. URL https://doi.org/10.1063/5.0007228.

[60] David Hasson and Ralph E. Peck. Thickness distribution in a sheet formed by impinging jets. *Aiche Journal*, 10:752–754, 1964. URL https://api.semanticscholar.org/CorpusID:97682124.

[61] Takao Inamura and Minori Shirota. Effect of velocity profile of impinging jets on sheet characteristics formed by impingement of two round liquid jets. *International Journal of Multiphase Flow*, 60:149–160, 2014. ISSN 0301-9322. doi: https://doi.org/10.1016/j.ijmultiphaseflow.2013.12.006. URL https://www.sciencedirect.com/science/article/pii/S0301932213002188.

[62] H B Squire. Investigation of the instability of a moving liquid film. *British Journal of Applied Physics*, 4(6):167, jun 1953. doi: 10.1088/0508-3443/4/6/302. URL https://dx.doi.org/10.1088/0508-3443/4/6/302.

[63] N. Dombrowski, R. P. Fraser, and Dudley Maurice Newitt. A photographic investigation into the disintegration of liquid sheets. *Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences*, 247(924):101–130, 1954. doi: 10.1098/rsta.1954.0014. URL https://royalsocietypublishing.org/doi/abs/10.1098/rsta.1954.0014.

[64] Wolfgang Sander and Bernhard Weigand. Direct numerical simulation and analysis of instability enhancing parameters in liquid sheets at moderate Reynolds numbers. *Physics of Fluids*, 20(5):053301, 05 2008. ISSN 1070-6631. doi: 10.1063/1.2909661. URL https://doi.org/10.1063/1.2909661.

[65] Yong cui Ma, Fu qiang Bai, Qing Chang, Ji ming Yi, Kui Jiao, and Qing Du. An experimental study on the atomization characteristics of impinging jets of power law fluid. *Journal of Non-Newtonian Fluid Mechanics*, 217:49–57, 2015. ISSN 0377-0257. doi: https://doi.org/10.1016/j.jnnfm.2015.01.001. URL https://www.sciencedirect.com/science/article/pii/S0377025715000038.

[66] Ri Li and Nasser Ashgriz. Characteristics of liquid sheets formed by two impinging jets. *Physics of Fluids*, 18(8):087104, 08 2006. ISSN 1070-6631. doi: 10.1063/1.2338064. URL https://doi.org/10.1063/1.2338064.

[67] Navish Wadhwa, Pavlos Vlachos, and Sunghwan Jung. Noncoalescence in the oblique collision of fluid jets. *Phys. Rev. Lett.*, 110:124502, Mar 2013. doi: 10.1103/PhysRevLett.110.124502. URL https://link.aps.org/doi/10.1103/PhysRevLett.110.124502.

[68] Daniel Weihs. Stability of thin, radially moving liquid sheets. *Journal of Fluid Mechanics*, 87:289 – 298, 1978. URL https://api.semanticscholar.org/CorpusID:123522792.

[69] H. M. Ryan, W. E. Anderson, S. Pal, and R. J. Santoro. Atomization characteristics of impinging liquid jets. *Journal of Propulsion and Power*, 11(1):135–145, 1995. doi: 10.2514/3.23851. URL https://doi.org/10.2514/3.23851.

[70] V. Nagabhushana Rao and K. Ramamurthi. Internal instability of thin liquid sheets. *Physics of Fluids*, 21(9):092106, 09 2009. ISSN 1070-6631. doi: 10.1063/1.3234190. URL https://doi.org/10.1063/1.3234190.

[71] Akira Watanabe, Hiroshi Saito, Yuzo Ishida, Masafumi Nakamoto, and Tatsuo Yajima. A new nozzle producing ultrathin liquid sheets for femtosecond pulse dye lasers. *Optics Communications*, 71(5):301–304, 1989. ISSN 0030-4018. doi: https://doi.org/10.1016/0030-4018(89)90012-6. URL https://www.sciencedirect.com/science/article/pii/0030401889900126.

[72] Alexandra M.F.R. Pinto, Vânia B. Oliveira, and Daniela S. Falcão. 7 - miniaturization of direct alcohol fuel cells: Microfabrication techniques and microfluidic architectures. In Alexandra M.F.R. Pinto, Vânia B. Oliveira, and Daniela S. Falcão, editors, *Direct Alcohol Fuel Cells for Portable Applications*, pages 245–264. Academic Press, 2018. ISBN 978-0-12-811849-8. doi: https://doi.org/10.1016/B978-0-12-811849-8.00007-3. URL https://www.sciencedirect.com/science/article/pii/B9780128118498000073.

[73] G. V. Kaigala, R. D. Lovchik, U. Drechsler, and E. Delamarche. A vertical microfluidic probe. *Langmuir*, 27(9):5686–5693, 2011. doi: 10.1021/la2003639. URL https://doi.org/10.1021/la2003639. PMID: 21476506.

[74] Franz Laermer, Sami Franssila, Lauri Sainiemi, and Kai Kolari. Chapter 16 - deep reactive ion etching. In Markku Tilli, Mervi Paulasto-Krockel, Matthias Petzold, Horst Theuss, Teruaki Motooka, and Veikko Lindroos, editors, *Handbook of Silicon Based MEMS Materials and Technologies (Third Edition)*, Micro and Nano Technologies, pages 417–446. Elsevier, third edition edition, 2020. ISBN 978-0-12-817786-0. doi: https://doi.org/10.1016/B978-0-12-817786-0.00016-5. URL https://www.sciencedirect.com/science/article/pii/B9780128177860000165.

[75] Gary S Settles. *Schlieren and shadowgraph techniques: visualizing phenomena in transparent media.* Springer Science & Business Media, 2001.

[76] Richard Goldstein. Fluid mechanics measurements. *Taylor and Francis*, 2017.

[77] G. S. Settles. Review of flow visualization techniques in heat and mass transfer. *Journal of Heat Transfer*, 111(4):848–859, 1989. doi: 10.1115/1.3248971.

[78] K Okamoto, S Nishio, T Saga, and T Kobayashi. Standard images for particle-image velocimetry. *Measurement Science and Technology*, 11(6):685, jun 2000. doi: 10.1088/0957-0233/11/6/311. URL https://dx.doi.org/10.1088/0957-0233/11/6/311.

[79] Mohd-Zulhilmi Ismadi, Simon Higgins, Chaminda Samarage, David Paganin, Kerry Hourigan, and Andreas Fouras. Optimisation of a stirred bioreactor through the use of a novel holographic correlation velocimetry flow measurement technique. *PloS one*, 8:e65714, 06 2013. doi: 10.1371/journal.pone.0065714.

[80] Velocimetry.net. Principle of particle image velocimetry, 2024. URL https://velocimetry.net/principle.htm. Accessed: 2024-07-23.

[81] Cospheric. Flow visualization (piv) tracers. URL https://www.cospheric.com/particles_for_flow_visualization_PIV/density_1g.htm.

[82] J G Santiago, S T Wereley, C D Meinhart, D J Beebe, and R J Adrian. A particle image velocimetry system for microfluidics. *Experiments in Fluids*, 25(4):316–319, September 1998.

[83] A K Prasad. Stereoscopic particle image velocimetry. *Experiments in Fluids*, 29(2):103–116, August 2000.

[84] C D Meinhart, S T Wereley, and J G Santiago. PIV measurements of a microchannel flow. *Experiments in Fluids*, 27(5):414–419, October 1999.

[85] A Melling. Tracer particles and seeding for particle image velocimetry. *Measurement Science and Technology*, 8(12):1406, dec 1997. doi: 10.1088/0957-0233/8/12/005. URL https://dx.doi.org/10.1088/0957-0233/8/12/005.

[86] Willert C. E. Wereley S. T. Kompenhans J. Raffel, M. *Particle Image Velocimetry: A Practical Guide*. Springer, 2007.

[87] Ronald J Adrian and Jerry Westerweel. *Particle image velocimetry*. Number 30. Cambridge university press, 2011.

[88] Ansys® Academic Research Mechanical. Ansys. *Ansys User Manual*, 2018.

[89] Vatsal Sanjay and Arup Das. Formation of liquid chain by collision of two laminar jets. *Physics of Fluids*, 29, 11 2017. doi: 10.1063/1.4998288.

[90] Araz Sarchami, Nasser Ashgriz, and Honghi Tran. Initial perturbation amplitude of liquid sheets produced by jet-impingement nozzles. *AIAA Journal*, 47(11):2775–2779, 2009. doi: 10.2514/1.37830. URL https://doi.org/10.2514/1.37830.

[91] David J Hoffman, Tim B Van Driel, Thomas Kroll, Christopher J Crissman, Elizabeth S Ryland, Kacie J Nelson, Amy A Cordones, Jake D Koralek, and Daniel P DePonte. Microfluidic liquid sheets as large-area targets for high repetition XFELs. *Front Mol Biosci*, 9:1048932, December 2022.

[92] J. C. P. Huang. The break-up of axisymmetric liquid sheets. *Journal of Fluid Mechanics*, 43(2):305–319, 1970. doi: 10.1017/S0022112070002392.

[93] GD Crapper, Norman Dombrowski, and GAD Pyott. Large amplitude kelvin-helmholtz waves on thin liquid sheets. *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences*, 342(1629):209–224, 1975.

[94] Xiaodong Chen, Dongjun Ma, Vigor Yang, and Stephane Popinet. High-fidelity simulations of impinging jet atomization. *Atomization and sprays*, 23 (12), 2013.

[95] Christopher J. Crissman, Mianzhen Mo, Zhijiang Chen, Jie Yang, Diego A. Huyke, Siegfried H. Glenzer, Kathryn Ledbetter, J. Pedro F. Nunes, May Ling Ng, Hengzi Wang, Xiaozhe Shen, Xijie Wang, and Daniel P. DePonte. Sub-micron thick liquid sheets produced by isotropically etched glass nozzles. *Lab Chip*, 22:1365–1373, 2022. doi: 10.1039/D1LC00757B. URL http://dx.doi.org/10.1039/D1LC00757B.

[96] Kuo-Shen Chen, A.A. Ayon, Xin Zhang, and S.M. Spearing. Effect of process parameters on the surface morphology and mechanical performance of silicon structures after deep reactive ion etching (drie). *Journal of Microelectromechanical Systems*, 11(3):264–275, June 2002. ISSN 1941-0158. doi: 10.1109/JMEMS.2002.1007405.

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חיבור על מחקר

לשם מילוי חלקי של הדרישות לקבלת התואר
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פריינקה סינה

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